

FUNDAMENTALS AND APPLICATIONS OF CHEMISTRY

EXAM PREPARATION GUIDE

Dr. R. S. S. Srikanth Vemuri

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Fundamentals and Applications of Chemistry: Exam Preparation Guide

First Edition

Author

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Dedicated To

This book is dedicated to my parents and my wife, whose unwavering support, love, and encouragement have been my greatest sources of strength. Their belief in me has made this journey possible, and for that, I am eternally thankful.



Preface

This book, "Fundamentals and Applications of Chemistry: Exam Preparation Guide," is designed to serve as a comprehensive resource for first-year Bachelor of Technology (B. Tech) students specializing in Electrical and Electronics Engineering (EEE), Electronics and Communication Engineering (ECE), Computer Science and Engineering and allied branches (CSE, CAI, CSM, CSC & CSD) and Information Technology (IT). It aims to provide a thorough understanding of key concepts in chemistry, tailored to meet the curriculum requirements of JNTU - GV.

Chemistry is a foundational science that plays a crucial role in various engineering disciplines. From the principles governing molecular interactions to the applications of advanced materials, a solid grasp of chemistry is essential for engineering students. This guide is structured to facilitate both learning and revision, ensuring students are well-prepared for their examinations.

The book comprises a rich collection of questions that cover a wide array of topics, including quantum mechanics, electrochemistry, polymers, and spectroscopy. These questions are crafted to assess and enhance students' understanding of basic principles and their ability to apply this knowledge in practical scenarios.

Each chapter delves into a specific area of chemistry, providing opportunities for detailed exploration and critical thinking. By engaging with these questions, students will develop a deeper understanding of both the theoretical and practical aspects of chemistry. The content is designed to not only prepare students for their exams but also to cultivate a comprehensive understanding that will be beneficial throughout their engineering careers.

This book emphasizes the applications of chemistry in real-world scenarios, demonstrating the relevance of chemical principles in engineering and technology. By integrating theoretical knowledge with practical examples, students can appreciate the importance of chemistry in their chosen field.

Chemistry is more than an academic subject; it is a lens through which we can understand the world and innovate for the future. The ability to analyze materials, comprehend reaction mechanisms, and apply this knowledge to solve practical problems is crucial in various engineering disciplines. This book aims to bridge the gap between theoretical principles and practical applications, fostering a deeper appreciation and understanding of the subject.

I have meticulously compiled this guide to ensure it aligns with the latest curriculum and pedagogical standards. Each chapter is structured to build on previous knowledge, gradually advancing to more complex concepts. The inclusion of diverse question types aims to enhance critical thinking and problem-solving skills, preparing students not only for exams but also for real-world engineering challenges.

I hope this book will serve as a valuable tool for students, helping them build a strong foundation in chemistry and excel in their examinations. Whether used as a primary study resource or a supplementary revision guide, it is designed to support students in their academic journey and foster a lifelong appreciation for the fascinating world of chemistry.

- Dr. R. S. S. Srikanth Vemuri

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I would like to express my deepest gratitude to the Principal of Vignan's Institute of Engineering for Women for their continuous support and encouragement throughout the development of this book. Their dedication to fostering an environment of academic excellence has been truly inspiring.

I am also profoundly grateful to the Head of the Basic Sciences and Humanities Department for their invaluable guidance and assistance. Their expertise and commitment have been instrumental in shaping the content and direction of this work.

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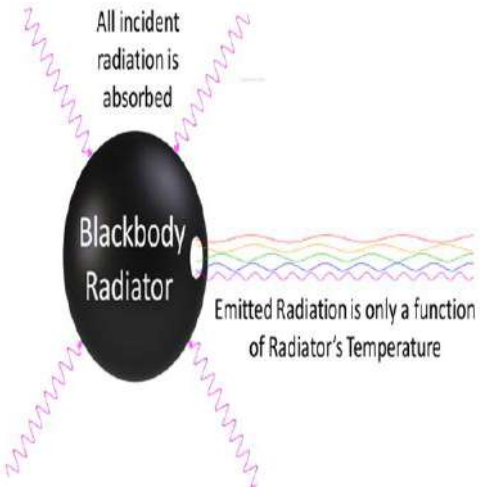
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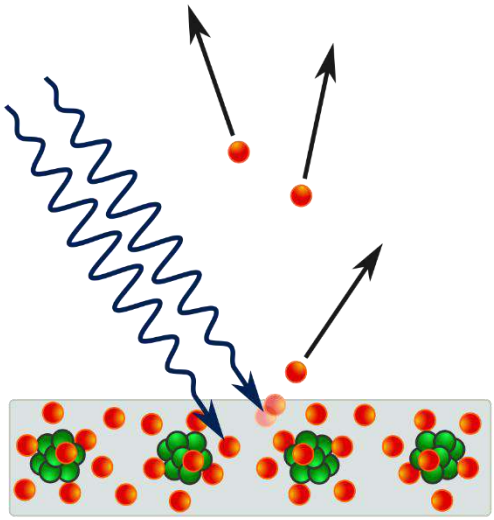
Structure and Bonding Models



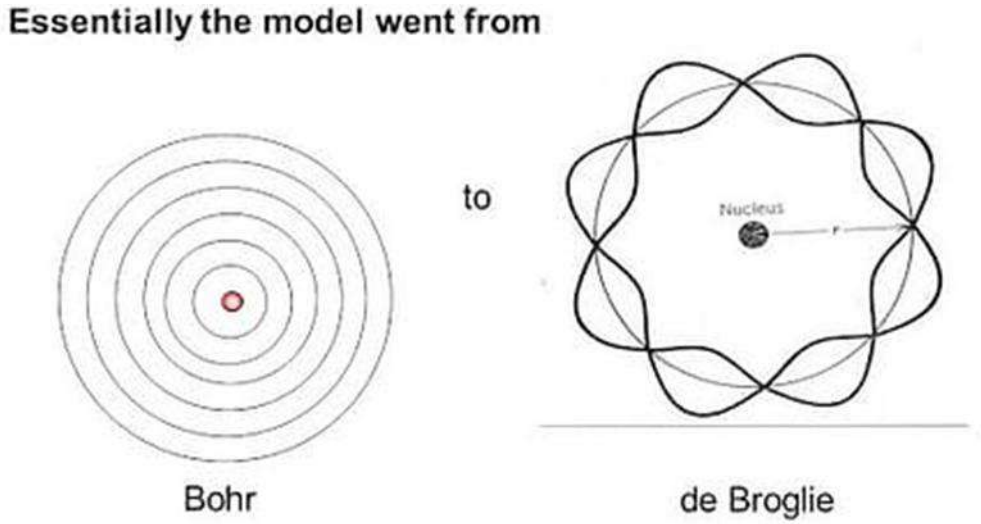
Introduction

- ✓ The history of quantum mechanics is a fundamental part of the history of modern physics.
- ✓ The main chapters begin with
  - Black body radiation
  - Photo electric effect
  - Solar emission spectra
- ✓ These are old quantum theories.
- ✓ The invention of Schrodinger wave equation around 1925 is the beginning of modern quantum theories.
- ✓ The phrase "Quantum mechanics" was coined by the group of physicists including Max Born, Werner Heisenberg & Wolfgang Pauli in 1920.
- ✓ Wave theory of light developed by Newton (1670) and later light nature explained by Youngs Double slit experiment (1801). Then Maxwell developed Maxwells equation, these are the beginning of quantum mechanics.
- ✓ In early 19th century
  - Dalton, Avogadro developed by Atomic Theory of matter
  - Boltzmann developed a kinetic theory of gases.
  - Max Planck developed quantum theory.
- ✓ Old quantum theory failed to understand the classical mechanics.





- ✓ In modern quantum mechanics, Max Planck introduces quanta to explain black body radiation then Einstein applied quanta to explain the photoelectric effect.
- ✓ Bohr model of atom for the quantization of matter.
- ✓ 1924, DE Broglie's findings indicates that matter have wave properties.
- ✓ The end of the first era of quantum mechanics was triggered by De Broglie's equation leading to Schrodinger's wave equation to give way to modern quantum theory.



1.1 Engineering Applications of Quantum Mechanics

- ✓ Laser
- ✓ Super fluid Helium
- ✓ Superconductors
- ✓ Semiconductors
- ✓ Flash memory chips found in USB devices also use quantum tunnelling, to erase their memory cells.

1.2 Differences Between Classical and Quantum Mechanics

Classical Mechanics	Quantum Mechanics
Newton's law Maxwell equations	Schrodinger wave equation De Broglie's wave equation Collection of postulates
Macroscopic phenomena (Kinematics)	Microscopic phenomena
Classical Point of view	Quantum point of view

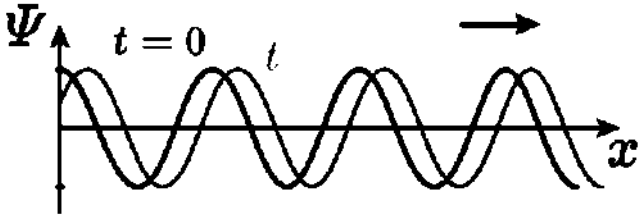
1.3 Schrodinger Wave Equation

1. Schrodinger's equation (1925) refers to a fundamental equation of quantum physics.
2. In classical physics, it is parallel to Newton's laws of motion.
3. This equation does not tell the position of the sub atomic particles at any future point in time,
4. It will tell only the possible position and probabilities of being in the possible positions.
5. For example, if we use a laser to shoot some photons towards a photographic plate, this equation can help you calculate the overall pattern of pixels that will form on plate but not which photon would light up.
6. This equation helps at statistical level but not at the individual particle level.
7. This equation expressed in terms of mathematical expression. It explains the transformation of the physical quantity with respect to time, where the quantum effects are like a wave – particle duality.

8. The equation has two forms, **time independent and time dependent** Schrodinger's equation. Time independent equation can be easily derived from time dependent equation. However, these are independent equations.
   The time – Dependent Equation: The wave function of a particle with fixed energy E was naturally expressed as a linear combination of wave function

$$\boldsymbol{\psi(x,t) = Ae^{(i(kx-\omega t))} = A[Cos(kx-\omega t)+isin(kx-\omega t)]}$$

   Where A is the normalization constant
   ω is angular frequency
   k is the wave number or Propagation constant is $2\pi/\lambda$
   t is taken by the particle.
   To create a standing wave and satisfy the boundary criteria,



9. This is a representation of a wave moving the positive x direction and a similar wave moving the opposite direction.
10. This indicates that we select the wave function described above as the proper wave function for a free particle because it naturally relates to our traditional idea of particle bouncing back and forth boundaries of the potential wall.
11. The free particle has momentum p = ħk and energy E = ħω
    where ħ is the correction to Planck's constant and ħ = h/2π and ω is angular frequency.

$$\boldsymbol{\psi(x,t) = Ae^{(i(kx-\omega t))}} \ldots (1)$$

Differentiate w. r. t time 't'

$$\frac{\partial \psi}{\partial t} = -i\omega A e^{i(kx-\omega t)}$$

$$\boxed{\frac{\partial \psi}{\partial t} = -i\omega\psi(x,t) \qquad \ldots (2)}$$

Double differentiate eq (1) w. r. t time 'x'

$$\frac{\partial \psi}{\partial x} = ikAe^{i(kx-\omega t)}$$

$$\frac{\partial^2 \psi}{\partial x^2} = i^2k^2Ae^{i(kx-\omega t)}$$

$$\boxed{\frac{\partial^2 \psi}{\partial x^2} = -k^2\psi(x,t) \qquad \ldots (3)}$$

We also know that, p = ħk, then
k = p/ħ
k is substitute in eq (3)

$$\frac{\partial^2\psi}{\partial x^2} = -\frac{p^2}{\hbar^2}\psi(x,t) \quad \ldots (4)$$

Energy equation: $E = \frac{p^2}{2m} + V(x)$ … (5)

After multiplying energy equation (5) with ψ (x, t) we will get,

$$E\psi(x,t) = \frac{p^2}{2m}\psi(x,t) + V(x)\psi(x,t) \quad \ldots (6)$$

From equation (4), $p^2 = -\frac{\hbar^2}{\psi(x,t)}\frac{\partial^2\psi}{\partial x^2}$ …. (7)

Substitute equation 7 in equation 6

$$E\psi(x,t) = -\frac{\hbar^2}{2m\psi(x,t)}\frac{\partial^2\psi}{\partial x^2}\psi(x,t) + V(x)\psi(x,t) \quad \ldots (8)$$

We know that E = hν or **E = ħω** from Planck's energy relation. Substitute E value in above equation (8)

$$\hbar\omega\,\psi(x,t) = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t) \quad \ldots (9)$$

From previous equation $\frac{\partial\psi}{\partial t} = -i\omega\psi(x,t)$, can be rearrange as $\psi(x,t) = -\frac{1}{i\omega}\frac{\partial\psi}{\partial t}$

$$\psi(x,t) = \frac{i^2}{i\omega}\frac{\partial\psi}{\partial t}$$

$$\psi(x,t) = \frac{i}{\omega}\frac{\partial\psi}{\partial t} \quad \ldots (10)$$

Substitute equation 10 in equation 9

$$\hbar\omega\frac{i}{\omega}\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t)$$

$$i\hbar\frac{\partial\psi}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t) \quad \ldots (11)$$

Substitute ħ = $\frac{h}{2\pi}$ in equation (11)

$$i\frac{h}{2\pi}\frac{\partial\psi}{\partial t} = -\frac{h^2}{8\pi^2 m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t) \qquad \ldots (12)$$

This is the Schrodinger's time dependent wave equation.

Time-Independent Equation

Now we solve time independent wave equation, we use wave function as $\psi(x,t) = \psi(x)e^{-iEt/\hbar}$

Substitute $\psi(x,t) = \psi(x)e^{-iEt/\hbar}$ in Schrodinger's time dependent wave equation (11)

$$i\hbar\frac{\partial\psi(x)e^{-iEt/\hbar}}{\partial t} = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}e^{-iEt/\hbar} + V(x)\psi(x)e^{-iEt/\hbar}$$

$$i\hbar.\frac{-iE}{\hbar}.e^{-iEt/\hbar.}\,\psi(x) = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2}e^{-Et/\hbar} + V(x)\psi(x)e^{-Et/\hbar}$$

Take common $e^{-iEt/\hbar}$ on both sides and cancel it, then

$$E\psi(x) = -\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} + V(x)\psi(x)$$

Re arrange as

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x)}{\partial x^2} + (V(x) - E)\psi(x) = 0 \qquad \ldots (13)$$

Substitute ħ = $\frac{h}{2\pi}$ in equation (13)

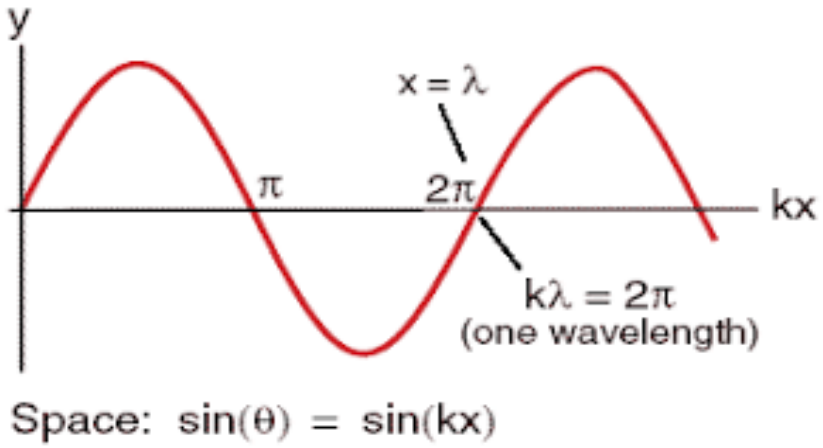
$$-\frac{h^2}{8\pi^2 m}\frac{\partial^2\psi(x)}{\partial x^2} + (V(x) - E)\psi(x) = 0 \qquad \ldots (14)$$

This is the Schrodinger's time independent wave equation

1.4 Significance of Ψ and Ψ^2

1. In the case of light or sound, the square of the amplitude of the wave at any point will give us the intensity of light or sound at that particular point.
2. Similarly, the square of the amplitude given electron wave at any point will give us the intensity of electron wave at that point which due to Heisenberg uncertainty principle means the probability of finding the electron at that point.
3. The square of the amplitude of electrons at any point gives the probability of finding the electron at that point is electron density at that point.

4. The region around the nucleus representing the electron density at different points is called an orbital.
5. The wave function for an electron in an atom is called the orbital wave function or atomic orbital.
6. Since electrons can have any wave function, there are many atomic orbitals in an atom.
7. The probability will provide the best description of any such situation which will be unable to describe with certainly.
8. Most of the time electrons are used to spend in the volume of the sphere which is bounded by the distance and for the rest of the time, the electrons can be found outside of the volume of sphere.
9. The electron cloud is the region of space around the nucleus that can describe the probability of finding of electron of given energy which is given in terms of dots.
10. An atomic orbital can be defined as the three-dimensional space around the nucleus with in which the probability of finding an electron of a given energy is the maximum.
11. Normalization of wave function means, the probability of finding a particle if it is existing in the system becomes 1. This explains that the exact form wave function ψ is found.



Physical Significance of ψ and ψ^2

1. As the wave function is not a quantity that can be observed there is no physical meaning to the wave function. Instead, it is a complex.
2. It is expressed as ψ (x,y,z) = a + ib and conjugate of wave function ψ*(x,y,z) = a – ib
3. The product of these two terms indicates the probability density of finding a particle in a space at a time.
4. Ψ^2 is a physical interpretation of wave function as it is providing the probability information for locating a particle in a given time.

1.5 Fundamentals of Quantum Mechanics

First Postulate:

Every physically – realizable state of the system is described in quantum mechanics by a state function ψ that contains all accessible physical information about the system in that state.

State function ψ consists position, momentum, energy which can be studied in laboratory.

If ψ_1 & ψ_2 represent two physically – realizable states of the system, then the linear combination

$$\boldsymbol{\psi = C_1\psi_1 + C_2\psi_2}$$

Where C_1 and C_2 are arbitrary complex constants.

Second Postulate:

If a system is in quantum state represented by a wave function ψ then $\boldsymbol{\hat{H}\psi = E\psi}$
Another equation can be $\boldsymbol{pdv} = |\boldsymbol{\psi}^2|\boldsymbol{dv}$ here $\boldsymbol{\psi^2}$ is the probability of the matter that in a position measurement at time t, the particle will be detected in the infinitesimal volume dv.
According to second postulate, the **integrated probability density** can be interpreted as a probability that in a measurement at time t, we will find the particle everywhere in space.

$$\int \boldsymbol{pdv} = \int |\boldsymbol{\psi}(x,y,z)|^2 \boldsymbol{dv}$$

Normalization condition for the wave function

$$\int \boldsymbol{pdv} = \int \boldsymbol{\psi}^*(x,y,z).\boldsymbol{\psi}(x,y,z).\ \boldsymbol{dv} = \mathbf{1}$$

Third Postulate:

Every observable in quantum mechanics is represented by an operator which is used to obtain physical information about the observable from the state function.
For an observable that is represented in classical physics by a function.
Q (x,p) the corresponding operator $Q(\hat{x}, \hat{p})$

Observable	**Operator**
Position	$\hat{x}$
Momentum	$\hat{p} = \frac{\hbar}{i}\frac{\partial}{\partial x}$
Energy	$\hat{E} = \frac{\hat{p}^2}{2m} + V(\hat{x})$ $= \frac{\hbar^2}{2m}\frac{\partial 2}{\partial x 2} + V(\hat{x})$

Fourth Postulate:

The time development of the state functions of an isolated quantum system is governed by the time – dependent Schrodinger wave equation

$$\hat{H}\psi = i\hbar\frac{\partial\psi}{\partial t}$$

Where $\hat{H} = \hat{T} + \hat{V}$ is the Hamiltonian of the system

Schrodinger wave equation which represents quantum equations of motion, and is of the form

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x,t) = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x,t)$$

$$= i\hbar\frac{\partial\psi}{\partial t}$$

$$= \hat{E}$$

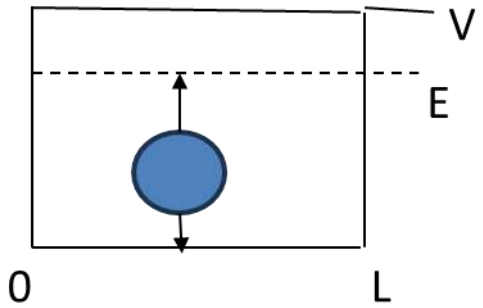
Fifth Postulate:

When a great many measurements of any observable represented by an operator $\hat{A}$ are made on a system characterised by a function ψ, then the average result is obtained by

$$\lambda = \frac{\int \psi^* \hat{A}\psi d\tau}{\int \psi^* \psi d\tau}$$

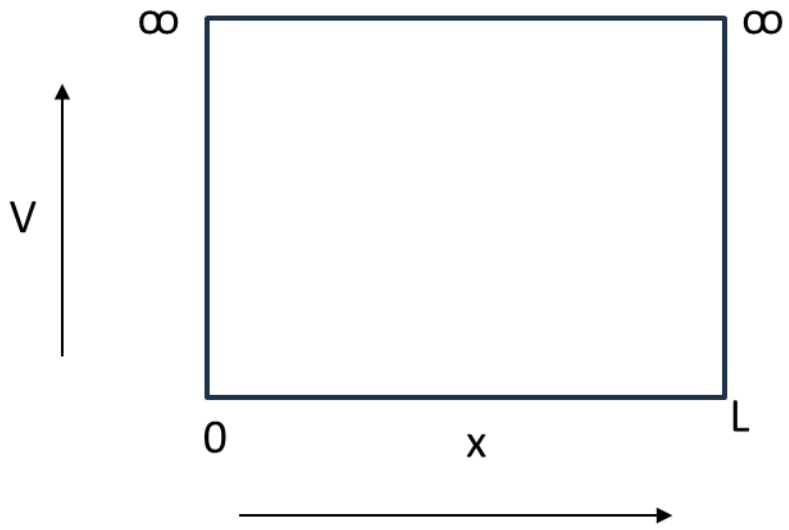
1.6 Particle in One Dimensional Box

1. The best way to understand Schrodinger's equation is to solve it for various potentials.
2. The simplest of these involving forces is a particle confinement (a particle in a box).
3. Consider a particle confined (restricted) in along with X – axis between the points x = 0 and x = L



E is the total energy of the particle
V is the potential energy of the particle
The potential difference of the walls
E<V

4. Inside the box it free but at the edges it experiences strong forces which keep it confined.
5. **Example:** A ball is bouncing between two impermeable walls.
6. Inside the well, the particle is free because U(x) is zero indie the well.
7. Increasing V(x) to infinity as the width is reduced to zero. We have idealization of an infinite potential square well.
8. Classically there is no restriction on the energy or momentum of the particle.
9. However, from quantum mechanics we have energy quantization



10. We are interested in the time independent wave function ψ (x) of the particle.

11. The particle can never be found outside the well.
i.e., ψ (x) = 0 in the region 0>x>L

12. Since V(x) = 0 we get that,

$$-\frac{\hbar^2}{2m}\frac{\partial^2\psi}{\partial x^2} + V(x)\psi(x) = E\psi(x)$$

$$E = \frac{p^2}{2m} + V(x)$$

since **p = ħk and ħ = h/2π**

$$E = \frac{(\hbar k)^2}{2m}$$

$$k^2 = \frac{2mE}{\hbar^2}$$

$$-\frac{E}{k^2}\frac{\partial^2\psi}{\partial x^2} = E\psi(x)$$

$$\frac{\partial^2\psi}{\partial x^2} = -k^2\psi(x)$$

13. The solutions to this equation are of the form ψ (x) = A sin kx + B cos kx for 0<x<L
(A linear combination of cosine and sine waves of wavenumber k)

14. The wave must be continuous which means the interior wave must match the exterior wave at the boundaries of the wall.

15. Therefore, the wave **ψ(x) must be zero** at the boundaries. **x = 0 and x = L**

At x = 0, ψ (x) = A sin kx + B cos kx → ψ (x) = A sin 0 + B cos 0 → ψ (x) = A.0 + B.1 = 0

ψ (x) = 0 + B = 0

ψ (x) = B = 0

16. At x =L, ψ (x) = A sin kx + B cos kx → ψ (x) = A sin k.L + B cos k.L → ψ (x) = A sin kx = 0
Since A ≠ 0 so sin k.L = 0 (since sin nπ = 0)
k.L = nπ where n =1, 2…….

17. From this we find the particle energy is quantized.

18. The restricted values are $E = \frac{(\hbar k)^2}{2m}$
k = nπ/L

$$E = \frac{(\hbar n\pi)^2}{2mL^2}$$

$$E = \frac{\hbar^2 n^2 \pi^2}{2mL^2}$$

$$\boldsymbol{E = \frac{\hbar^2 n^2 \pi^2}{8\pi^2 mL^2}}$$

E = 0 not allowed
n = 1 ground state
n = 2, 3, … excited states
Finally, given k and B are write the waveform as

$$\boldsymbol{\psi_n = Asin(\frac{n\pi x}{L})}$$

We need to determine, A
To do this we need to normalization
Normalizing

$$\int_0^L \psi_n^*(x)\psi_n(x)dx = 1$$

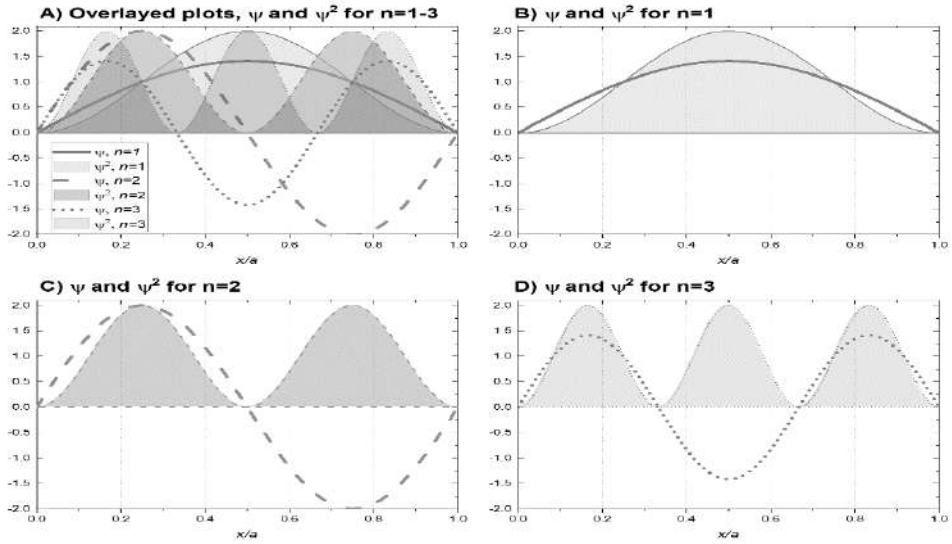
$$\int_0^L A^2 \sin 2\left(\frac{n\pi x}{L}\right) dx = 1$$

$$A^2 \int_0^L sin2\left(\frac{n\pi x}{L}\right) dx = 1$$

$A^2 . \frac{L}{2} = 1$
$A = \sqrt{\frac{2}{L}}$

$$\boldsymbol{\psi_n(x) = \sqrt{\frac{2}{L}}\ sin(\frac{n\pi x}{L})}$$

19. For each value of the quantum number **"n"** there is a specific wave function **ψ(x)** describing the state of a particle with energy E_n.



1.7 Summary of Particle in One Dimensional Box

1. The energy of a particle is quantized. This means it can only take on discreet energy values.
2. The lowest possible energy for a particle is NOT zero (even at 0 K). This means the particle always has some kinetic energy.

$$E = \frac{h^2 n^2 \pi^2}{8\pi^2 m L^2}$$

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$$

3. The square of the wavefunction is related to the probability of finding the particle in a specific position for a given energy level.
4. The probability changes with increasing energy of the particle and depends on the position in the box you are attempting to define the energy for
5. In classical physics, the probability of finding the particle is independent of the energy and the same at all points in the box

1.8 Molecular Orbital Theory

1. First proposed by F. Hund and R. S. Mullikan in 1932.
2. It is an alternative model for valence bond theory and explains most of the theories which Valence Bond theories could not.
3. In MOT, the valency electrons are considered as particles or waves.
4. The atomic orbitals from different atoms are combined to produce molecular orbitals.
5. The wave function describing a molecular orbital by one of the two procedures.
   Linear combination of atomic orbitals (LCAO)
   United atom method
   LCAO Method:
6. Consider two atoms A and B which have atomic orbitals described by the wave functions ψ_A and ψ_B

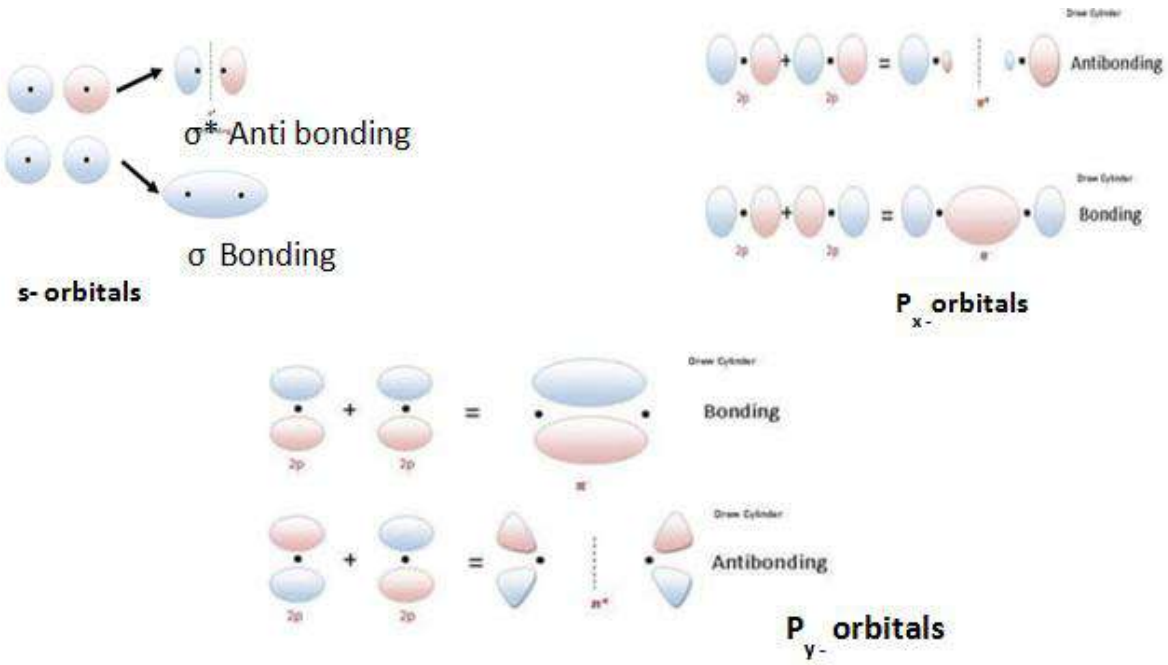
$$\psi_{AB} = N(C_1\psi_A + C_2\psi_B)$$

7. The probability of finding an electron in a volume of space dv is $\psi^2 dv$, so the probability density for the combination of two atoms is
$$\psi_{AB}^2 = (C_1^2\psi_A^2 + 2C_1\psi_A C_2\psi_B + C_2^2\psi_B^2)$$

8. Two combinations of the wave functions ψ_A and ψ_B are possible.

$$\psi_g = N(\psi_A + \psi_B) \quad \text{(for bonding)}$$

$$\psi_u = N(\psi_A + (-\psi_B)) \quad \rightarrow \quad \psi_u = N(\psi_A - \psi_B) \quad \text{(for antibonding)}$$



Rules for Linear Combination of Atomic Orbitals

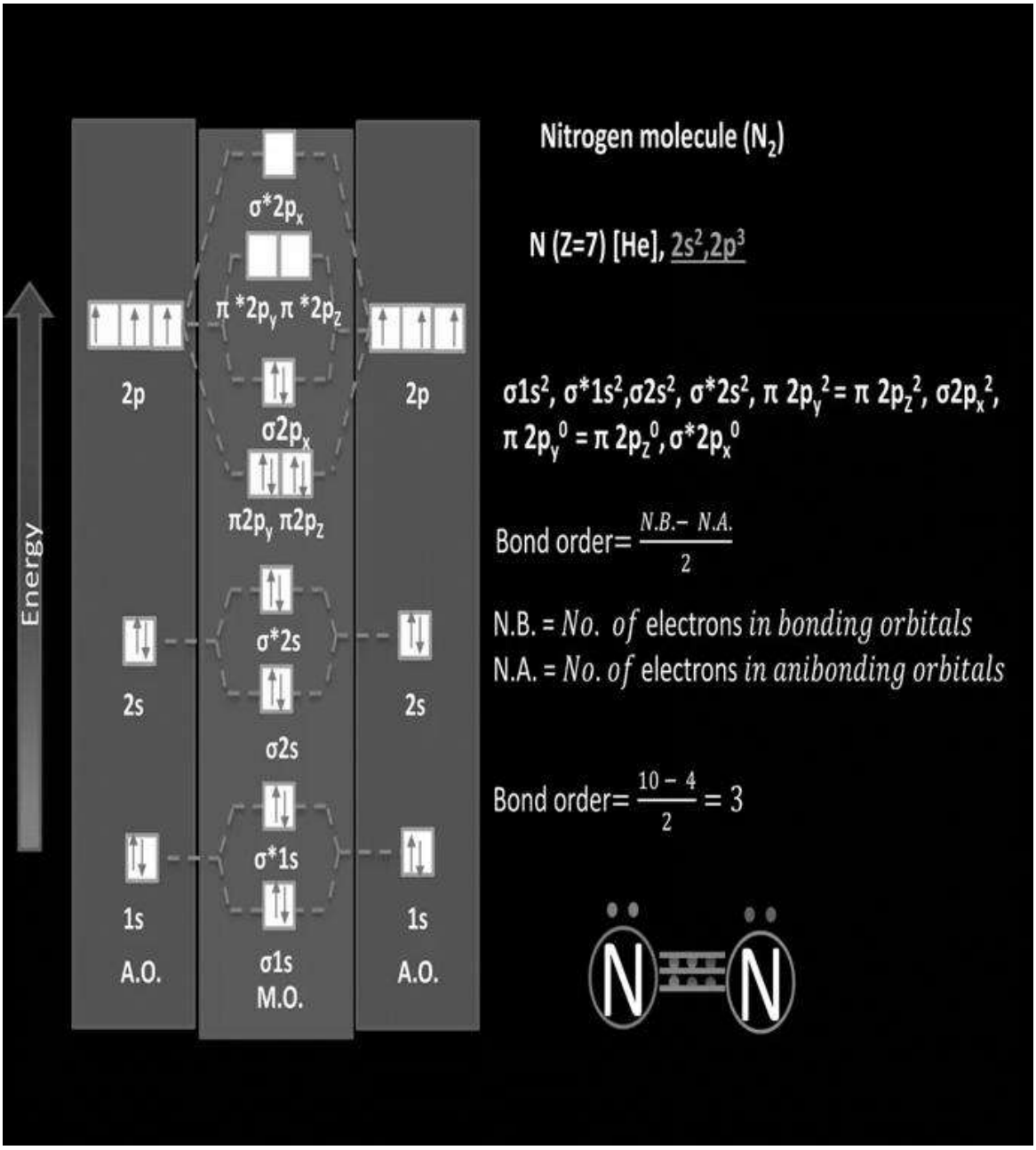
1. The atomic orbitals must be roughly of the same energy.
2. The orbitals must overlap one another as much as possible.
3. In order to produce bonding and antibonding molecular orbitals, either the symmetry of the two atomic orbitals must remain unchanged when rotated about the internuclear line (axis), or both atomic orbitals must change symmetry in an identical manner.
4. Each atomic and molecular orbital has a definite energy and is defined by four quantum numbers.
5. All the molecular orbitals obey Aufbau principle, Hund's rule and Pauli exclusion principle.
6. **Aufbau principle** states that an electron occupies orbitals in order from lowest energy to highest. The Aufbau (German for *building up, construction*) principle is sometimes referred to as the "building up" principle.
7. **Hund's rule** states that orbitals of equal energy are each occupied by one electron before any orbital is occupied by a second electron, and that each of the single electrons must have the same spin.
8. **Pauli exclusion principle** states that in a single atom, no two electrons will have an identical set or the same quantum numbers (n, l, ml, and ms).

9. The order of energies of molecular orbital (for N or more than): σ1s<σ1s*<σ2s<σ2s*<σ2px<(π2py=π2pz) <(π*2py=π*2pz) < σ*2px

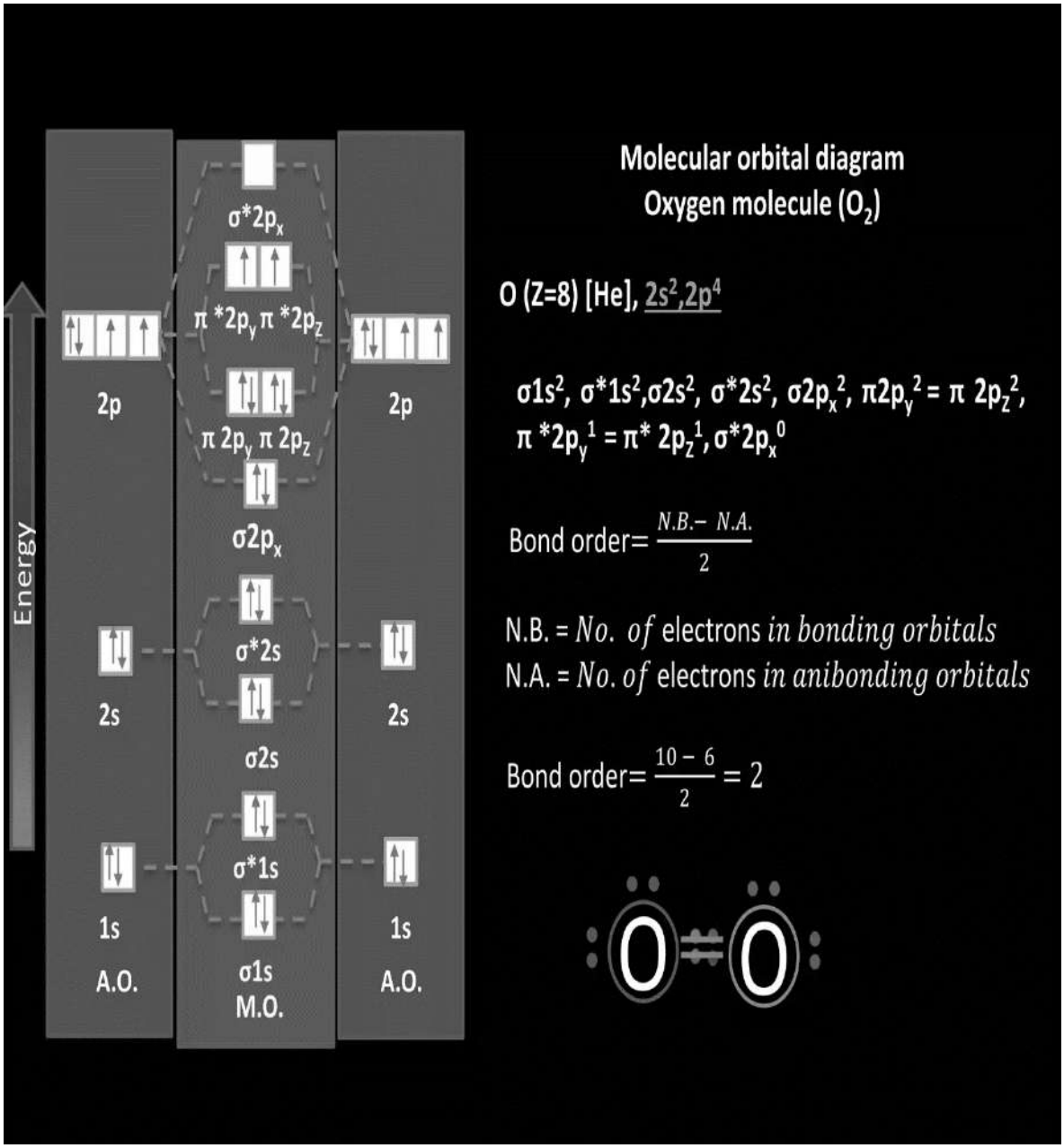
10. For O or less than σ1s<σ1s*<σ2s<σ2s*<(π2py=π2pz) <σ2px<(π*2py=π*2pz) < σ*2px

11. The bond order is given by the half of the difference between the number of electrons present in bonding and antibonding molecular bonding. Bond order = ½ [N.B – N.A.B]

1.9 MO-Diagram for N_2

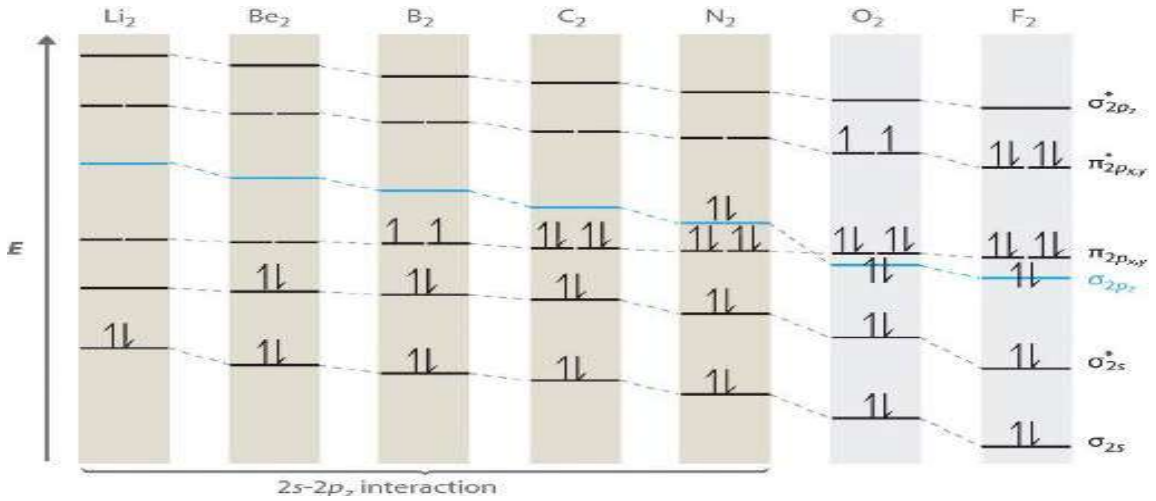


1.10 MO Diagram for O_2



1.11 Why Energy Levels are Differing for O_2 and N_2?

1. Molecular Orbital Energy-Level Diagrams for the Diatomic Molecules of the Period 2 Elements.
2. Unlike earlier diagrams, only the molecular orbital energy levels for the molecules are shown here.
3. For simplicity, the atomic orbital energy levels for the component atoms have been omitted.
4. For Li_2 through N_2, the $\sigma2p_z$ orbital is higher in energy than the $\pi2p_{x,\ y}$ orbitals.
5. In contrast, the $\sigma2p_z$ orbital is lower in energy than the $\pi2p_{x,\ y}$ orbitals for O_2 and F_2 due to the increase in the energy difference between the 2s and 2p atomic orbitals as the nuclear charge increases across the row.



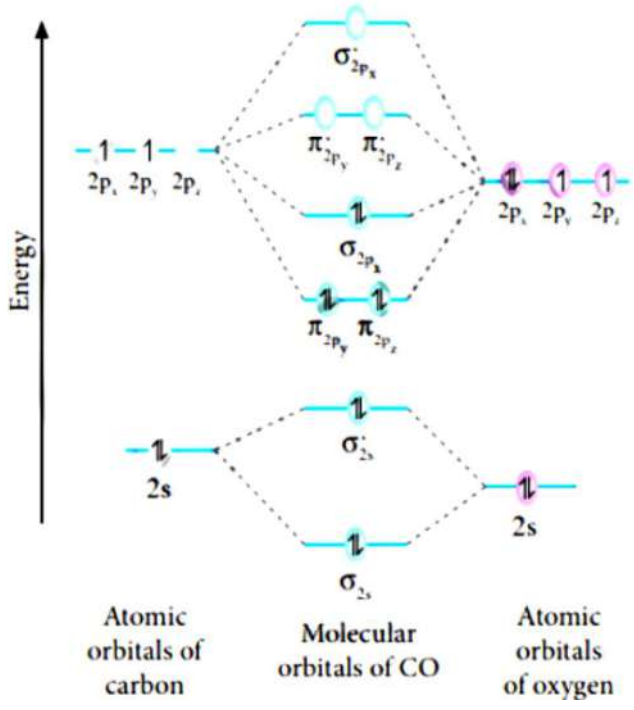
1.12 MO Diagram for CO (Bonding in some Hetero Nuclear di-atomic Molecules)

1. Molecular orbital diagram of Carbon monoxide molecule (CO**)**
2. Electronic configuration of C atom: $1s^2\ 2s^2\ 2p^2$
3. Electronic configuration of O atom: $1s^2\ 2s^2\ 2p^4$
4. Electronic configuration of CO molecule:
5. $\sigma 1s^2, \sigma 1s^{*2}, \sigma 2s^2, \sigma 2s^{*2}, \pi 2p_y^2, \pi 2p_z^2\ \sigma 2p_x^2$
6. Bond order = $(N_b - N_a)/ 2$

   $= (10 - 4)/2$

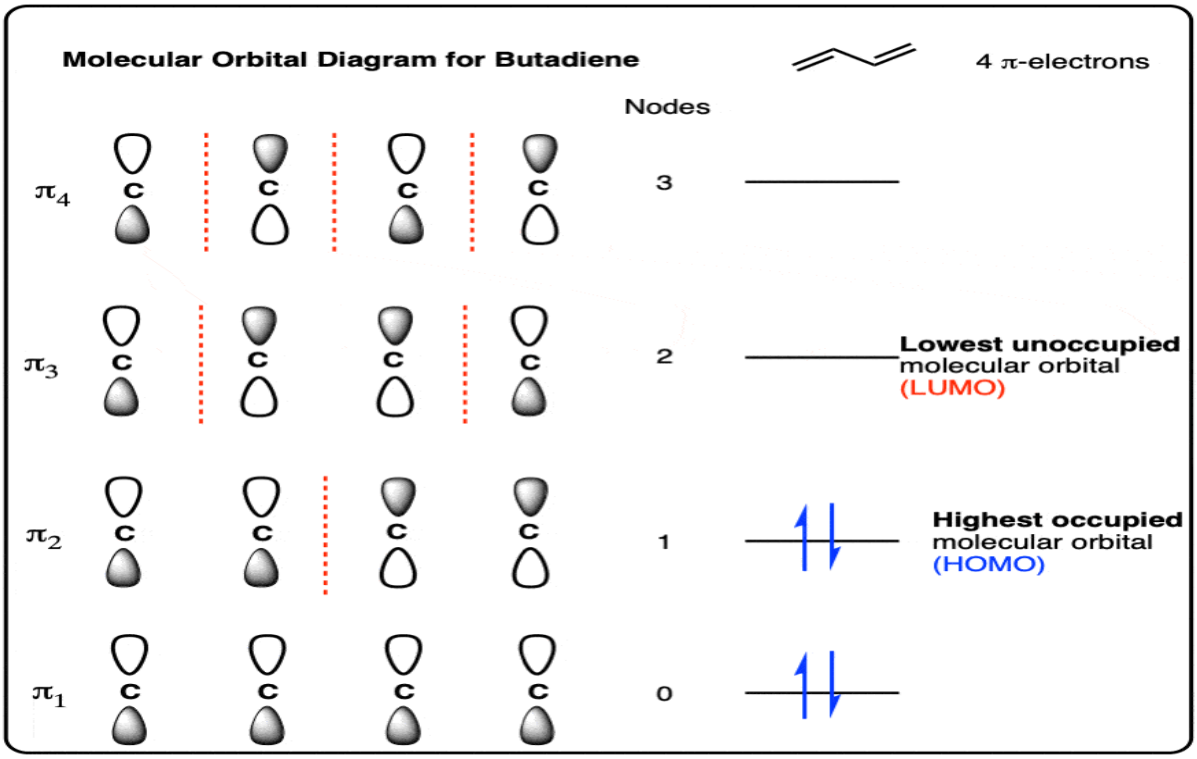
   $= 3$

The molecule has no unpaired electrons hence it is diamagnetic.



1.13 π Molecular Orbital Diagram for Butadiene

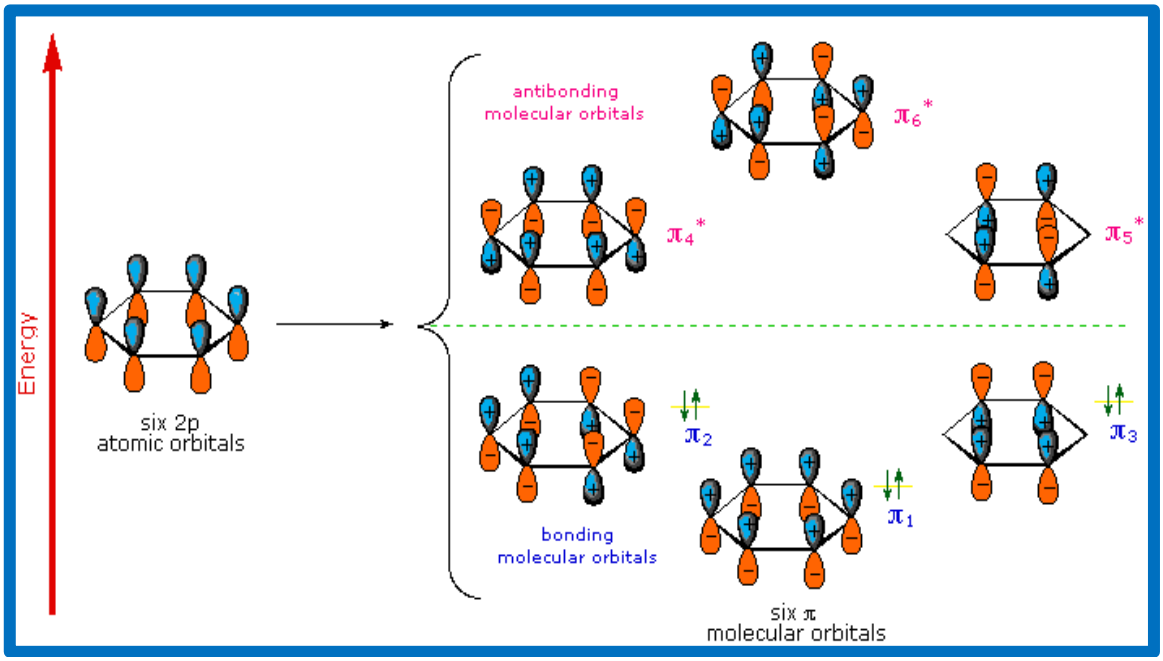
1. Butadiene is a conjugated diene consisting of two adjacent pi-bonds and comprised of 4 p-orbitals and 4 pi-electrons.
2. To draw the molecular orbital diagram of butadiene, start by drawing 4 p-orbitals all aligned with the same phase. This has zero nodes and is the lowest energy pi-orbital (π_1)
3. As the number of nodes in an orbital increase, so does its energy. The highest-energy molecular orbital has three nodes and has all p-orbitals with opposite phases (π_4)
4. Intermediate orbitals (π_2) and (π_3) have one and two nodes, respectively.
5. Once the molecular orbital diagram is built, the next step is to add the 4 pi-electrons. This will fill up the lowest-energy orbital (π_1) and the second-lowest-energy orbital (π_2)
6. The highest-occupied molecular orbital (HOMO) of butadiene is the highest-energy orbital that contains pi-electrons. This is π_2
7. The lowest-unoccupied molecular orbital (LUMO) of butadiene is the lowest-energy orbital that has zero pi-electrons. This is π_3



1.14 π Molecular Orbital Diagram for Benzene

1. Benzene has a planar hexagonal structure in which all the carbon atoms are sp^2 hybridised and all the carbon bonds are equal in length.
2. As shown in figure, the remaining cycle array of six p-orbitals (one on each carbon) overlap to generate six molecular orbitals, three bonding and three antibonding.
3. The plus(blue) and minus (black) signs shown in diagram do not represent electrostatic charge, but refer to phase signs.
4. The orbitals overlap to generate a common region of like phase, with those orbitals having the greatest overlap being lowest in energy.
5. The remaining carbon valence electrons then occupy these molecular orbitals in pairs resulting in a fully occupied (6 electrons) set of bonding molecular orbitals.

6. It is this completely filled set of bonding orbitals or closed shells that gives the benzene ring its thermodynamic and chemical stability.



1.15 Additional Information

1. Derivation of ψ(x,t) for time independent equation

Note: How to write $\psi(x,1) = \psi(x)e^{-iEt/h}$

We know that $ih\frac{\partial\psi(x,t)}{\partial t} = E \longrightarrow \frac{\partial\psi(x,t)}{\partial t} = \frac{1}{ih}E \longrightarrow \frac{\partial\psi(x,t)}{\partial t} = \frac{-i}{h}E$

By the method of separating variables, $\psi(x,1) = \psi(x).f(t)$ is substitute in above equation

$\frac{\partial\psi(x).f(t)}{\partial t} = \frac{-i}{h}E \longrightarrow \partial\psi(x).f(t) = \frac{-i}{h}E\partial t$

$\psi(x)\partial f(t) = \frac{-i}{h}E\partial t$ is divided by $\psi(x).f(t)$ on left side

$\frac{\partial f(t)}{f(t)} = \frac{-i}{h}E\partial t$ Integrating on both sides $\int\frac{\partial f(t)}{f(t)} = \int\frac{-i}{h}E\partial t \longrightarrow \ln f = \frac{-i}{h}Et + C$

$f = e^{-Et/h} + C$ Where C = 0 at f(t=0)

2. Operators

Additional information about operators

- ❑ Operator is an instruction
- ❑ A symbol which can tells us to perform one or more mathematical acts on a function f(x)

Operator rule	Example
Addition rule	$(\hat{Q}_1 + \hat{Q}_2)f(x) = \hat{Q}_1 f(x) + \hat{Q}_2 f(x)$
Subtraction rule	$(\hat{Q}_1 - \hat{Q}_2)f(x) = \hat{Q}_1 f(x) - \hat{Q}_2 f(x)$
The product of two operators implies successive operation	$\hat{Q}_1\hat{Q}_2 f(x) = \hat{Q}_1(\hat{Q}_2 f(x)$
The product of two operators is a third operator	$\hat{Q}_3 = \hat{Q}_1\hat{Q}_2$
Two operators commute if they obey the simple operator expression	$[\hat{Q}_1\hat{Q}_2] = \hat{Q}_1\hat{Q}_2 - \hat{Q}_1\hat{Q}_2 = 0$ $\hat{Q}_1\hat{Q}_2 = \hat{Q}_2\hat{Q}_1$

Short Answer Questions

1. Interpret the significance of Ψ and Ψ^2.
2. Compare the energy and wave functions of one-dimensional box for different energy levels.
3. Demonstrate the total wave functions for Bonding and Anti bonding states of Hydrogen molecule ion.
4. Summarize the postulates of quantum mechanics.
5. Outline the MO diagram of 1, 3 Butadiene.
6. Outline the MO diagram of Benzene.
7. Illustrate the bond order for CO molecule from molecular orbital diagram.
8. Interpret the bond order for O_2^+, O_2, O_2^-.
9. Compare time dependent and time independent Schrodinger wave equation.
10. Compare the magnetic behavior of O_2 and N_2 from MO theory.

Descriptive Questions

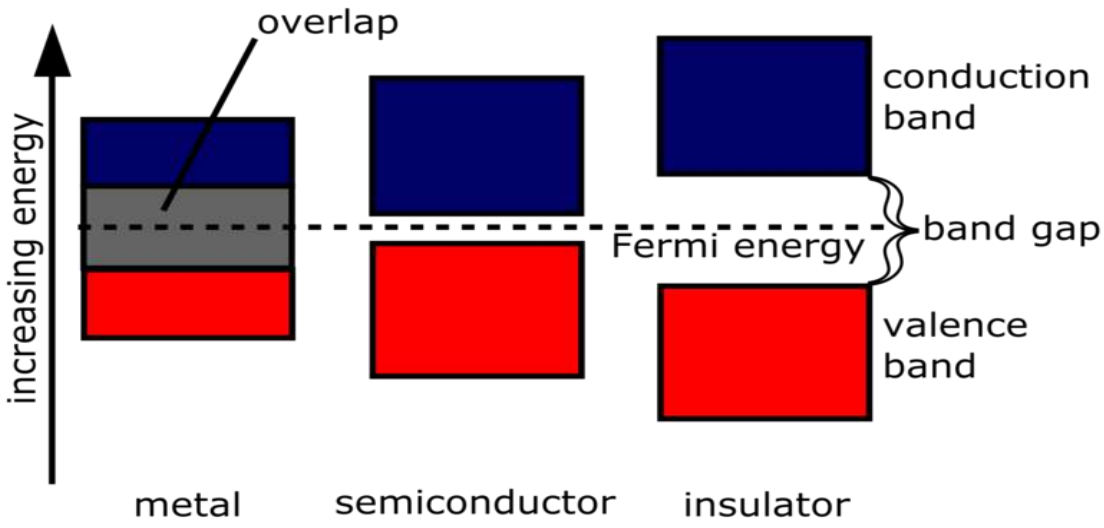
1. Construct the Schrodinger wave equation for time independent wave mechanics.
2. a) Interpret the significance of Ψ and Ψ^2 with examples. b) Illustrate the properties of a well-behaved wave function.
3. Demonstrate the wave mechanics of one-dimensional box.
4. Explain MO theory using LCAO approximation.
5. Sketch out the MO diagrams of π molecular orbital containing system 1,3 Buta Diene and interpret its energy levels.
6. Demonstrate the MO diagram of hetero nuclear diatomic molecule (CO).
7. Illustrate the Molecular orbital theory for homo nuclear diatomic molecules (N_2 and O_2).
8. Interpret the fundamental postulates of quantum mechanics with various examples.
9. Explain the MO diagram of Benzene using various energy levels.
10. Compare the relative stability, magnetic behavior and bond order for O_2, O_2^+ and O_2^- ions.

Unit – 2

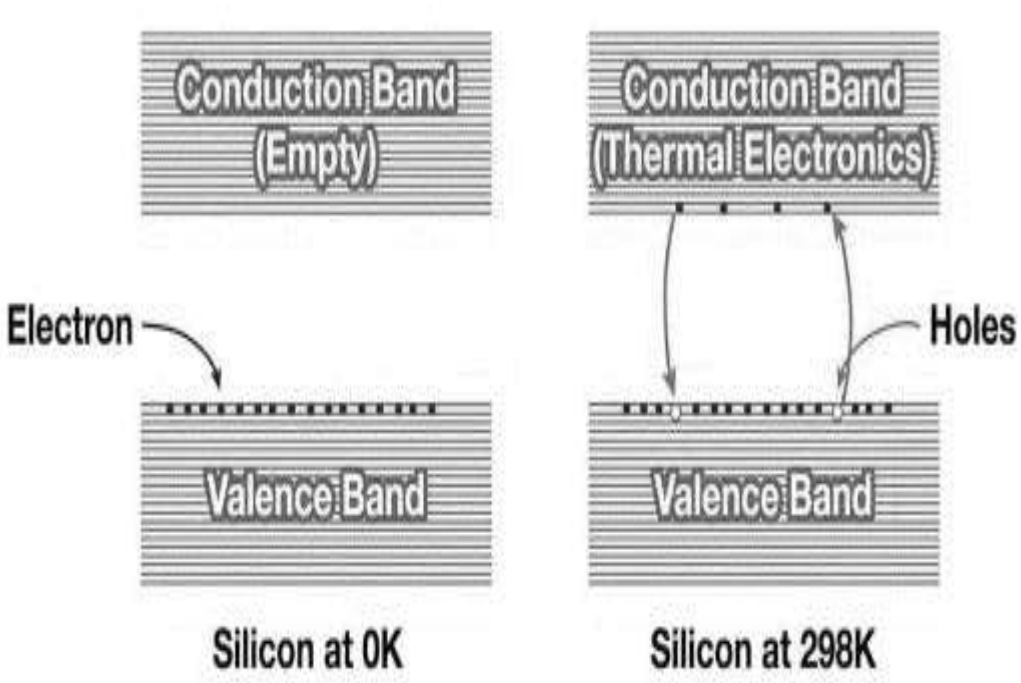
Modern Engineering Materials



2.1 Band Gap Diagram for Conductors, Semiconductors and Insulators



Electron hole transfer diagram in Semiconductors:



2.2 Types of Semiconductors

Elemental semiconductors	Non-elemental semiconductors
IV A group elements Ge and Si	1. Stoichiometric 2. Non-stoichiometric 3. Controlled valency 4. Chalcogen photo semiconductors

2.2.1 Stoichiometric Semiconductors

These compounds are intermediate compounds with an average of 4 valence electrons per atom and shows similar crystal and band structure of Si and Ge.

These are prepared by combining of

III A & VA	Fermi gap (eV)	II A & VI A	Fermi gap (eV)
GaP	2.24	CdS	2.42
GaAS	1.35	CdSe	1.74
GaSb	0.67	PbS	0.37

2.2.2 Non-Stoichiometric Semiconductors

Many non-stoichiometric compounds act as semiconductors.

Metal oxides and sulphides containing stoichiometric defects in their crystal structure.

p – type non stoichiometric semiconductor	n – type non stoichiometric semiconductor
FeO contains Fe^{+2} and O^{2-} . On heating FeO, some Fe^{2+} ions oxidized to Fe^{3+}.	ZnO on reduction, $Zn_{1.1}O$ formed due to partial removal of oxygen.
In $Fe_{0.9}O$ crystal $Fe_{0.7}{}^{2+}$ and $Fe_{0.2}{}^{3+}$ exists.	In $Zn_{1.1}O$ crystal $Zn_{0.9}{}^{2+}$ and $Zn_{0.2}{}^{+}$ exists.
Fe^{3+} ions short by one electron with respect to Fe^{2+}. So Fe^{3+} in FeO crystal lattice considered as hole.	Zn^{+} ion is rich by one electron with respect to Zn^{2+}.
In the presence of electrical field, the holes can move from Fe^{2+} to Fe^{3+}. Therefore crystal producing electrical conductivity.	So each Zn^{+} ion is capable of donating one electron to conduction band. In the presence of electrical field, ZnO act as n-type semiconductor.
These crystals can be represented $M_{(1-x)}Y$ known as metal ion deficient compounds	These crystals can be represented as $M_{(1+x)}Y$
Examples: FeO, Cu_2O, NiO, FeS	Examples: ZnS, PbS, CdS, Fe_2O_3, Cr_2O_3

2.2.3 Controlled Valence Semiconductors

- Non-stoichiometric semiconductors are generally difficult to obtain and the compositions are difficult to reproduce exactly.
- Difficulties overcome by "Controlled Valence Semiconductors"
- Li_2O heated with NiO in presence of air at 1200^0C $Li_xNi_{1-x}O$ is formed.

$$\frac{x}{2}Li_2O + (1-x)NiO + \frac{x}{4}O_2 \rightarrow Li_xNi_{1-x}O$$

- Li^+ ion occupy interstitial site and substitute Ni^{2+} ions in the lattice.
- One Ni^{2+} can formed for every Li^+ ion to maintain neutrality.
- The resulting compound can be represented by the $\boldsymbol{Li_xN_x^{III}Ni_{1-2x}^{II}O}$
- The positive holes in the Ni^{2+} can move from one Ni to another thus give to p-type semiconductor.
- The conductivity of pure NiO is about 10^{-10} Scm^{-1}.
- If 10% Ni atoms replaced by Li atoms has a conductivity about 10^{-1} SCm^{-1}.
- n-type semiconductor $\boldsymbol{La_xBa_{1-x}Ti_x^{III}Ti_{1-x}^{IV}O_3}$

Ni^{2+} O^{2-} $Ni^{(3+)}$ O^{2-} $Ni^{(2+)}$ O^{2-}

O^{2-} Ni^{2+} * Ni^{2+} O^{2-} Ni^{2+}

Ni^{2+} O^{2-} $Ni^{(3+)}$ O^{2-} Ni^{2+} O^{2-}

O^{2-} Li^+ O^{2-} Ni^{2+} O^{2-} Ni^{2+}

[Li^+ ion doped NiO lattice]

2.2.4 Chalcogen Photo Semiconductor

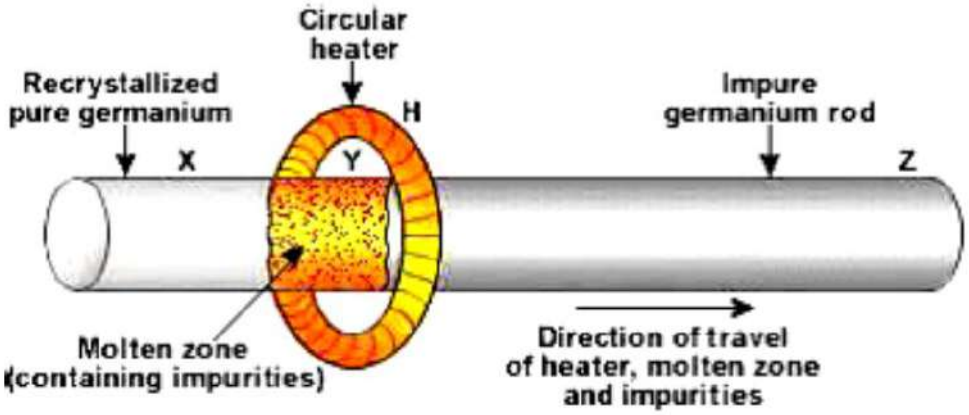
- Oxides or sulphides of VI A group elements.
- They behave as semiconductors or photoconductors either alone or by combining with other elements.
- Conductance increases with increase in atomic mass of chalcogens.
- As_2Se_3 is a good photo conductor.
- Used in photo copying process (Xerox).

2.3 Preparation of Semiconductors

2.3.1 Zone Refining

1. This method is particularly used when metals are required in high degree of purity.
2. It is based on the principle that when an impure metal in a molten state is allowed to cool, only the metal crystallizes while the impurities remain present in the molten state (mass) or melt.
3. The zone refining technique involves melting a narrow zone of long cylindrical rod of the material to be processed.
4. The impure metal converted into a rod which is heated at one end with a circular heater. A narrow zone of metal is melted.
5. The heater is slowly moved along the cylindrical rod.

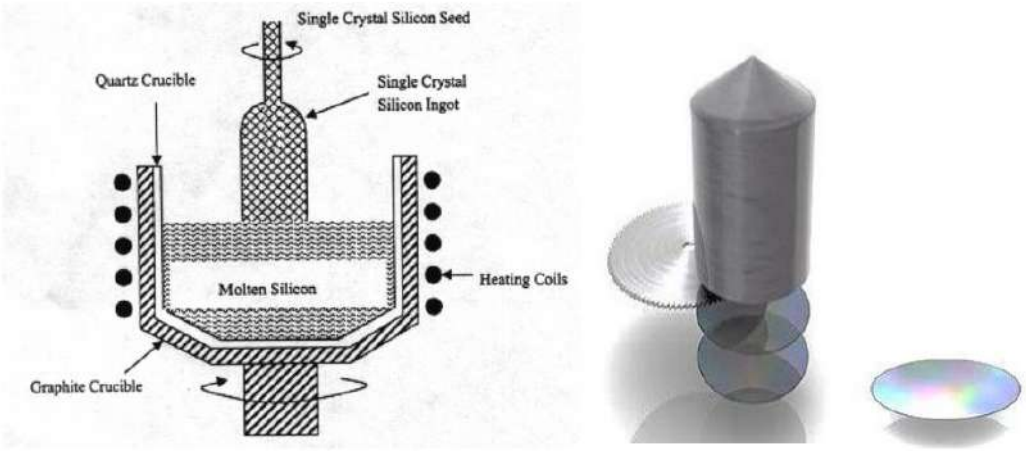
6. The pure metal recrystallizes out of melt while the impurities remain in the melt which moves along with the melted zone of the rod with the movement of heater.
7. The process is repeated several times. The end of rod where impurities have collected is cut off.
8. This method is employed for the purification of germanium, silicon, gallium etc, which are used in semiconductors.



Zone refining of germanium metal.

2.3.2 Chzocralski Crystal Pulling Technique

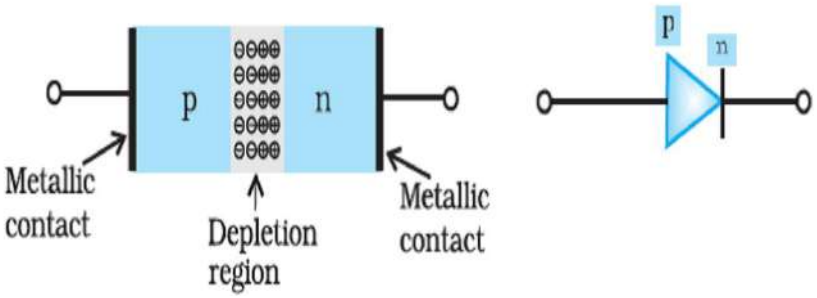
1. Ge or Si obtained by zone refining method is polycrystalline, i.e., there is no regularity in their crystal structure and contain crystals of different sizes.
2. But fabrication of most of the semiconductor devices requires single crystal of semi conducting materials.
3. Consequently, they are grown in the form of single crystals.
4. In this process, solid polycrystalline semiconducting material is taken in a quartz chamber and heated by using RF (radio frequency) power source, till it melts.
5. Then the temperature is lowered to just above the melting point of Ge.
6. At this stage, properly oriented single crystal of Ge attached to a puller rod is carefully lowered into the melt, till it is in contact with the surface of the molten Ge.
7. Then the crystal is rotated slowly and then gradually pulled out so that the melt in contact with the seed crystal starts solidifying producing the orientation identical as that of the seed crystal.
8. Finally, an intrinsic crystal of Ge is obtained.
9. This crystal is tested for its conductivity and cut into wafers with the help of a diamond-tipped saw.



2.4 Semiconductor devices and Applications

2.4.1 PN-junction Diode

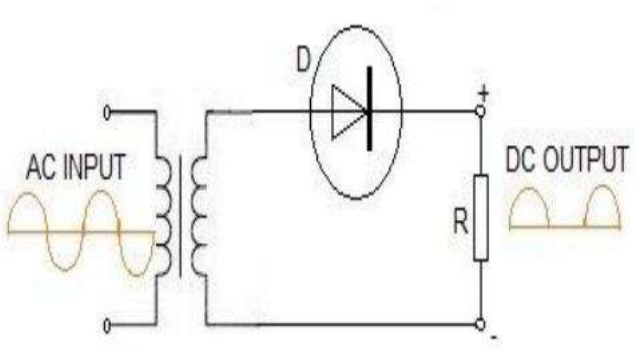
1. A p–n junction diode is formed by joining P-type and N-type semiconductors together in very close contact.
2. Diode is a two-terminal electronic component that conducts electric current in only one direction.
3. When an external voltage V is applied across a semiconductor diode such that p-side is connected to the positive terminal of the battery and n-side to the negative terminal, it is said to be forward biased.
4. When an external voltage (V) is applied across the diode such that n side is positive and p-side is negative, it is said to be reverse biased.



2.4.2 PN Junction Diode as Rectifier

A device which converts alternating current or voltage into direct current or voltage is known as rectifier. The process of converting AC into DC is called rectification.

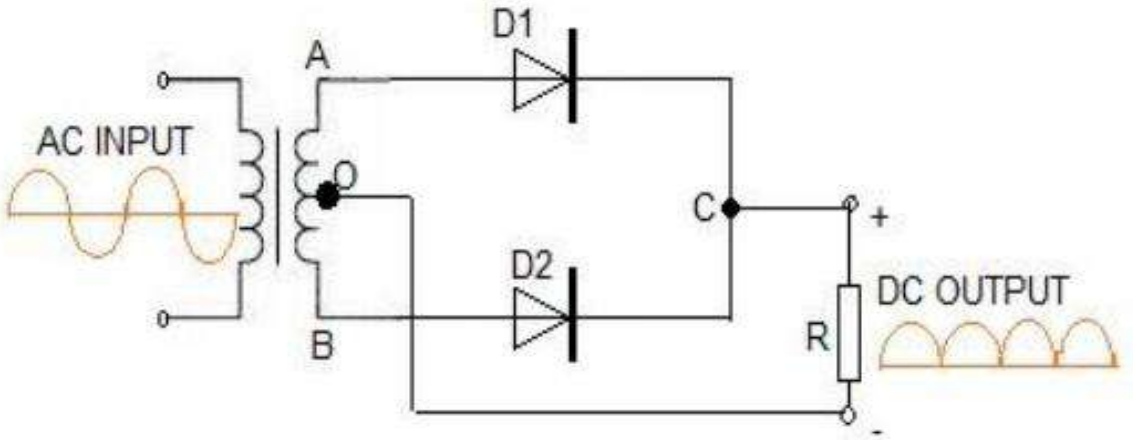
Half-Wave Rectifier: A half-wave rectifier converts the half cycle of applied AC signal into DC signal. Ordinary transformer may be used here



Half Wave Rectifier

Full-Wave Rectifier: A full-wave rectifier converts the whole cycle of applied AC signal into DC signal. Centre tap transformer is used here.

Half-wave rectifier converts only one-half of AC into DC while full wave rectifier rectifies both halves of AC input.

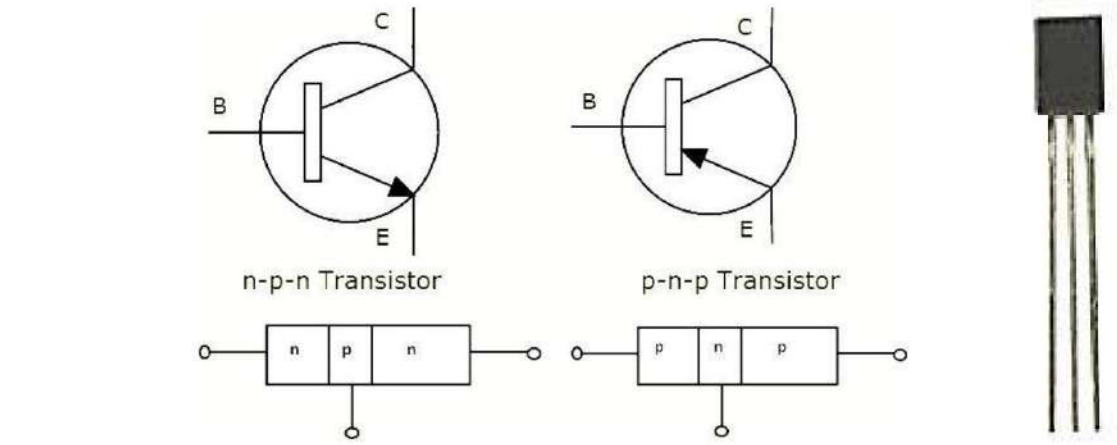


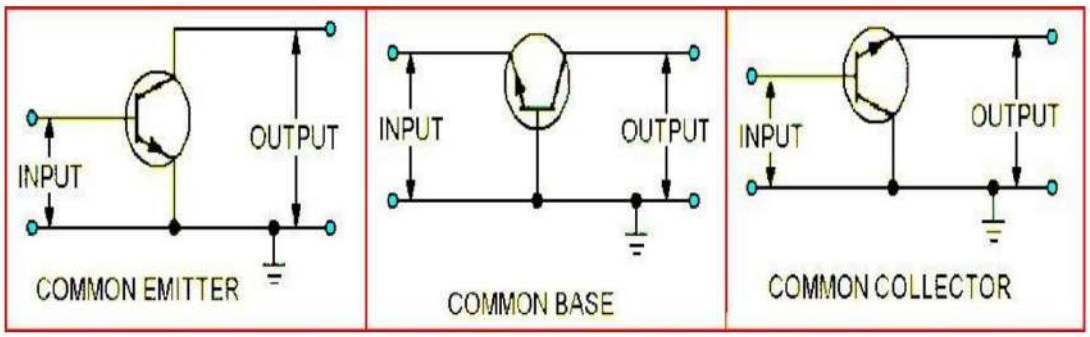
Full Wave rectifier (Centre tap)

2.4.3 Transistors

1. A transistor is an arrangement obtained by growing a thin layer of one type of semiconductor between two thick layers of other similar type semiconductor.
2. A bipolar junction transistor (BJT) is a three-terminal electronic device constructed of doped semiconductor material.
3. It may be used in amplifying or switching applications.
4. Bipolar transistors are so named because their operation involves both electrons and holes.
5. Charge flow in a BJT is due to bidirectional diffusion of charge carriers across a junction between two regions of different charge concentrations.
6. 3 adjacent regions of doped Si (each connected to a lead) 1. Base. (thin layer, less doped). 2. Collector. 3. Emitter

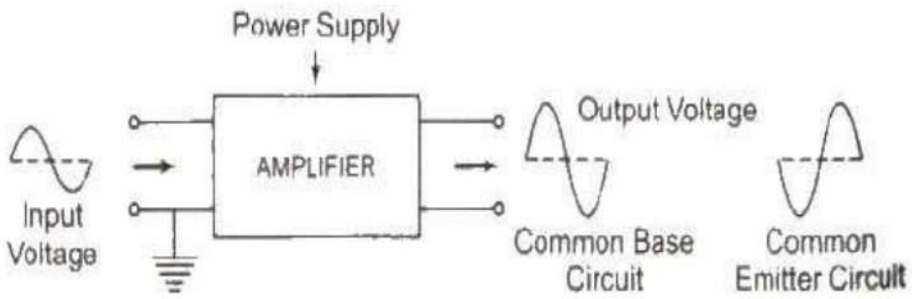
Types of transistors





2.4.4 Transistor as an Amplifier

1. An amplifier is a device which is used for increasing the amplitude of variation of alternating voltage or current or power.
2. The amplifier thus produces an enlarged version of the input signal. The general concept of amplification is represented in figure.
3. There are two input terminals for the signal to be amplified and two output terminals for connecting the load; and a means of supplying power to the amplifier.



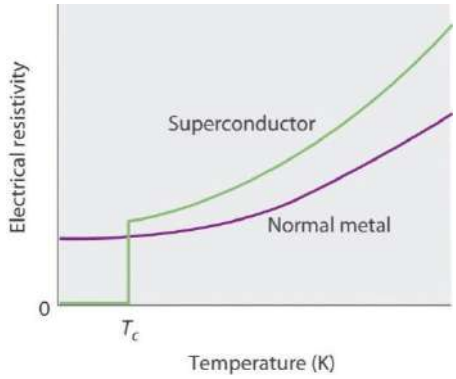
2.5 Superconductors

2.5.1 Introduction

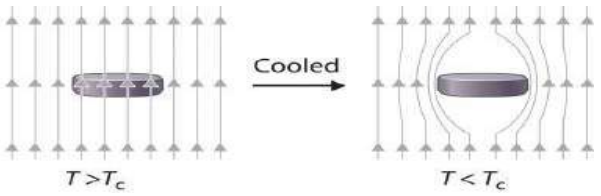
- **Superconductors** are solids that at low temperatures **exhibit zero resistance** to the flow of electrical current, a phenomenon known as **superconductivity**.
- The temperature at which the electrical resistance of a substance drops to zero is its **superconducting transition temperature (T_c) or critical temperature**.
- **Superconductors** also **expel a magnetic field** from their interior, a phenomenon known as the **Meissner effect**.
- Superconductivity can be explained by the **BCS theory**, which says that electrons are able to travel through a solid with no resistance because they couple to form **pairs of electrons** (Cooper pairs).
- **High-temperature superconductors** have T_c values greater than 30 K.

The temperature dependence of the electrical resistivity of a normal metal and a superconductor.

The superconducting transition temperature (T_c) is the temperature at which the resistivity of a superconductor drops to zero.



2.5.2 Meissner Effect

Below its T_c, a superconductor completely expels magnetic lines of force from its interior. In magnetic levitation, a small magnet "floats" over a disk of a high-temperature superconducting material ($YBa_2Cu_3O_{7-x}$) cooled in liquid nitrogen.	

2.5.3 BCS Theory

According to the BCS theory, as the temperature of the solid increases, the vibrations of the atoms in the lattice increase continuously, until eventually the electrons cannot avoid colliding with them. The collisions result in the loss of superconductivity at higher temperatures.

2.5.4 Disadvantages of Superconductors

1. Unfortunately, for many years the only superconductors known had serious limitations, especially the need for very low temperatures, which required the use of expensive cryogenic fluids such as liquid He.
2. In addition, the superconducting properties of many substances are destroyed by large electrical currents or even moderately large magnetic fields, making them useless for applications in power cables or high-field magnets.

2.5.5 Applications of Superconductors

1. Using superconducting wires in power cables would result in zero power losses, even over distances of hundreds of miles.
2. Additionally, because superconductors expel magnetic fields, a combination of magnetic rails and superconducting wheels (or vice versa) could be used to produce *magnetic levitation* of, for example, a train over the track, resulting in friction-free transportation.
3. The ability of materials such as NbTi, NbSn, Nb_3Si, and Nb_3Ge to tolerate rather high magnetic fields.
4. They led to a number of commercial applications of superconductors, including high-field magnets for nuclear magnetic resonance (NMR) spectrometers and magnetic resonance imaging (MRI) instruments in medicine, which can detect small changes in soft tissues in the body.

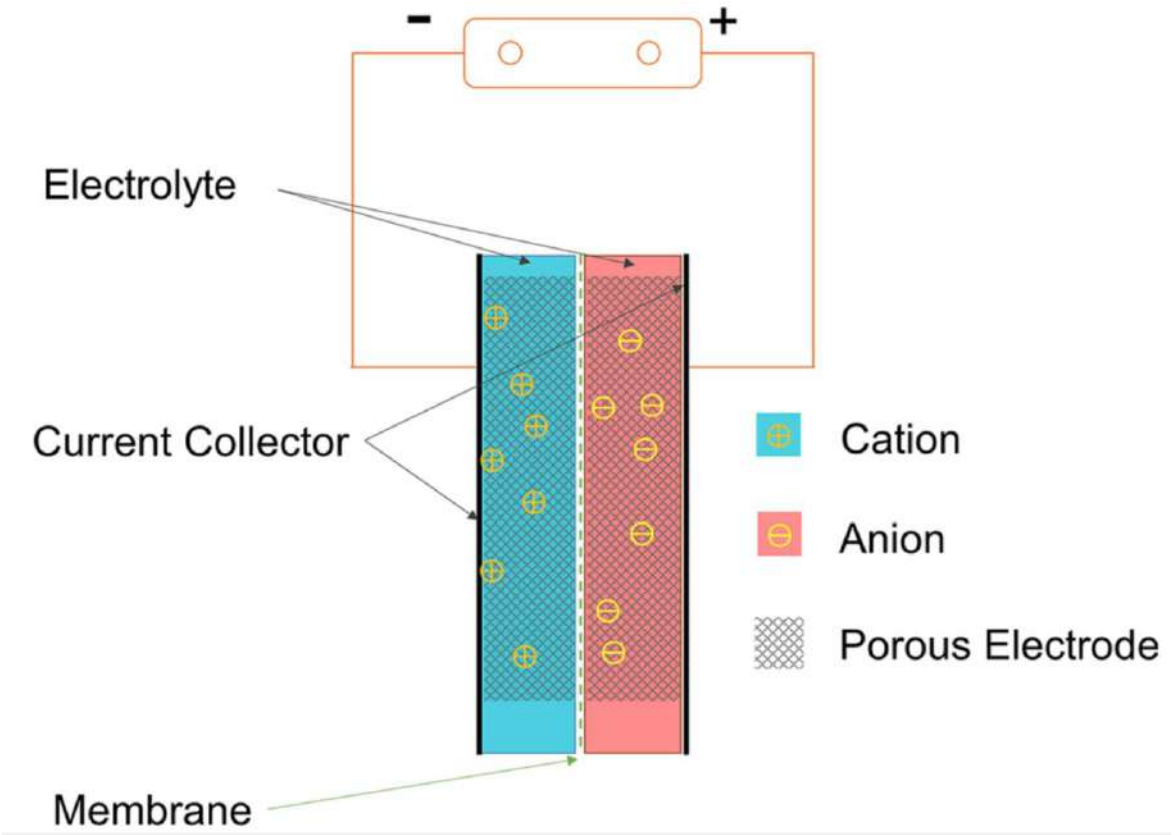
2.5.6 Type – I Superconductors

1. This kind of superconductor includes basic conductive parts and these are utilized in different fields from electrical cabling to microchips on the computer.
2. These types of superconductors **lose their superconductivity** very simply when it is placed in the magnetic field **at the critical magnetic field (Hc).** After that, it will become like a conductor.
3. These types of semiconductors are also named as soft superconductors due to the reason of loss of superconductivity.
4. These superconductors **obey the Meissner effect completely**. The **superconductor examples** are Zinc and Aluminum.

2.5.7 Type – II Superconductors

1. This kind of superconductor will **lose their superconductivity slowly but not simply** as it is arranged within the **exterior magnetic field**. When we observe the graphical representation between magnetization vs. the magnetic field, when the second type semiconductor is placed within a magnetic field, then it will lose its superconductivity slowly.
2. This kind of semiconductors will start to lose their superconductivity on the less significant magnetic field & **totally drop their superconductivity** at the **higher critical magnetic field**. The condition between the slighter critical magnetic field & higher critical magnetic field is called an **intermediate state** otherwise vortex state.
3. This type of superconductor is also named as **hard superconductors** due to the reason they lose their superconductivity slowly but not simply.
4. These semiconductors **will obey the effect of Meissner but not totally**. The best examples of these are NbN and $BaBi_3$. These superconductors are applicable for **strong field superconducting magnets**.

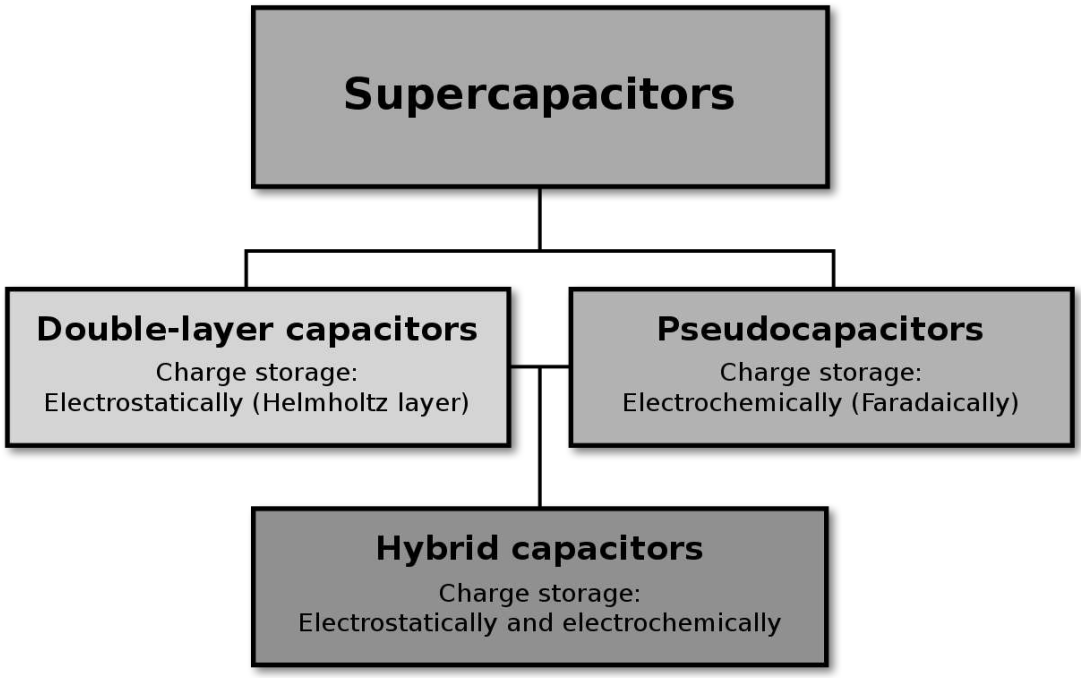
2.6 Supercapacitors



2.6.1 Introduction

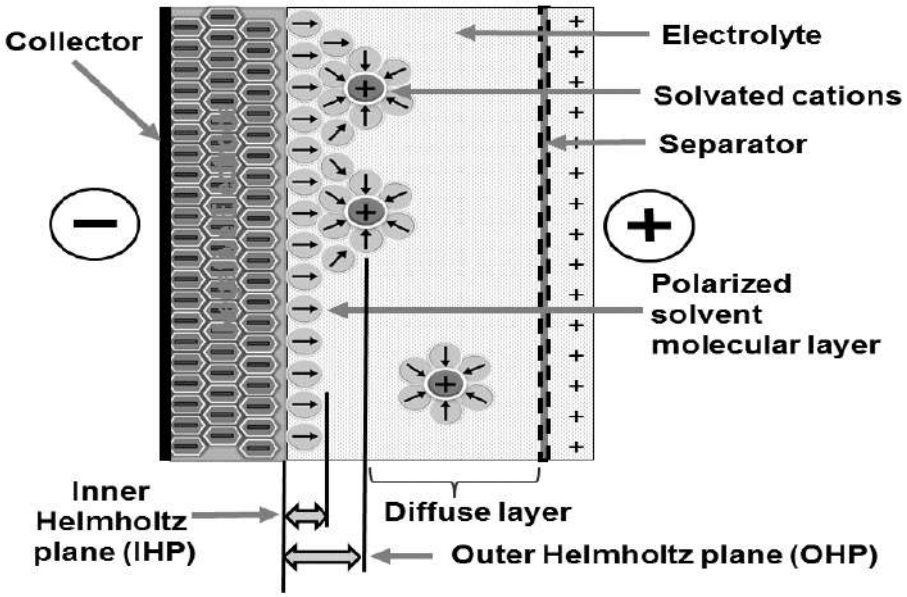
- Supercapacitor is an electrochemical capacitor that has high energy density and better performance efficiency as compared to the common capacitor, the reason why it has the prefix ‘super ’attached to it.
- **It stores and releases energy by reversible desorption and adsorption of ions at the electrode-electrolyte interface.**
- Conventional capacitors have low energy density with wider cell voltage and higher specific power. On the other hand, supercapacitors have high capacitance over a lower limit of cell voltage.

2.6.2 Types of Supercapacitors



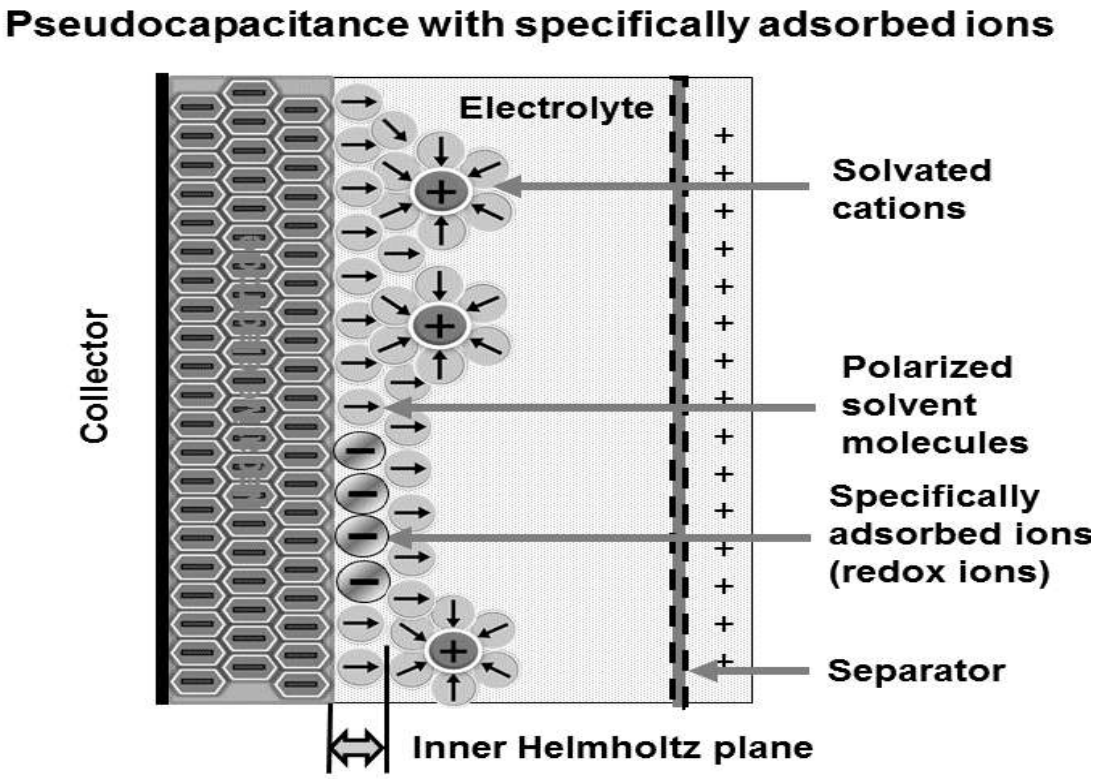
2.6.3 Electrostatic Double Layer Capacitors

1. Electrostatic Double Layer Capacitors comprises two electrodes, a separator, and an electrolyte.
2. Electrolyte is a mixture comprising positive and negative ions dissolved in water.
3. The two electrodes are separated from each other through a separator.
4. The supercapacitors use carbon electrodes with much higher electrostatic double-layer capacitance.
5. The separation of charge in electrostatic double-layer capacitors is much less than in a conventional capacitor which ranges from 0.3–0.8 nm.



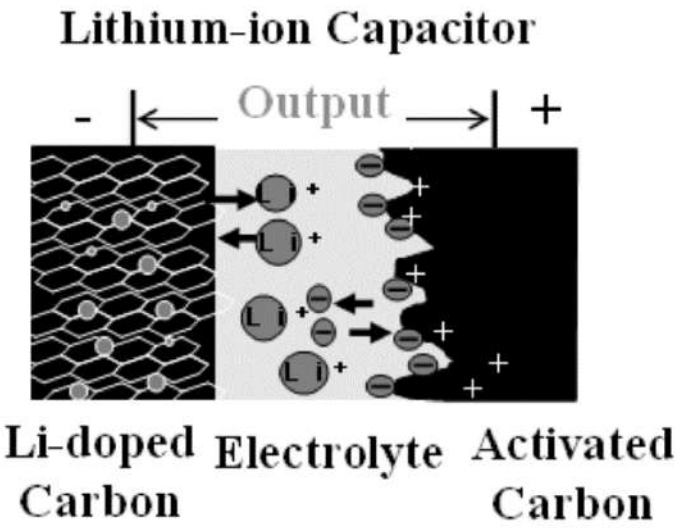
2.6.4 Pseudo Capacitors

1. Pseudo Capacitors are also referred to as electrochemical pseudo-capacitors.
2. They make use of metal oxide or conducting polymer electrodes that have a high amount of electrochemical pseudocapacitance.
3. They store electrical energy by electron charge transfer between electrode and electrolyte.
4. It is done through the oxidation and reduction reaction commonly known as a redox reaction.



2.6.5 Hybrid Capacitors

1. Hybrid Capacitors are made by using the techniques of double-layer capacitors and pseudo-capacitors.
2. In these capacitors, electrodes with different characteristics are used.
3. One electrode has the capacity to display electrostatic capacitance and the other electrode showcases electrochemical capacitance.
4. An example of a hybrid capacitor is the lithium-ion capacitor.



2.6.6 Characteristics of Supercapacitors

Charge Time

- The charge and discharge times for supercapacitors are equal to those of regular capacitors.
- High charge and discharge currents are achievable owing to the low internal resistance.
- A good example is a mobile phone battery that often takes several hours to fully charge, however, supercapacitors can reach the same charge state in less than two minutes.

Specific Power

- Specific power of a battery or supercapacitor is used to compare various technologies through the maximum power output divided by the total mass of the device.
- The specific power of supercapacitors is 5-10 times greater than that of batteries.
- A typical supercapacitor has a specific power of about 10 kW/kg, while Li-ion batteries have a specific power of 1 to 3 kW/kg.

Cycle Life of Supercapacitors

- Supercapacitors are devices that can be charged and discharged millions of times.
- They have a virtually limitless cycle life in comparison to batteries that only have a cycle life of 500 times or more.
- They are extremely beneficial in applications that frequently need to store and release energy.

Safety of Supercapacitors

- Supercapacitor batteries are considered to be safer in comparison to conventional batteries when mishandled.
- Supercapacitors do not heat up, unlike batteries because of their low internal resistance, whereas batteries explode owing to excessive heating when short-circuited.

2.6.7 Working of Supercapacitors

1. Positively and negatively charged ions are present between the two plates of the supercapacitor's electrolyte solution.
2. One of the plates develops a positive charge whereas the other plate develops a negative charge when a voltage is placed across the plates.
3. The positively charged plate attracts the positively charged ions in the electrolyte solution, whereas the negatively charged metal plate attracts the negatively charged ions towards itself.
4. A thin coating of ions is deposited on the inner surface of both plates.
5. An electrostatic double layer is formed as a result, which is like connecting two capacitors in series.
6. Since the space between their charge layers is relatively small, each of the two resulting capacitors has a high capacitance value.
7. A supercapacitor's total capacitance can be calculated by $C_{tot} = \frac{C_1 C_2}{C_1 + C_2}$

2.6.8 Applications of Supercapacitors

- Backup for uninterrupted power supplies (UPS)
- Light weight power supplies for small aircraft
- Provide short duration power for various vehicle systems such as breaking on steering
- Used to absorb power during short periods such as Regenerative Braking
- Extend range and Battery life in Hybrid Electric Vehicles (HEV)

2.6.9 Disadvantages of Supercapacitors

1. Supercapacitors discharge themselves much more frequently which is significantly higher than a battery.
2. There are low voltages that exist within individual cells due to which series connections are necessary to attain greater voltages.
3. There is a significantly lesser amount of energy stored per unit weight in comparison to an electrochemical battery.
4. Supercapacitors deliver a poor energy density compared to batteries. It is equal to between one-fifth and a tenth of the battery's energy.
5. They cannot be utilized in circuits with AC or higher frequencies.

2.7 Carbon Nanotubes (CNT)

2.7.1 Properties of CNT

1. These are allotropes of carbon with a cylindrical nanostructure.
2. The chemical bonding of nanotubes is composed entirely of sp^2 bonds.
3. Mechanical *Properties*: Young's modulus of carbon nano tubes is 10 times greater than that of steel. CNTs have very structural defects in their walls, and hence, do not fracture on bending. The tensile strength of CNTs is about 20 times that of steel.
4. Electrical Properties: The electrical properties of CNTs vary between metallic to semiconducting materials. The very high electrical conductivity of CNT is due to the minimum defects in the structure

5. Thermal Conductivity: The thermal conductivity of CNT is very high due to the vibration of covalent bonds due to minimum defects in the structure.

2.7.2 Applications of CNTs

1. CNTs find use in water purification.
2. They are used in stain- resistant textiles.
3. Electrically conducting CNT films are used in LCD's, touch screens, computers, cell phones and photovoltaic devices.
4. Used to prepared high tensile strength fibers.
5. CNTs are used for hydrogen storage owing to their capillary effect.
6. CNT fibers are used to prepare bullet proof jackets.
7. CNTs are used for treatment of cancer.
8. CNTs are used to build ultra-capacitors in which energy storage capacity is much higher than the traditional capacitors.
9. CNT may be able to replace steel in suspension and other bridges.
10. CNTs can be used for drug delivery within the body by placing the drugs within the tubes or by attaching the drug to the sides of the tubes.

2.8 Fullerenes

2.8.1 Properties of Fullerenes

1. Its behaviour and structure depend on the temperature. As the temperature is increased fullerene gets converted into the C_{70}.
2. The structure of fullerenes can change under different pressures.
3. Fullerene has an ionization enthalpy of 7.61 electron volts.
4. Its electron affinity is 2.6 to 2.8 electron volts.
5. Fullerenes are stable, but not totally unreactive.
6. In chemical reactions, fullerene can act as an electrophile.
7. It acts as an electron-accepting group and is characterised as an oxidizing agent.
8. Fullerenes when doped or crystallized with alkali or alkaline earth metals showcases superconductivity properties.
9. Fullerene is ferromagnetic.
10. Some fullerenes are inherently chiral.
11. It is soluble in organic solvents such as toluene, chlorobenzene, and 1,2,3-trichloropropane.

2.8.2 Applications of Fullerenes

1. As a catalyst for hydrogenation, since these have ability to transfer H_2
2. Act as potential inhibitor to HIV.
3. Takes part in catalytic processes.
4. Used in rechargeable batteries, in microelectronic devices, solar cells.
5. Used in cosmetics for skin protection.
6. Used as MRI agents in medical world.
7. Used as proton exchange membrane for fuel cells.
8. Used as lubricants since act as miniature ball bearings.
9. Used to prepare soft ferro magnets with zero remanence (TDAE C60) (tetrakis dimethyl amino ethylene)

2.9 Graphene

2.9.1 Properties of Graphene

1. **Electronic Properties:** Graphene has a delocalized pi-electron system across the entirety of its surface, the movement of electrons is fluid. Due to overlapped of pi-electrons no band gap in graphene. Electron mobility is very high (15,000 $cm^2V^{-1} s^{-1}$).
2. **Thermal Properties:** The repeating structure of graphene makes it an ideal material to conduct heat in plane. Graphene can exhibit two types of thermal conductivity – in-plane and inter-plane.
3. **Mechanical Strength:** Graphene is one of the strongest materials ever discovered with a tensile strength of 1.3 x 10^{11} pa and also lightweight (0.77 mg/m^2). The mechanical strength of graphene is unmatched and as such can significantly enhance strength in many composite materials.
4. **Flexibility/Elasticity:** The repeating sp^2 hybridized backbone of graphene molecules allow for flexibility, as there is rotation around some of the bonds, whilst still providing enough rigidity and stability that the molecule can withstand changes in conformation and support other ions. Elasticity of graphene is 1-5 Nm^{-2} with a Young's modulus of 0.5TPa.

2.9.2 Applications of Graphene

1. **Sensors**: Due to the planar and consistent arrangement of atoms in a graphene sheet, every atom within the sheet is exposed to the surrounding environment it can be applicable in sensor applications like bio-sensors, diagnostics, field effect transistors, DNA sensors and gas sensors with high degree of sensitivity.
2. **Batteries**: Graphene can be incorporated into either the anode or the cathode in various battery systems to increase the efficiency of the battery and improve the charge/discharge cycle rate. Due to its versatile nature, graphene has been incorporated into lithium-ion batteries, lithium-sulphur batteries, supercapacitors and fuel cells.
3. **Electron Emission Displays**: Graphene uses as ideal material for use in electronic emission displays as it exhibits a high aspect ratio.
4. **Structural Composites**: Graphene is being incorporated into materials to make the existing material stronger and more lightweight. It can be used in aviation industry.
5. **Catalyst Supports**: Catalysts in the form of metal ions can sit in these cavities and be supported. In addition to providing mechanical support, the excellent charge carrier ability of graphene assists the charge transfer reactions involving the catalyst.
6. **Polymer Masterbatches**: Graphene can be incorporated into polymeric materials to form graphene-polymer composite materials. Incorporating graphene into polymers can also give polymers electrical conductivity properties.
7. **Functional Inks**: Graphene can be used in functional inks for electronic, heat resistant and anti-corrosion purposes. By incorporating graphene into ink formulations, the conductivity properties associated with graphene influence the ink, causing it to become conductive.

Short Answer Questions

1. Sketch out the band energy diagram for conductors, semiconductors and insulators along with Fermi energy gap representation.
2. Classify the semiconductors with a neat representation.
3. What is a p-n junction diode?
4. Write any two technological applications of superconductors.

5. Explain Meissner effect.
6. Give illustration about super capacitors.
7. Write the differences between type I and Type II super conductors.
8. Sketch out the diagrams related to zone refining process and Czochralski Crystal pulling technique.
9. Write the applications of graphenes and fullerenes.
10. Write the applications of semiconductors.

Descriptive Questions

1. Demonstrate the use of p-n junction diode as rectifier.
2. Explain the classification of Super Capacitors with a neat sketch.
3. Write about Type –I and Type –II super conductors.
4. Illustrate about Nano materials and give their various applications in engineering.
5. List out the properties and applications of fullerenes.
6. Explain the properties and applications of carbon nano tubes.
7. Explain the zone refining Process for the preparation of semiconductors.
8. Write about the properties and applications of graphene.
9. Explain the Czochralski crystal pulling technique for the preparation of semiconductors.
10. Write the working principle of transistor.

Unit - 3

Electro Chemistry and Applications



3.1 Electrochemical Cell

- **An electrochemical cell is a device for converting chemical energy into electrical energy.**
- A redox reaction is utilised to get electrical energy.
- The EMF (electromotive force) of a cell is directly proportional to intensity of reaction taking place in it.
- Electrochemical cell is commonly known as voltaic cell or galvanic cell. Daniel cell is an example for galvanic cell.
- The representation of Daniel cell is Zn | $ZnSO_4$ || $CuSO_4$ | Cu and E = 1.09 volt.
- The negative electrode (anode) is extreme left and positive electrode (cathode) is extreme right. The double vertical between the two liquids signifies salt bridge separating the two half cells.
- The electrons leave the cell at anode and enters at cathode. Hence current flows from cathode to anode.

  Cell reactions:
  Anode: $Zn \rightarrow Zn^{2+} + 2e^{-}$

  Cathode: $Cu^{2+} + 2e^{-} \rightarrow Cu$

3.2 Nernst Equation

Cell potential can be calculated in non-standard circumstances using the Nernst Equation. It makes it possible to accurately determine equilibrium constants by connecting the measured cell potential to the reaction quotient.

The Nernst equation is derived from the Gibbs free energy under standard conditions.
$E^0 = E^0_{reduction} - E^0_{oxidation}$ – (1)

ΔG is also related to E under general conditions

$\Delta G = -nFE$ – (2)

Where n is number of electrons transferred in the reaction, F is the Faraday constant (96,500 C/mol) and E is potential difference.

Under standard conditions, equation 2 is then

$\Delta G^0 = -nFE^0$ – (3)

Hence, when E^0 is positive, the reaction is spontaneous and when E^0 is negative, the reaction is non-spontaneous.

From thermodynamics, the Gibbs energy change under non-standard conditions can be related to the Gibbs energy change under standard conditions
ΔG = $\Delta G^0 + RT \ln Q$ – (4)

Substituting equation 3 into equation 4, we have
$-nFE = -nFE^0 + RT \ln Q$ – (5)

Divide both sides of the equation above by -n F, we have

$$E = E^0 - \frac{2.303RT}{nF} \ln Q - (6)$$

Equation 6 can be rewritten in the form of $\log_{10}$

$$E = E^0 - \frac{2.303RT}{nF} \log Q - (7)$$

At standard temperature T = 298 K, the $\frac{2.303RT}{F}$ term equals to 0.0592 V and equation 7 can be rewritten

$$E = E^0 - \frac{0.0592}{n} \log Q - (8) \text{ (where } Q = \frac{[product\ ]}{[reactant\ ]}\text{)}$$

3.3 Problems

1. The potential of the cell Zn | Zn_{aq}^{2+} (1M) || Cu_{aq}^{2+} (1M) | Cu at 25^0 C is 1.1 volt. Calculate the potential of cell Zn | Zn_{aq}^{2+} (0.5 M) || Cu_{aq}^{2+} (0.1M) | Cu at the same temperature.
   Sol. $E = E^0 - \frac{0.0592}{n} \log Q$
   Given E^0 = 1.1 V

$$E = E^0 - \frac{0.0592}{n} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$= 1.1 - \frac{0.0592}{2} \log \frac{[0.5]}{[0.1]}$$

$$= 1.1 - (0.0296 * 0.699)$$

$$= 1.079 \text{ V}$$

2. What is the emf of the following cell at 25^0 C Zn | Zn_{aq}^{2+} (0.2M) || Ag_{aq}^{+} (0.002 M) | Ag. The standard emf of the cell is E^0 is 1.54 V.

$$Zn \rightarrow Zn^{2+} + 2e^-$$

$$2\,Ag^+ + 2e^- \rightarrow Ag$$

$$E = E^0 - \frac{0.0592}{n} \log \frac{[Zn^{2+}]}{[Ag^+]^2}$$

$$= 1.54 - \frac{0.0592}{2} \log \frac{[0.2]}{[0.002]^2}$$

$$= 1.48 \text{ V}$$

3. A voltaic cell has an E_{cell} value is 1.536 V. What is the concentration of Ag^+ in the cell?
   Zn | Zn_{aq}^{2+} (2.0 M) || Ag_{aq}^{+} (x M) | Ag
   Sol.

$$Zn \rightarrow Zn^{2+} + 2e^-$$

$$2\,Ag^{+} + 2e^{-} \rightarrow Ag$$

$$E = E^{0} - \frac{0.0592}{n}\log\frac{[Zn^{2+}]}{[Ag^{+}]^{2}}$$

First find $E_{cell}^{0} = E_{cathode}^{0} - E_{anode}^{0} = E_{Ag}^{0} - E_{Zn}^{0} = 0.800 - (-0.763) = 1.563$ V

$$1.536 = 1.563 - \frac{0.0592}{2}\log\frac{[2]}{[x]^{2}}$$

x = 0.495 M

4. The EMF of the following cell at 25^{0} C is 0.445 V.
   Pt, $H_{2(g)}$ (1 atm) | H^{+} (unknown) || KCl (sat soln) | Hg_2Cl_2, Hg

   Calculate the pH of the unknown solution given that oxidation potential of saturated calomel electrode is – 0.2415 V
   Sol. Ecell = $E_{cathode} - E_{anode}$

   $0.445 = 0.2415 - (-0.0591\times pH)$

   $0.445 = 0.2415 + 0.591pH$

   $pH = (0.445-0.2415)/0.0591 = 3.64$

5. What is the single electrode potential of a half cell for zinc electrode dipping in a 0.01 M zinc solution? (E^{0} = 0.763 V)

   Sol. $E = E^{0} - \frac{0.0592}{n}\log[ion]$

   $$= 0.763 - \frac{0.0592}{2}\log[0.01]$$

   = 0.822 V

6. A nickel electrode (E^{0} = -0.13 V) was combined with an aluminium electrode (E^{0} = 1.66 V) to form a cell. What would be the potential of the cell?
   Sol. $E_{cell} = E_{cathode}^{0} - E_{anode}^{0}$

   = 1.66 – (-0.13)

   = 1.66 + 0.13

   = 1.79 V

7. Calculate the EMF of the cell Zn | $ZnSO_4$ || AgCl | Ag. Given $E_{Zn2+|Zn}^{0}$ = -0.763 V and $E_{Ag+|Ag}^{0}$ = 0.22V.
   Sol. $E_{cell} = E_{cathode}^{0} - E_{anode}^{0}$

   = 0.22 – (-0.763)

   = 0.985 V

3.4 Potentiometric Titrations (Redox Titrations)

A titration in which the end point of a reaction is determined with the help of the measurement of potentials of the reaction mixture is known as potentiometric titration.
Potassium dichromate oxidises ferrous sulphate is an example for redox titration of potentiometric titrations.

Here the end point is detected using a potentiometer.

From the Nernst equation, you know that the potential of a given reaction will depend on the relative concentration of oxidized /reduced form.

During their titration, the solution potential changes due to the change in the concentration of oxidized /reduced form.

At one stage, where either of the form is absent i.e., at the end point, there is a sharp change in potential.

Potassium dichromate is an oxidizing agent and in acid medium; it follows the half reaction to give Cr (III) as the reduction product.

$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightleftharpoons 2\ Cr^{3+} + 7\ H_2O$ ... (1)

While Fe^{2+} which is used to titrate $K_2Cr_2O_7$ gets oxidized to Fe^{3+} as per the reaction

$Fe^{2+} \rightleftharpoons Fe^{3+} + e^-$ … (2)

The overall ionic equation of this titration can be obtained by adding the above two:

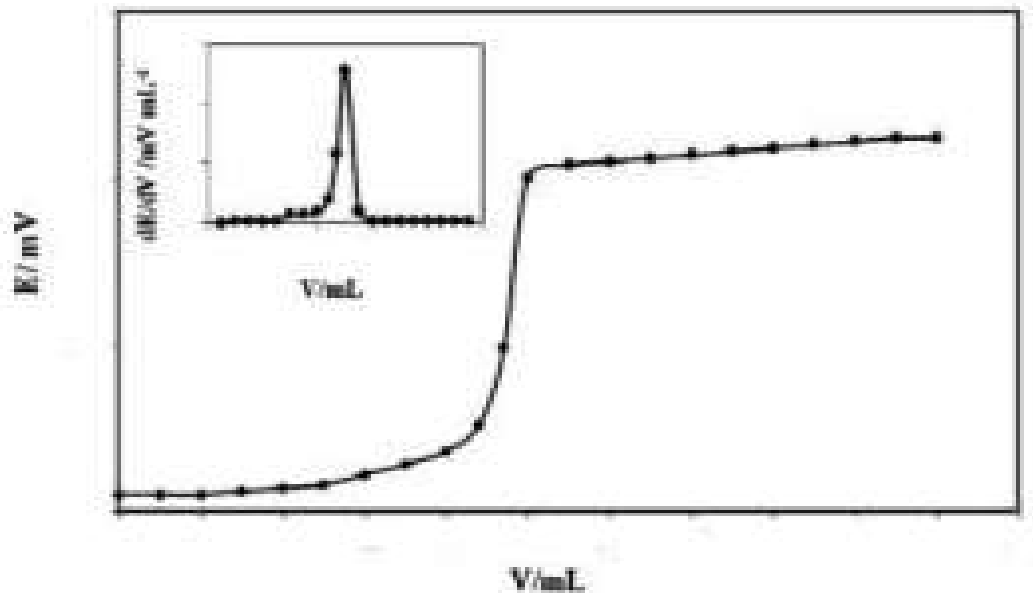
$Cr_2O_7^{2-} + 6\ Fe^{2+} + 14H^+ \rightarrow 6\ Fe^{3+} + 2Cr^{3+} + 7\ H_2O$ ... (3)

Procedure

1. Using 100 ml volumetric flasks prepare of 0.02 M potassium dichromate solution and 0.10 M ammonium iron (II) sulphate solution. You may have to add sufficient amount of dilute acid to prepare ammonium iron (II) sulphate solution.
2. Take 25 ml of given Fe^{2+} solution and add 25 ml dilute H_2SO_4 acid and 50 ml of distilled water in a 250 cm3 beaker.
3. Standard Calomel Electrode is used as the Reference electrode. Platinum metal foil, dipped in Fe^{2+}solution is used as the indicator electrode.
4. Standardize the potentiometer using a standard cell before replacing it with the working cell.
5. Add 2 ml of 0.02 M $K_2Cr_2O_7$ solution from burette, operate the magnetic stirrer for 2 minutes, stop it, wait it for 1 minute and measure the E.M.F.
6. Repeat the above step, each time adding two more ml of $K_2Cr_2O_7$ at a time and go on noting the E.M.F. after each addition.
7. When the volume reached near about 1 ml of the expected equivalence point (approximate), add the solution from burette in 0.5 ml intervals and note the potential each time.
8. Continue adding these instalments even after the equivalence point (This can be easily observed from the change in measured potential). The change becomes very small. Continue for another 5-6 additions. Note the potential readings.

9. Record the observations in the Observation Table 1.Follow the same procedure for plotting the graphs and locate the equivalence point.
10. Observations

S. No	Volume (V) of $K_2Cr_2O_7$ (ml)	EMF (E) of the cell (milli volts)
1 2		



Calculations

The vol. of K2Cr2O7 used corresponding to the equivalent point = _____

Apply the dilution law to calculate concentration of potassium dichromate solution

$N_1V_1 = N_2V_2$

Where Concentration of Fe^{2+} solution (N_1) = _____ M/lit

Volume of Fe^{2+} solution (V_1) = _____mL

Concentration of $K_2Cr_2O_7$ solution (N_2) = _____M/lit

Volume of $K_2Cr_2O_7$ solution (V_2) = _____mL

The concentration of Fe^{2+} solution = N_1

= N_2 V_1/V_2

= N

The strength of the given Ferrous Ammonium Sulphate solution = N_1*Equivalent weight of Fe^{2+} ions

=N_1* _____

= _____ gm./mole

Result: The concentration of Fe^{+2} solution = N The strength of the given Ferrous Ammonium Sulphate solution=_____ gm/mole.

3.5 Concept of Conductivity

Conductivity is a measure of a solution's ability to conduct electricity. Since electricity needs charged particles in order to flow, there is generally a positive relationship between the concentration of ions and the ability of a solution to conduct electricity.

The specific conductance of a solution is the conductance of a solution contain two parallel electrodes of one cm^2 cross section and which are kept a part of one cm. If R is the resistance of a solution measured by a cell having electrodes of cross-sectional area "a" cm^2 and "l" cm apart, the specific conductance K of the solution is given by

$$K = \frac{1}{R} \times \frac{l}{a}$$

$$K = c \times \frac{l}{a}$$

Where "c" is observed conductance and "*l/a*" is cell constant

Cell constant = specific conductance / observed conductance

3.6 Conductometric Titrations

The titration in which the end point is determined by measuring the conductivity of sample solution without using any indicator is called conductometric titrations.

A solution of an electrolyte conducts electricity due to its dissociation in to ions which depends on their number and mobility.

Since specific conductance of a solution is proportional to the concentration of ions in it, conductance of the solution is measured during titration.

Strond Acid with Strong Base

A reaction between strong acid and strong base, leading to the following neutralization.

$$HCl + NaOH \rightarrow NaCl + H_2O + NaOH \text{ (excess)}$$

This reaction is followed conductometric titration in a conductivity meter using a conductivity cell.

When a strong base of NaOH is added slowly from the burette to the solution of HCl, the fast-moving H^+ ions are progressively replaced by slow moving Na^+ ions.

As a result, conductance of the solution decreases.

This decrease in conductance will take place until the end point is reached.

Further addition of NaOH increases the conductance sharply as there is an excess of fast moving OH^- Ion, because it is a strong electrolyte.

$$NaOH \rightarrow Na^+ + OH^-$$

Procedure

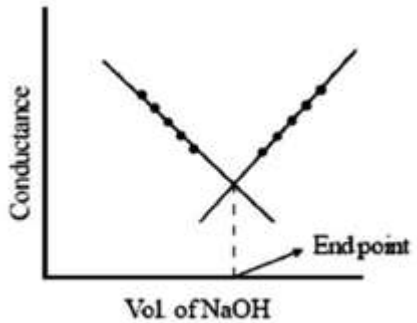
The burette is filled with standard 0.1N NaOH solution.
Make up the given HCl sample solution up to the mark of100 ml volumetric flask.
Take 20 ml of the given HCL in to a clean 100 ml beaker.
The conductivity cell is placed in it. Initial conductance is read in the instrument for the acid alone without the addition. of NaOH.
Now 0.5 ml of NaOH solution from the burette is added to the solution taken in the beaker, stirred and then conductivity is measured.
Repeat the procedure of adding 0.5 ml NaOH by noting the conductivity of resulting solution in a tabular form each time.
Take 12-15 readings and plot the graph.

Observations:

S. NO.	Volume of NaOH added (ml)	Conductance (mʊ)
1.		
2.		
3.		

Graph

A plot is making between volume of NaOH added along X-axis and the conductance of sample solution along Y- axis. The end point is intersection of the two lines.



Calculation

Step -1: Calculation of normality of HCl:
Volume of NaOH (V_1) = _____ ml (titer value from graph)

Normality of NaOH (N_1) = _____ N (given)

Volume of HCl (V_2) =

Normality of HCl (N_2) =

By volumetric principle $N_1V_1 = N_2V_2$

Normality of HCl (N_2) = _____N

Step – 2: Calculation of amount of HCl
The amount of HCl present in the given solution = N_2 X Eq. wt. of HCl X100/1000

= N_2X 36.45/10 gm/100cc

Report:

The amount of HCl present in the given sample solution _____ gm/100cc

Weak acid with Strong Base

A reaction between weak acid and strong base, leading to the following neutralization.

$$CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O + NaOH \text{ (excess)}$$

This reaction is followed conductometric titration in a conductivity meter using a conductivity cell.

When a strong base of NaOH is added slowly from the burette to the solution of CH_3COOH, the slow-moving H^+ ions are progressively replaced by slow moving Na^+ ions.

As a result, conductance of the solution constantly increases. This constant increase in conductance will take place until the end point is reached.

Further addition of NaOH increases the conductance sharply as there is an excess of fast moving OH^- Ion, because it is a strong electrolyte.

$$NaOH \rightarrow Na^+ + OH^-$$

Procedure

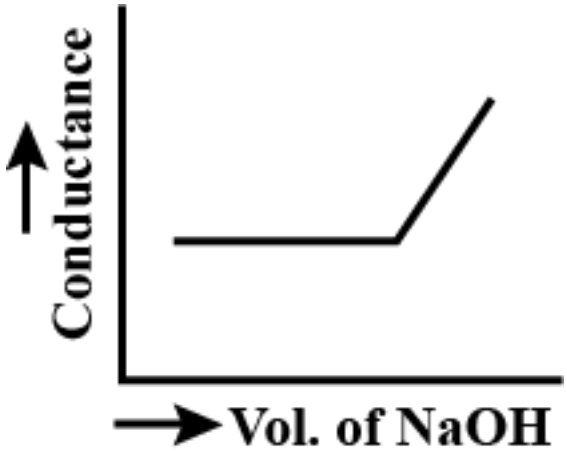
The burette is filled with standard 0.1N NaOH solution.
Make up the given CH_3COOH sample solution up to the mark of 100 ml volumetric flask.
Take 20 ml of the given CH_3COOH in to a clean 100 ml beaker.
The conductivity cell is placed in it. Initial conductance is read in the instrument for the acid alone without the addition. of NaOH.
Now 0.5 ml of NaOH solution from the burette is added to the solution taken in the beaker, stirred and then conductivity is measured.
Repeat the procedure of adding 0.5 ml NaOH by noting the conductivity of resulting solution in a tabular form each time. Take 12-15 readings and plot the graph**.**

Observation:

S.NO.	Volume of NaOH added (ml)	Conductance (mʊ)
1.		
2.		
3.		

Graph:

A plot is making between volume of NaOH added along X-axis and the conductance of sample solution along Y- axis. The end point is intersection of the two lines.



Calculation:

Step -1: Calculation of normality of CH_3COOH:

Volume of NaOH (V_1) = _____ ml (titer value from graph)

Normality of NaOH (N_1) = _____ N (given)

Volume of CH_3COOH (V_2) =

Normality of CH_3COOH (N_2) =

By volumetric principle $N_1V_1 = N_2V_2$

Normality of CH_3COOH (N_2) = _____N

Step – 2: Calculation of amount of CH_3COOH

The amount of CH_3COOH present in the given solution =

= N_2 X Eq. wt. of CH_3COOH X100/1000

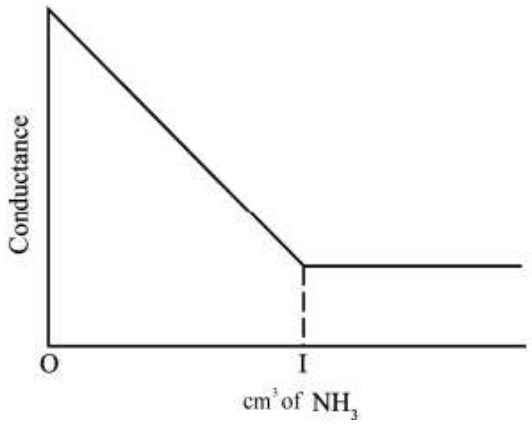
= N_2X 60.05/10 gm/100cc

Report

The amount of CH_3COOH present in the given sample solution _____ gm/100cc

Strong Acid with a Weak Base

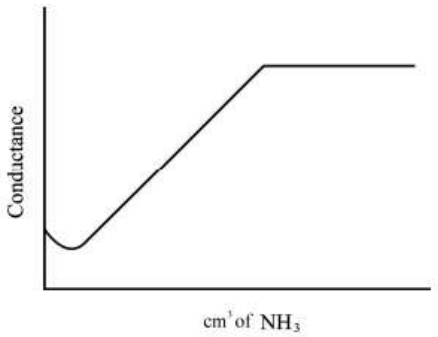
Example: Sulphuric acid with dilute ammonia:
Initially the conductance is high and then it decreases due to the replacement of H^+.
But after the endpoint has been reached the graph becomes almost horizontal, since the excess aqueous ammonia is not appreciably ionised in the presence of ammonium sulphate.



Weak Acid with a Weak Base

The nature of curve before the equivalence point is similar to the curve obtained by titrating weak acid against strong base.

After the equivalence point, conductance virtually remains same as the weak base which is being added is feebly ionized and, therefore, is not much conducting.



3.7 Electrochemical Sensors

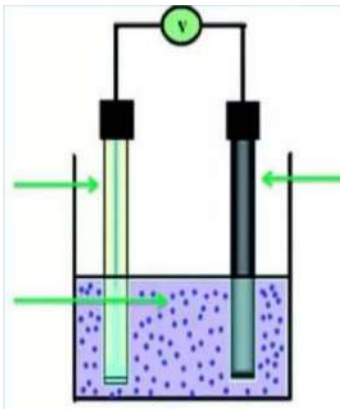
The electrochemical sensors are developed in 1962 by Naoyoshi Taguchi.
The devices based on the technology are called TGS (Taguchi gas sensors).
Electrochemical sensors are a class of chemical sensors in which an electrode is used as a transducer element in presence of analyte.
Transducer is a device that converts physical change into electrical signals and transducer element is a sensitive element.

3.7.1 Applications of Electrochemical Sensors

1. Environmental monitoring of air quality. Due to the expansion of expanding industries many countries are affected by poor air quality.
2. To detect explosive gases and toxic vapours.
3. Oxygen monitoring in medical sector.
4. In food quality control, petrochemical industries, detection of heavy metals and in transportation.

3.7.2 Potentiometric Sensors

A potentiometric sensor is a type of electrochemical sensor that may be used to measure potential difference of an electrode.



The signal is measured as the potential difference (voltage) between the working electrodes and the reference electrode.

The working electrode's potential must depend on the concentration of the analyte in the gas or solution phase.

In the potentiometric sensor, the ion selective electrode is coupled with the reference electrode to complete the circuit and the sensor measure the potential difference between two electrodes.

Applications:
The potentiometric sensors are used in analysis of glucose in blood.
The measuring system consists of a pH measuring electrode and reference electrode, the potential difference between these two is a function of the pH value of a measure solution.

Advantages:
Response is very rapid.
Simple and low cost.
Easy to prepare and use.
It is not affected by catalytic poisons.
Chemically resistant to oxidizing and reducing agents, dissolved gases, salts etc…

Disadvantages:
Extremely fragile.

Minute abrasions on the surface of the tip, damages the electrode.
It cannot be used with simple potentiometers, because of high resistance.
Upper limit of electrode pH is 8.
It needs to be used freshly.

3.7.3 Amperometric Sensors

Amperometric sensors are a class of electrochemical sensors that measure current resulting from a redox reaction at an electrode surface. These sensors are widely used in various fields, including environmental monitoring, biomedical applications, and industrial processes. The basic principle behind amperometric sensing involves the measurement of current at a constant applied potential.

1. **Working Principle**
   - Amperometric sensors rely on the measurement of current generated by the oxidation or reduction of electroactive species at an electrode surface.
   - A constant potential is applied to the working electrode, which facilitates the electrochemical reaction with the target analyte.

2. **Components**
   - **Working Electrode:** The electrode where the electrochemical reaction occurs.
   - **Reference Electrode:** Maintains a stable potential, ensuring the working electrode operates at a constant voltage.
   - **Counter Electrode:** Completes the electrical circuit and facilitates the movement of electrons.

3. **Selectivity**
   - The selectivity of amperometric sensors can be enhanced by using specific materials for the working electrode or by modifying the electrode surface to increase affinity for the target analyte.

4. **Applications**
   - **Environmental Monitoring:** Amperometric sensors are used for detecting pollutants, such as gases or heavy metals, in air and water.
   - **Biomedical Applications:** They play a crucial role in glucose monitoring for diabetes management and are employed in various medical diagnostics.
   - **Industrial Processes:** Amperometric sensors are used for real-time monitoring of chemical processes, ensuring optimal conditions and product quality.

5. **Advantages**
   - High sensitivity and detection limits.
   - Real-time monitoring capabilities.
   - Selectivity can be tailored through electrode modifications.

6. **Challenges**
   - Interference from other electroactive species.
   - Calibration and maintenance requirements.
   - Sensitivity to environmental conditions.

Application: Oxygen (O_2) Sensor

Working Principle

- The amperometric oxygen sensor typically consists of a working electrode, a reference electrode, and an electrolyte.
- The working electrode is often made of materials like gold, platinum, or other noble metals.
- The oxygen reduction reaction occurs at the working electrode surface, resulting in the generation of a current proportional to the oxygen concentration in the environment.

Reaction at the Working Electrode:

$$O_2 + 4e^- + 2H_2O \rightarrow 4OH-$$

Components

1. **Working Electrode:** Often made of noble metals like platinum.
2. **Reference Electrode:** Maintains a stable potential.
3. **Electrolyte:** Facilitates the electrochemical reaction.

Advantages

- High sensitivity to changes in oxygen concentration.
- Fast response time.
- Wide range of applications, including medical, environmental, and industrial settings.

Applications

1. **Medical Applications**
   - Used in medical devices such as blood gas analyzers to measure oxygen levels in blood.
   - Monitoring oxygen concentration in incubators for premature infants.
2. **Environmental Monitoring**
   - Monitoring oxygen levels in bodies of water to assess water quality.
   - Tracking oxygen concentration in air to assess pollution levels.
3. **Industrial Processes**
   - Control of oxygen levels in manufacturing processes, such as in the food and beverage industry.
   - Monitoring oxygen in inert gas environments to prevent combustion.
4. **Automotive**
   - Integrated into exhaust gas sensors to monitor oxygen levels for optimal combustion in vehicles.

3.8 Zinc Air Battery

In zinc air battery, the anode is made of zinc plates.
A perforated carbon plate treated with water repellents acts as a cathode.
Sodium hydroxide (5M) or potassium hydroxide is used an electrolyte.
The anode, cathode and the electrolyte are contained in an ebonite or polymer case.
At this anode, zinc reacts with electrolyte (NaOH or KOH) to form zincate ions which decay into zinc oxide and water.

The electrons released at the anode travel to the cathode where oxygen of the air accepts the electrons to form hydroxide ions.
The cell represented as Zn/NaOH(5M)/Air/C
The cell reactions are

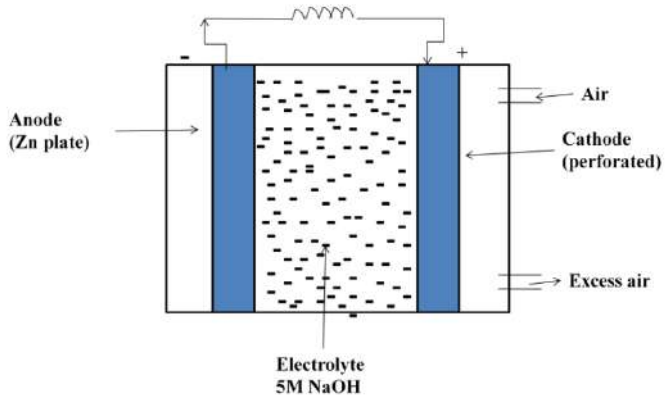
Anode: $2\ Zn + 4\ OH^- \rightarrow 2\ ZnO + 2\ H_2O + 4e^-$

Cathode: $O_2 + 2\ H_2O + 4e^- \rightarrow 4\ OH^-$

Net reaction: $2\ Zn + O_2 \rightarrow 2\ ZnO$

They produce 1.65V.
Zinc air batteries have the properties of both fuel cells and batteries.
Zinc air batteries are electrically non-rechargeable; however, they can be recharged mechanically by converting the zinc oxide produced in the battery into zinc metal.
The zinc metal can be used. They are very cheap and increasingly being used instead of mercury batteries.

Their size varies from small button cells that are used in the watches and in hearing aids to large batteries used in cameras and also for the electric propulsion of vehicles.



3.9 Li – ion Battery

In these batteries lithium metal or lithium compounds are used as anode.

The cathode is made of specially treated MnO_2 crystal obtained by special heat treatment.
Since, lithium is highly reactive with water and non-aqueous solvents, these solvents cannot be used as electrolyte.

Instead, lithium salt solution in propylene carbonate and dimethoxy salts is used the electrolyte solvents like thionyl chloride containing lithium compounds such as LiCl, LiBr, $LiAlCl_4$, $LiSO_3F_3$, $LiClO_4$ can also be used.

The cell can be represented as
Li / non-aqueous solvents containing lithium salts /MnO_2

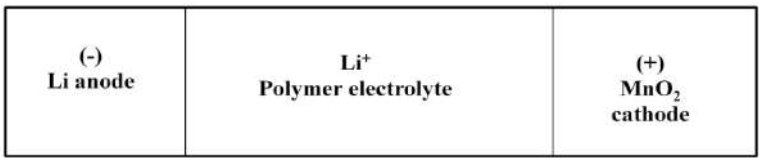
The cell reactions are

At anode: $Li \rightarrow Li^+ + e^-$

At cathode: $MnO_2 + Li^+ + e^- \rightarrow MnO2(Li^+)$

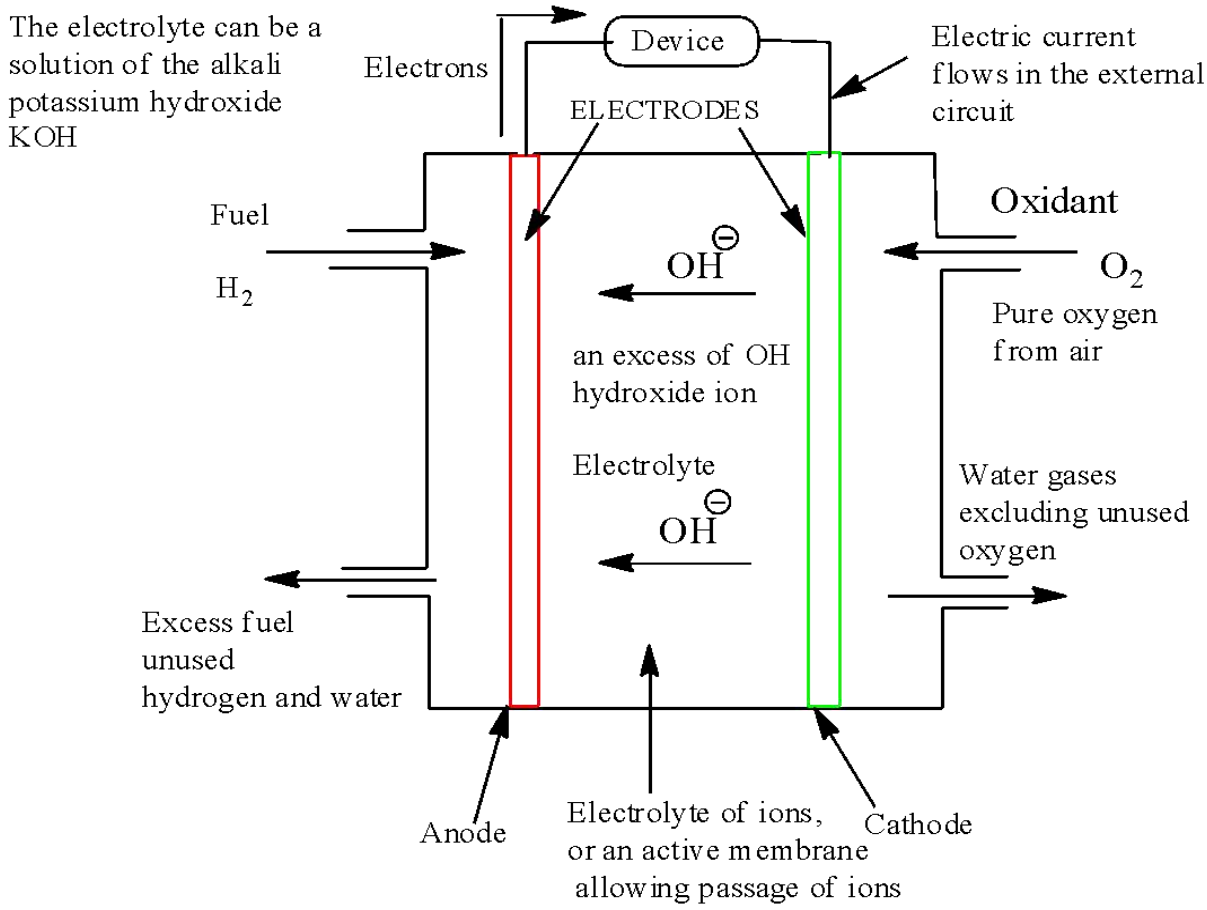
Net cell reaction: $Li + MnO_2 \rightarrow LiMnO_2$

Lithium batteries find use in clocks, calculators, toys, digital camera, watches, medical equipment like artificial pacemakers, thermometers and remote car locks etc.



3.10 H_2 -O_2 fuel cell

- Hydrogen - oxygen fuel cells consisting of two porous graphite electrodes, which is impregnated with a catalyst such as pt/pd/Ag.
- 25% KOH is used as an electrolyte.
- Hydrogen and oxygen are continuously supplied to the anode and cathode respectively.
- The hydrogen is oxidized and the electrons lost in this process migrate to the cathode through the external circuit.
- Oxygen at cathode receives the electrons and gets reduced to OH^- ions.
- The cell reaction, hydrogen and hydroxyl ions combine to form water which is removed from the cell.



Cell Reactions

At anode: $H_2 + OH^- \rightarrow 2H_2O + 2e^-$

At cathode: $\frac{1}{2} O_2 + H_2O \rightarrow 2OH^-$

Overall cell reaction: $H_2 + \frac{1}{2} O_2 \rightarrow H_2O$

Advantages

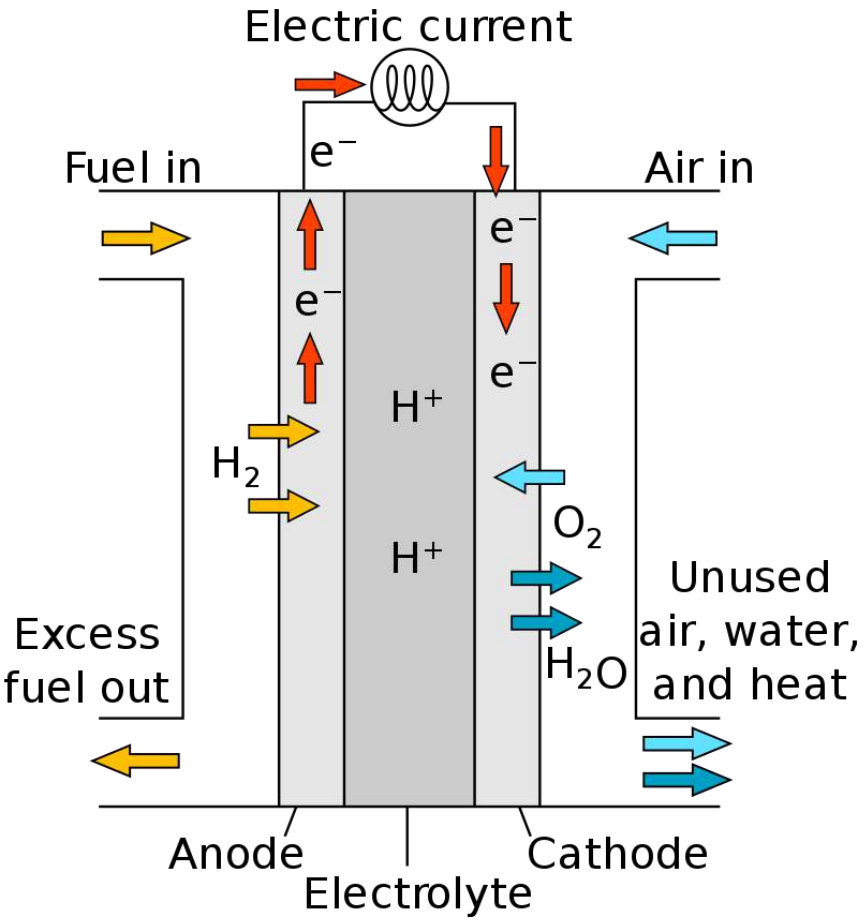
- The energy conversion is very high (75-82%).
- The product H_2O is drinking water for astronauts.
- Noise and thermal pollution are low.
- Maintenance cost is low for the fuels.
- The regenerative H_2-O_2 fuel cell is an energy storage system for space applications.

Limitations

- The life time of fuel cell is not accurately known.
- Initial cost is high.
- The distribution of hydrogen is not proper.

3.11 Polymer Electrolyte Membrane Fuel cells (PEMFC)

Polmer electrolyte membrane fuel cell is also known as proton exchange membrane fuel cell.



- These cells are also known as proton exchange membrane fuel cells (or PEMFCs).
- The temperature range that these cells operate in is between 50°C to 100°C
- The electrolyte used in PEMFCs is a polymer (Nafion, Polyether ketones, polyether sulphones) which has the ability to conduct protons.
- A typical PEM fuel cell consists of bipolar plates, a catalyst (ink of carbon), electrodes (platimum), and the polymer membrane.
- Each electrode is made up of with platinum or platinum alloy.
- Cell reaction

At anode: $H_2 \rightarrow 2H^+ + 2e^-$

At cathode: $O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$

Net reaction: $2\ H_2 + O_2 \rightarrow 2H_2O$

Applications

1. PEMFC is primarily used in transportation on because of their environmentally friendly nature because of the controlling of the emission of the greenhouse gases.
2. Distribution, stationary and portable power generation is another important application of PEMFC.

Short Answer Questions

1. Illustrate about electrochemical cell.
2. Interpret the significance of Nernst Equation.
3. Calculate the e.m.f. of the half-cell given below.
   Pt, H_2 | HCl at 1-atmosphere pressure and 0.1 M. Given, $E°_{(OP)} = 2$ V.
4. Represent the conductometric titration diagram for strong acid Vs strong base with a neat sketch.
5. Explain the principle of conductometric titrations.
6. Discuss the applications of electro chemical sensors.
7. Outline the diagram of a conductivity cell with a neat representation.
8. Discuss the applications of Amperometric sensors.
9. Write the applications of fuel cells.
10. Represent Li ion battery with its chemical equations.

Descriptive Questions

1. Construct the Nernst Equation for EMF measurements of a cell. The standard reduction potentials of given cell is 1.11V calculate the cell EMF of $Zn|Zn^{2+}(0.05M)||Cu^{2+}(0.005M)|Cu$.
2. Demonstrate the conductometric titrations of strong acid vs strong base and weak acid vs strong base with neat graphical representation.
3. Demonstrate the potentiometric titrations for a redox reaction with a neat graphical representation.
4. Illustrate the properties and applications of electro chemical sensors with examples.
5. Demonstrate the Potentiometric sensors with examples.
6. Demonstrate the Amperometric sensors with examples.
7. Explain Zinc air battery with a neat representation of chemical equations and diagram along with its applications.
8. Explain Lithium-ion battery with a neat representation of chemical equations and diagram along with its applications.
9. Explain Hydrogen – Oxygen fuel cell with a neat representation of chemical equations and diagram along with its applications.
10. Explain polymer electrolyte membrane fuel cell with a neat representation of chemical equations and diagram along with its applications.

Unit – 4

Polymer Chemistry



4.1 Introduction of Polymers

Polymers

Polymers are complex giant macro molecules made from the repeating units which are derived from small molecules called monomers. Polymers are high molecular mass macro molecules. Both inorganic as well as organic polymers are known.

Classifications of Polymers:

Classification based on source or origin:	• Natural polymers:	**Example:** Plant polymers, animal polymers
	• Synthetic Polymers:	**Example:** Fibres, Plastics.
	• Semi Synthetic Polymers:	**Example:** Varnishes, Cinema films.
Classification of polymers on the basis of structure:	• Linear or straight chain polymers:	**Example:** PVC, high density polythene.
	• Branched Chain polymers:	**Example:** low density polythene.
	• Cross linked polymers:	**Example:** Bakelite, Melamine.
Classification of polymers on the basis of mode of polymerisation:	• Addition polymers:	**Example:** polythene
	• Condensation polymers:	**Example:** polyester (Terylene)
	• Ring opening Polymers:	**Example:** Nylon 6.
Classification of polymers on the basis of intermolecular forces:	• Elastomers:	**Example:** Vulcanised rubber, Buna - S, Buna - N, Neoprene.
	• Fibres:	**Example:** Polyamide, Nylon 6, 6, Polyester (Terylene).
	• Thermo plastic polymers:	**Example:** Polyethene, Polystyrene, Poly vinyls.
	• Thermo setting polymer:	**Example:** Bakelite, urea formaldehyde resin.
Classification of polymers on the basis of type of different monomers	• Homopolymers:	**Example:** Polythene, PVC
	• Copolymers:	**Example:** Buna – S, Buna – N.
Classification of polymers on the basis of biodegradability	• Biodegradable Polymers:	**Example:** PLA, PGA
	• Non-Biodegradable Polymers:	**Example:** Plastic.

4.2 Functionality of Monomers

According to IUPAC, the functionality of a monomer is defined as the number of bonds that a monomer's repeating unit forms in a polymer with other monomers. Thus, in the case of a functionality of $f = 2$ a linear polymer is formed by polymerizing (a thermoplastic). Monomers with a functionality $f \geq 3$ lead to a branching point, which can lead to cross-

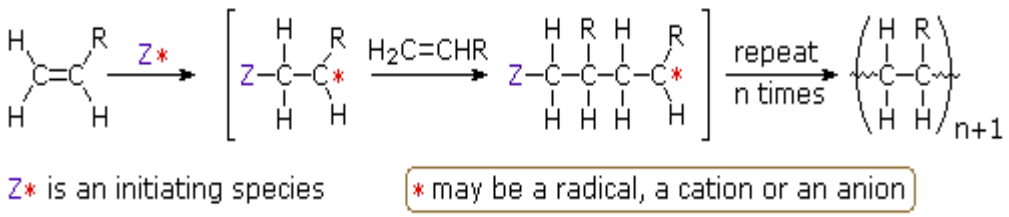
linked polymers (a thermosetting polymer). Monofunctional monomers do not exist as such molecules lead to a chain termination.

4.3 Mechanisms of Polymerization

4.3.1 Chain Growth Polymerization

Characteristics

1. The functionality of the monomer is a double bond and it is bifunctional.
2. The polymerization takes place by self-addition of the monomer molecules to each other through a chain reaction.
3. No biproducts like H2O, CH3OH etc are produced.
4. The polymer has the same chemical composition as that of monomer.
5. The mechanism is carried out in three steps i.e., initiation, propagation and termination.
6. The mechanism is rapid.
7. The conversion of a π bond into σ bond takes place during the polymerization, liberating 20 k cal / mol of energy. Hence highly exothermic reaction.
8. An initiator is required to start the polymerization reaction.

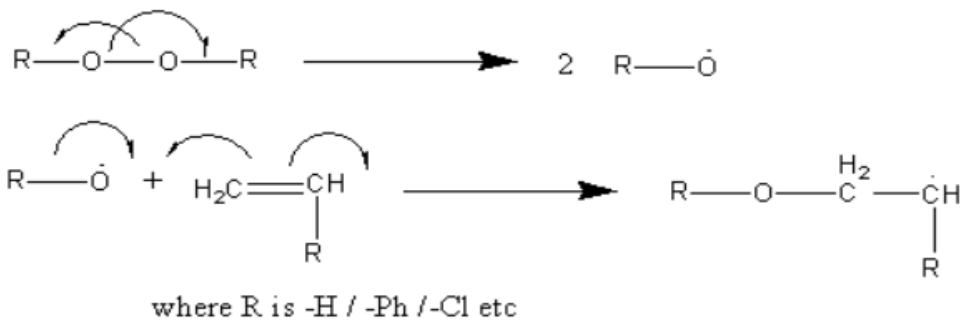


It is useful to distinguish four polymerization procedures fitting this general description.

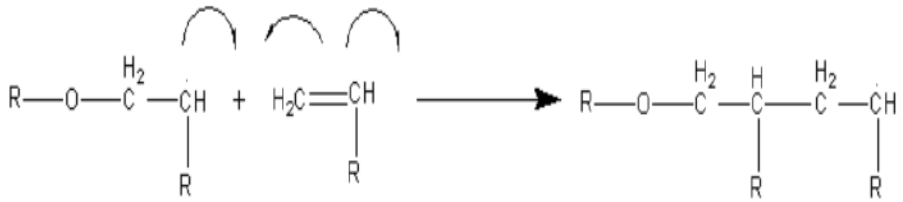
- Radical Polymerization: The initiator is a radical, and the propagating site of reactivity (*) is a carbon radical.
- Cationic Polymerization: The initiator is an acid, and the propagating site of reactivity (+) is a carbocation.
- Anionic Polymerization: The initiator is a nucleophile, and the propagating site of reactivity (-) is a carbanion.
- Coordination Catalytic Polymerization: The initiator is a transition metal complex, and the propagating site of reactivity (*) is a terminal catalytic complex.

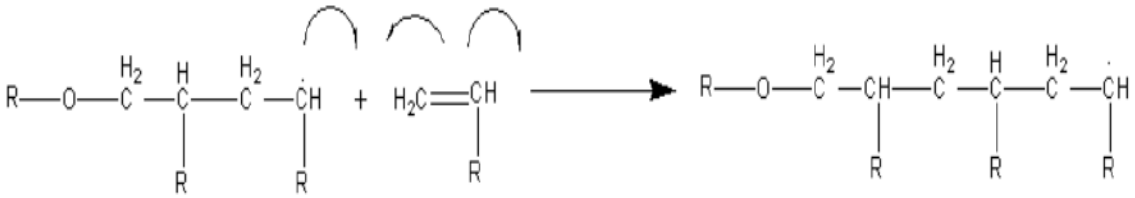
Free Radical Mechanism

1. Chain Initiation:



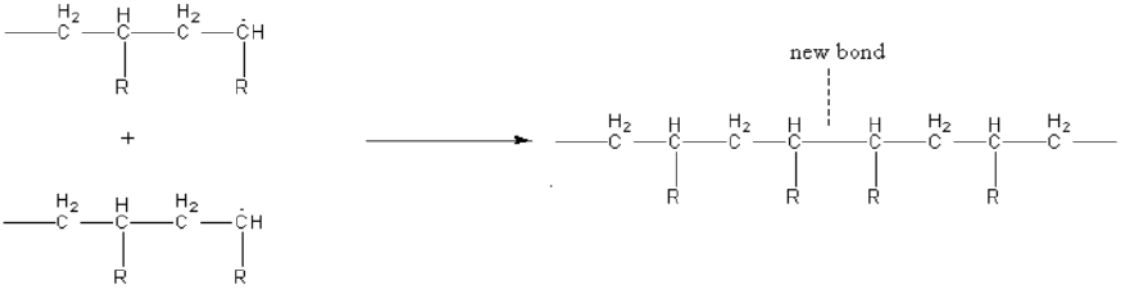
2. Chain Propagation



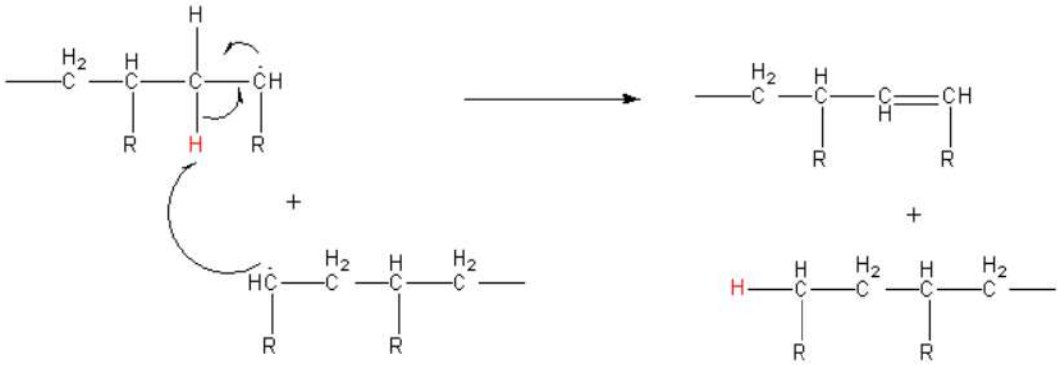


3. Chain Termination

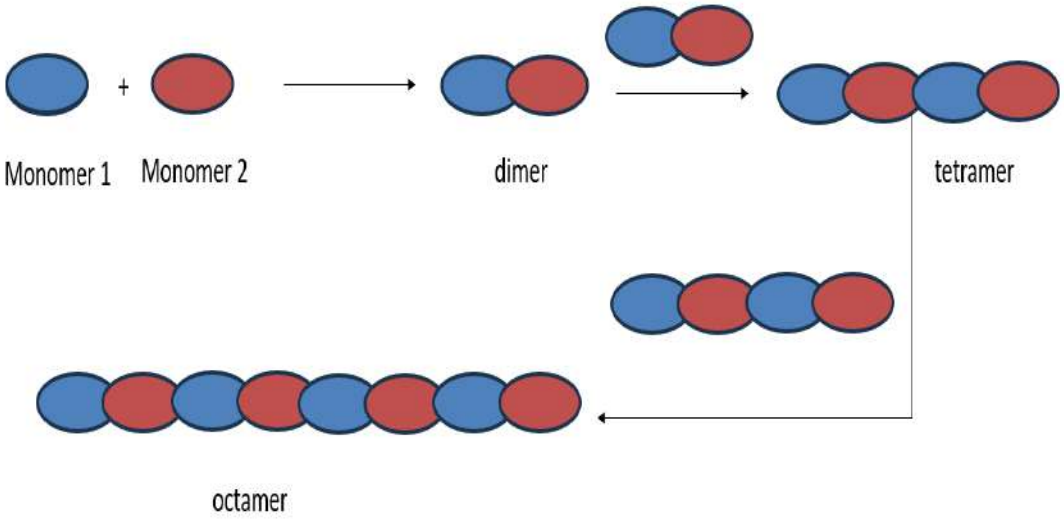
a. Combination

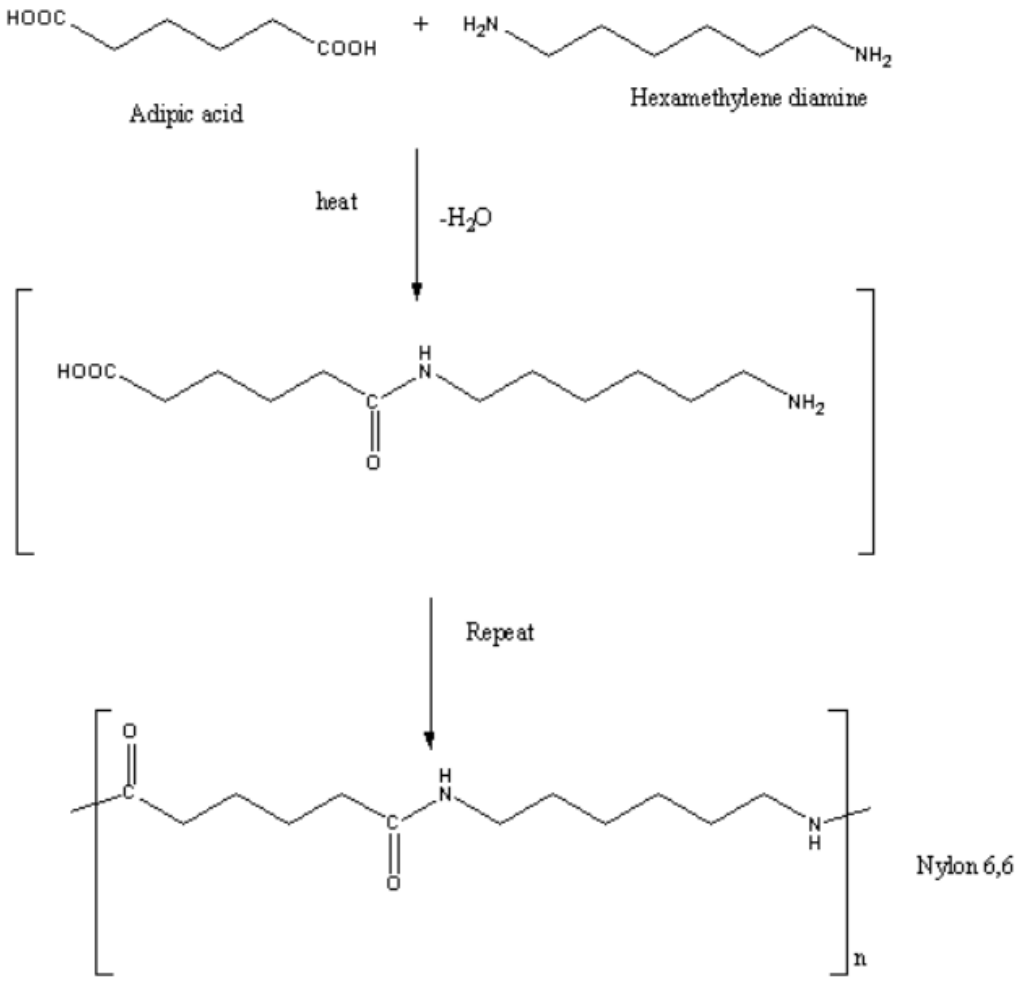


b. Disproportionation



4.3.2 Step Growth Polymerization



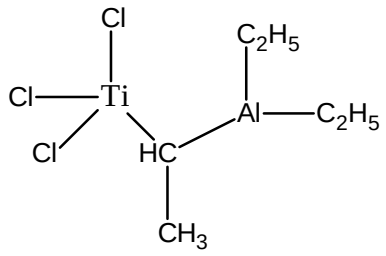


Characteristics

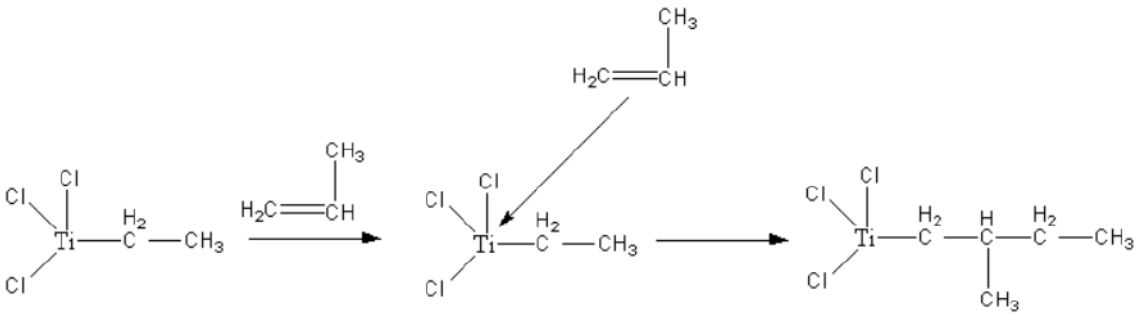
1. The monomers contains functional groups like -OH, -COOH, $-NH_2$, -RCOOR[1], halides etc.
2. The functionality of the monomer must be two or more than two. The monomers must de dibasic acids, diols, diamines or triols etc.
3. The polymer is built up by a slow step-wise condensation reaction of the functional groups of the monomer.
4. The polymerisation reaction is accompanied by the elimination of biproducts like H_2O, HCl, CH_3OH etc.
5. The reactions are not exothermic.
6. The molecular weight of the polymer is not the sum of the molecular weights of the monomers.
7. The polymers produced are living polymers containing functional groups at the end of the chain.

4.3.3 Coordination Polymerization

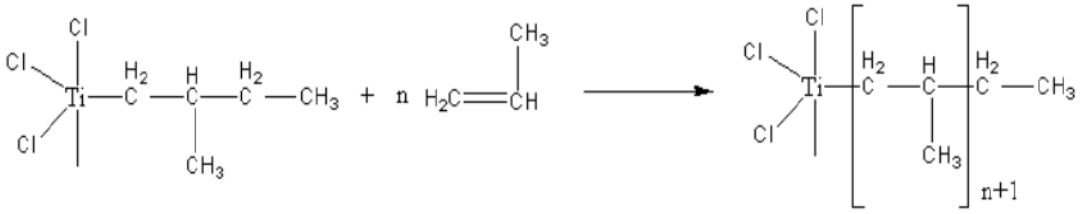
These polymerization reactions are catalysed by Zeigler Natta Catalyst formed between triethyl aluminium $Al(C_2H_5)_3$ and titanium chloride $TiCl_3$. A coordination complex is formed between these two in which the ethyl group is coordinated to titanium this is an active catalyst and its structure is shown below



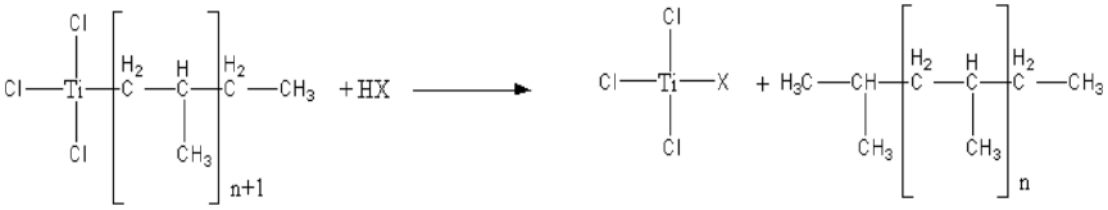
a. Initiation: In the initiation step π cloud of alkene overlaps with an empty orbital of titanium then there is insertion of alkene molecule between the Ti – C bond.



b. Propagation: As intramolecular rearrangement again generates a vacant site on Titanium each time one monomer molecule can be added in the same way.



c. Termination: It can be done by the addition of molecules containing active hydrogen like HX.



Significance:

1. It permits stereo chemical control. For example, propylene when polymerized by this method preferentially forms isotactic polymer, but the conventional polymerization results in atactic polypropylene.
2. Liner polymer, devoid of any branching, can be made. For instance, polyethylene made by Zeigler – Natta polymerization is virtually free from any branching.
3. Stereoselective synthesis is also possible with coordination catalysts since they permit stereochemical control about the carbon-carbon double bond.

4.4 Thermosetting and Thermoplastics

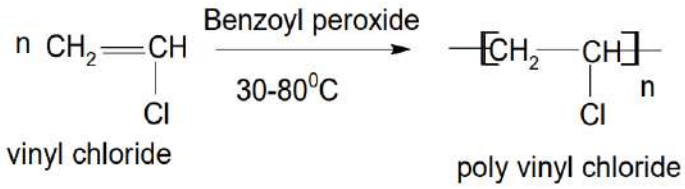
Thermoplastics polymer	Thermosetting polymer
1. They are formed by addition polymerisation.	1. They are formed by condensation polymerisation.
2. They have low molecular weight and total molecular weight of reactants equals to total molecular weight of products.	2. They have high molecular weight and total molecular weight of reactants not equals to total molecular weight of products.
3. They are reversible with temperature. Soft on heating and hard on cooling and vice versa.	3. They are irreversible with temperature. Soft on heating and never hard on cooling and vice versa.
4. They are usually soft, weak and less brittle.	4. They are usually hard, strong and mode brittle.
5. They are long chain linear polymer with negligible cross links.	5. They have three - dimensional network structure
6. They are usually soluble in some organic solvents.	6. Due to strong bonds and crosslinks, they are insoluble in almost all organic solvents.

7. They ger softened on reheating readily because secondary forces between the individual chain can break easily by heat or pressure.	7. They do not soften on reheating because the crosslinks and bonds retain their strength on heating.
8. **Example:** Polyethylene, Polystyrene, PVC, PVA	8. **Example:** Phenol formaldehyde, Urea formaldehyde, Nylon 6,6.

4.5 PVC

1. Polyvinyl chloride is a highly dense and durable thermoplastic polymer.
2. Polyvinyl chloride is the third most widely used plastic in the world.
3. Polyvinyl chloride was first prepared by Eugen Baumann in 1872.
4. It has high tensile strength and resists deformation as compared to other plastics.
5. Resin is the main component in the production of PVC.
6. PVC has a wide range of applications in the field of electrical wiring, construction, electronics, automobiles etc.

Poly Vinyl Chloride is produced by free radical chain polymerization of vinyl chloride in presence of a small amount of benzoyl peroxide or hydrogen peroxide.



Properties

- It is a colourless, non-inflammable, chemically inert material.
- It is having low density and high softening point 148°C.
- It is resistant to light, atmospheric oxygen, inorganic acids and alkalis.
- It is not stable to heat and UV radiation.
- Strong and brittle.

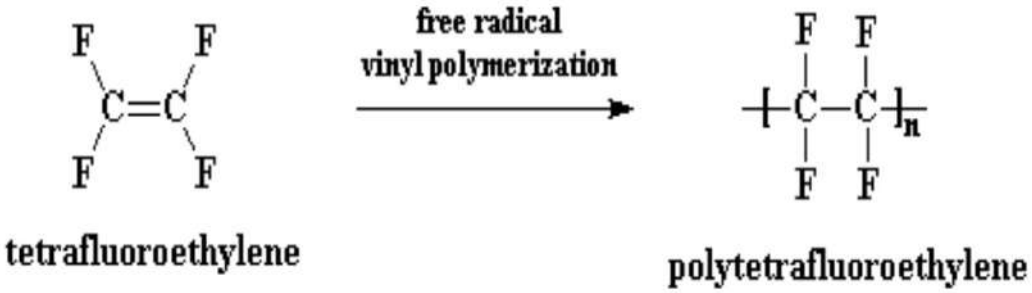
Uses

- It is mainly used as cable insulation, tank-lining, packing and toys.
- It is used for manufacturing of safety helmets, refrigerator components and floor covering.
- PVC pipes are used for carrying corrosive chemicals in petrochemical factories.
- Water carrying pipes, pipes for drainage and making frames for doors and windows.

4.6 Teflon

- Polytetrafluoroethylene or **PTFE** is known by the brand name **Teflon**.
- It was discovered by Roy J Plunkett in 1938.
- Mass production of PTFE was undertaken owing to its numerous advantages over other synthetic polymers in the market.

- A very high melting point, good thermal stability and low friction coefficient were a few of the many advantageous properties that made its use immensely wide.
- The free radical polymerisation reaction of tetrafluoroethylene can be represented as



Properties

- PTFE or Teflon is a thermoplastic polymer.
- It is ivory white solid at room temperature.
- The density of Teflon is 2200kg/m^3.
- The melting point of Teflon is 600K.
- Teflon has good strength and lubrication at low temperatures (5K). It is thermally stable at low temperatures.
- Teflon is a highly flexible, chemically resistant, non-stick and electrically resistant material.

Applications

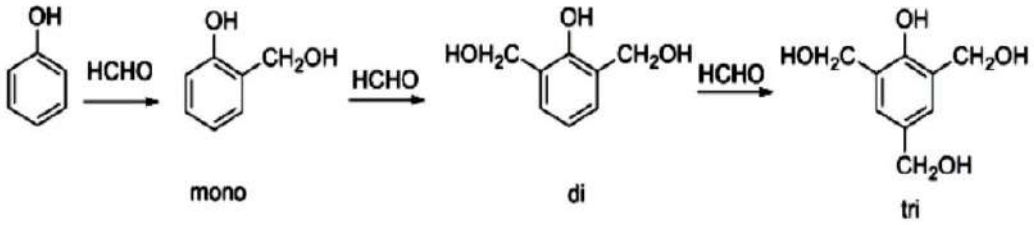
- In the manufacturing and engineering sector, Teflon is used for making tubes or liners for handling corrosive chemicals. PTFE is also used as a coating for bearings and screws due to its chemical and thermal inertness.
- PTFE coatings provide friction and corrosion-resistant surface for machinery parts thus increasing their work life. A longer lasting machinery means more efficiency and less maintenance cost.
- Teflon is also used in textile finishing as it repels oil and water, reducing the need for dry cleaning.
- Teflon is best known for its non-stick properties due to its high melting point and slippery nature. Non-stick kitchen cookware is made from Teflon.

4.7 Bakelite

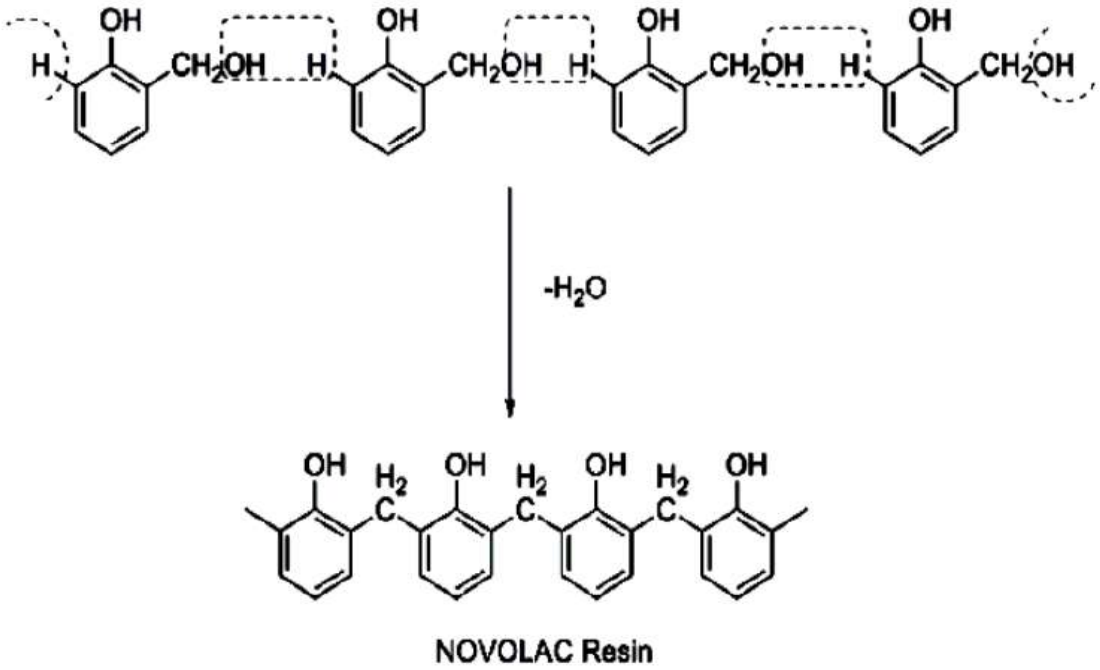
It is prepared by condensing phenol with formaldehyde in presence of acidic/alkaline catalyst.

Preparation

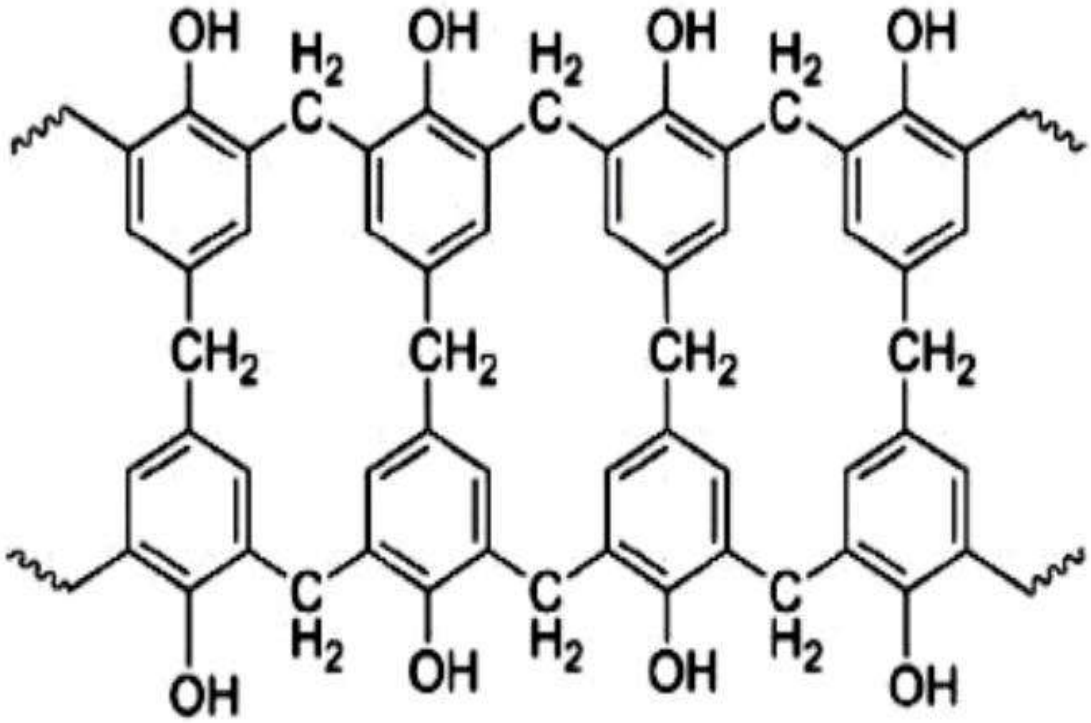
Step1: Phenol reacts with formaldehyde in basic condition to form mono, di or tri methanolated Phenol (methylol phenol), depending on the formaldehyde ratio taken (1:1, 1:2 or 1:3)



Step2: Preparation of PF resin Mono-methanolated phenol derivative condense under acid catalysis to give a low molecular weight polymer called resin (NOVOLAC).



Step 3: Bakelite formation This stage is done during moulding. NOVOLAC resin is mixed with additives and curing agent hexamethylenetetramine (HMTA) and heated. Methylene Bridge cross linking occurs between NOVOLAC resins. A hard insoluble material called Bakelite is formed.

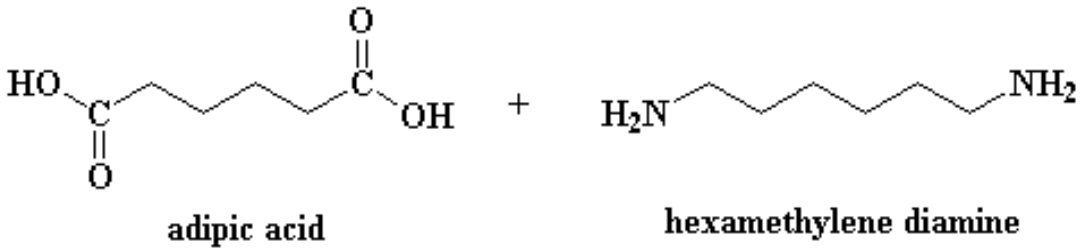


Properties and Uses

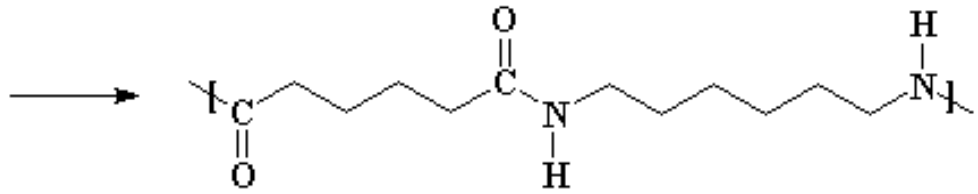
- It is electrically non-conducting, they are insulators. Used in the manufacture of circuit boards, sockets, plug points.
- They are heat resistant. Used in the manufacture of handles for kitchen ware, heaters etc.
- They are resistant to oxidizing acids. Used for the manufacture of acid storage cans. ¬ It is a good adhesive. Used as adhesive for grinding wheels.
- It is a good ion exchanging resin so used in water purification by ion exchange method
- It has good chemical and corrosive resistance.

4.8 Nylon 6, 6

Preparation:



adipic acid hexamethylene diamine



nylon 6,6

Uses of Nylon

- It is used for making strings of musical instruments and as a thread in bristles for toothbrushes.
- It is used in the synthesis of artificial fibres.
- It is also used to make fishnet.
- In the industries, nylons are used for making Conveyor and seat belts, parachutes, airbags, nets and ropes, tarpaulins, thread, and tents.

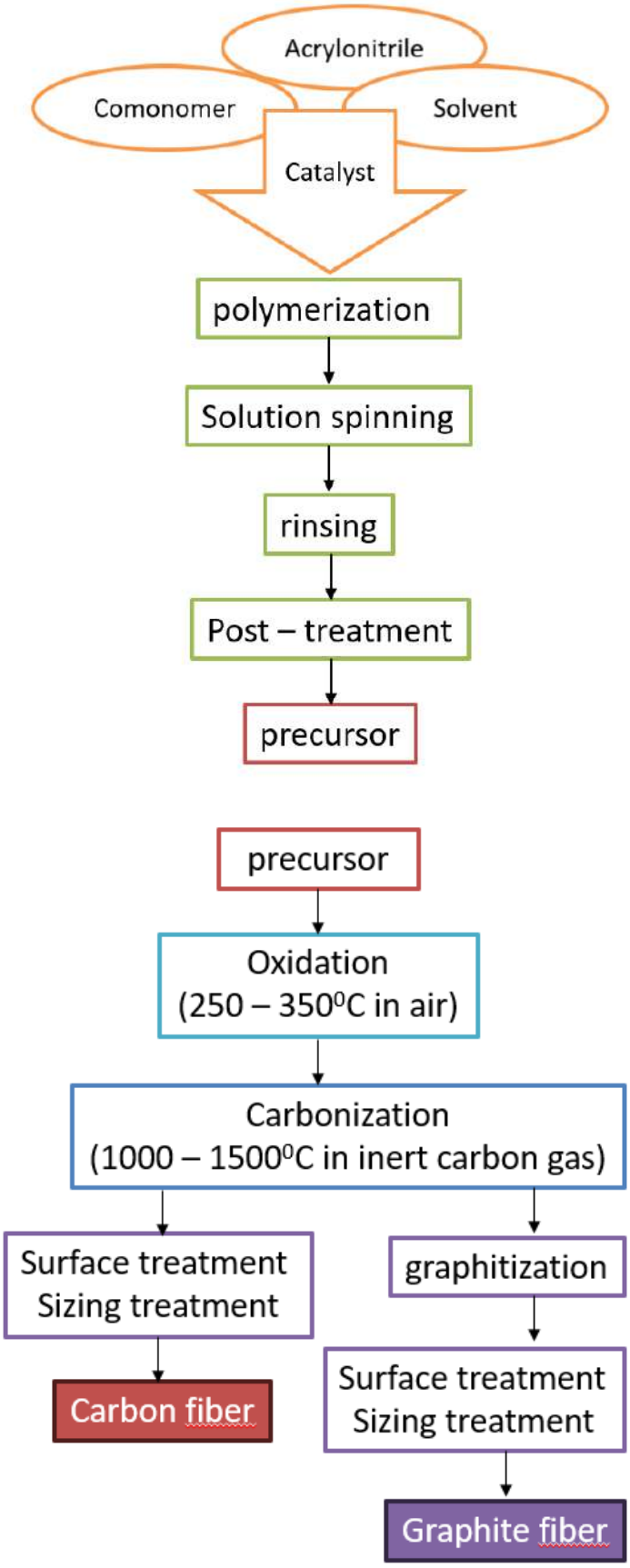
Properties of Nylon

- Nylons are Lustrous.
- They are Elastic and are very strong.
- They are Damage resistant to many oils and chemicals.
- Moreover, they are very Resilient and do not absorb water.
- Nylons dry quickly.
- They feel softer in hand.
- They have a high melting point at 256 °C or 492.8 °F.
- And they are great colour fastness.
- Additionally, they are Excellent abrasion-resistant.
- Nylons also have a high resistance to fungi, insects, animals, mildew, moulds, rot and various kinds of chemicals.

4.9 Carbon Fibres

"Carbon fibers" term is used for both carbon fibers and graphite fibers. The carbon content in carbon fibers is about 80-90 % and in Graphite fibers the carbon content is in excess of 99%. Fiber diameter ranges from 4 to10 micron (fineness around 0.4 ktex). The strongest carbon fibers are ten times stronger than steel and eight times that of aluminum.

Preparation:



1. Polymerization is performed in the presence of catalyst by adding acrylonitrile with comonomer and solvent.
2. After polymerization the components passing into solution spinning for making as a filament.
3. After spun into filament, that material taken into post treatment method by adding other ingredients for making fibre as strong and made as a precursor.
4. Precursor oxidized under 250^0 – 350^0C and send to carbonization chamber. In carbonization chamber pass inert carbon gas over the fibre at 1000 – 1500^0C.
5. After that surface treatment and sizing treatment done roll mills and made required carbon fibre.
6. Similarly after carbonization, graphite can be added for graphite fibre in graphitization process.

Propertied of Carbon Fibres

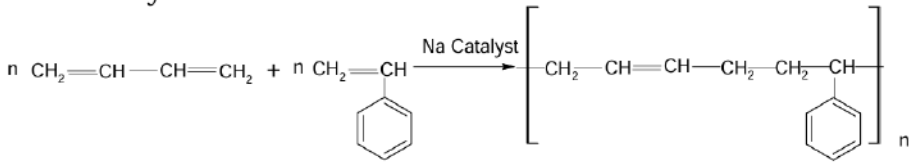
- High tensile strength
- Low thermal expansion
- Electrically and thermally conductors
- Light weight and low density
- Density 1.95 gm/cc
- Elongation at break 0.5%
- Elasticity not good.

Applications

- Civil engineering
- Sports equipment (Jerseys, Shoes, Tennis rocket, Head guards etc)
- Aerospace and aircraft industry
- Acoustics
- Portable power
- Building and construction materials
- Fibre reinforced plastics like fibre glass and bullet proof glass.

4.10 Buna – S (Styrene Butadiene Rubber / SBR / GR-S)

Buna – S prepared by the copolymerization of butadiene with styrene using sodium as a catalyst.
Bu stands for butadiene - monomer
Na stands for sodium –catalyst
S- Stands for styrene



Properties

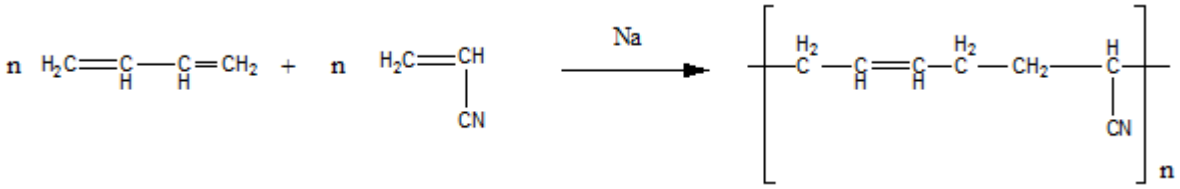
- It is a strong and tough polymer.
- It is vulcanized by sulphur monochloride or sulphur.

- It resembles natural rubber in processing characteristics and quality of finished products.
- It has excellent abrasion resistance and also good insulator.
- It is resistant to chemicals but swells in oils and attacked by even traces of ozone present in atmosphere.
- It has high load bearing capacity.

Applications

- Major applications of styrene rubber are in the manufacture of tyres.
- It is used in the foot wear industry for making shoe soles and foot wear components.
- It is also used for making wires and cables insulations.
- The other applications of styrene rubber are for the production of floor tiles, tank linings in chemical industries and as an adhesive.

4.11 Buna – N



Butadiene reacts with acrylonitrile in presence of sodium to form Buna N.

Properties

- It is highly resistant to oil and heat.
- It has high abrasion resistance.
- It is less resistant to alkalis.
- Vulcanized Buna – N rubber is resistant to ageing and high temperatures.

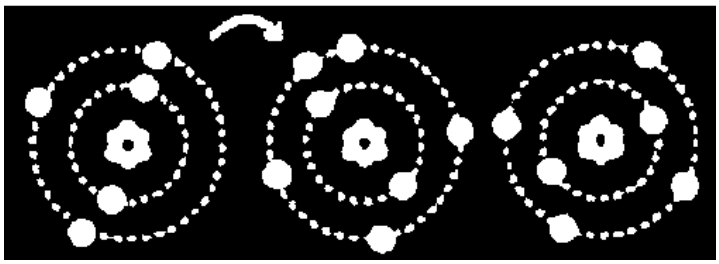
Applications

- It is used for making conveyor belts, aircraft components, automobiles parts etc.
- It is used in the manufacture of gaskets, hose pipes etc.
- It is also used for making tank linings and pipes for chemical industries.

4.12 Conducting Polymers

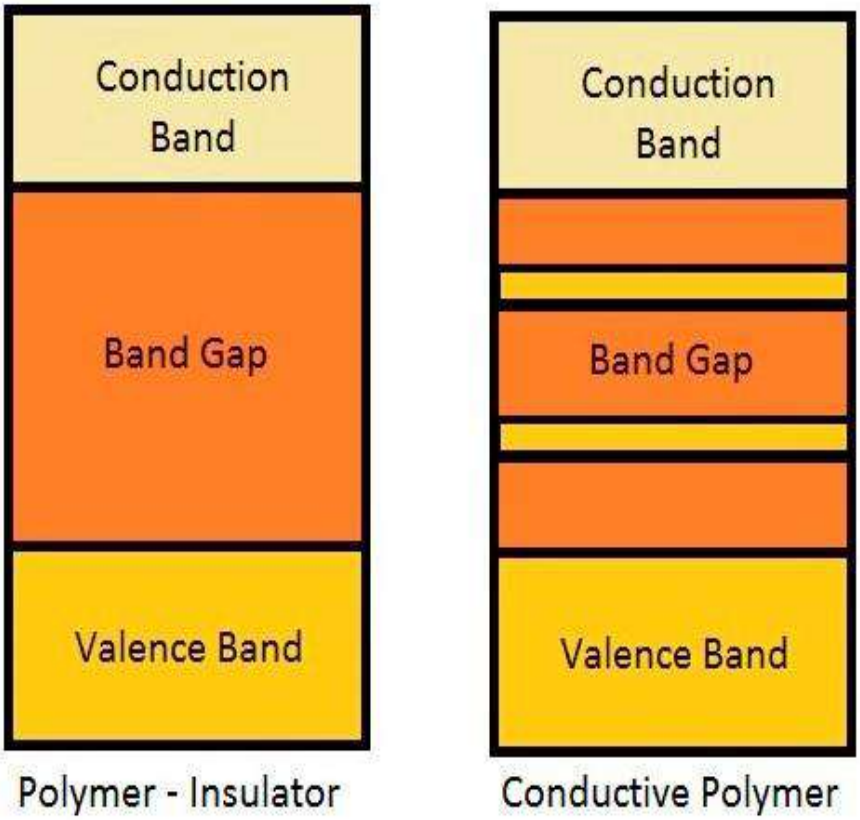
A polymer which can conduct electricity is known as conducting polymer. In future, there is a day, conducting polymers may be replace of metal in electrical wiring. They are light weight and do not corrode. Conducting polymers also called synthetic metals.

4.12.1 Mechanism of Conduction

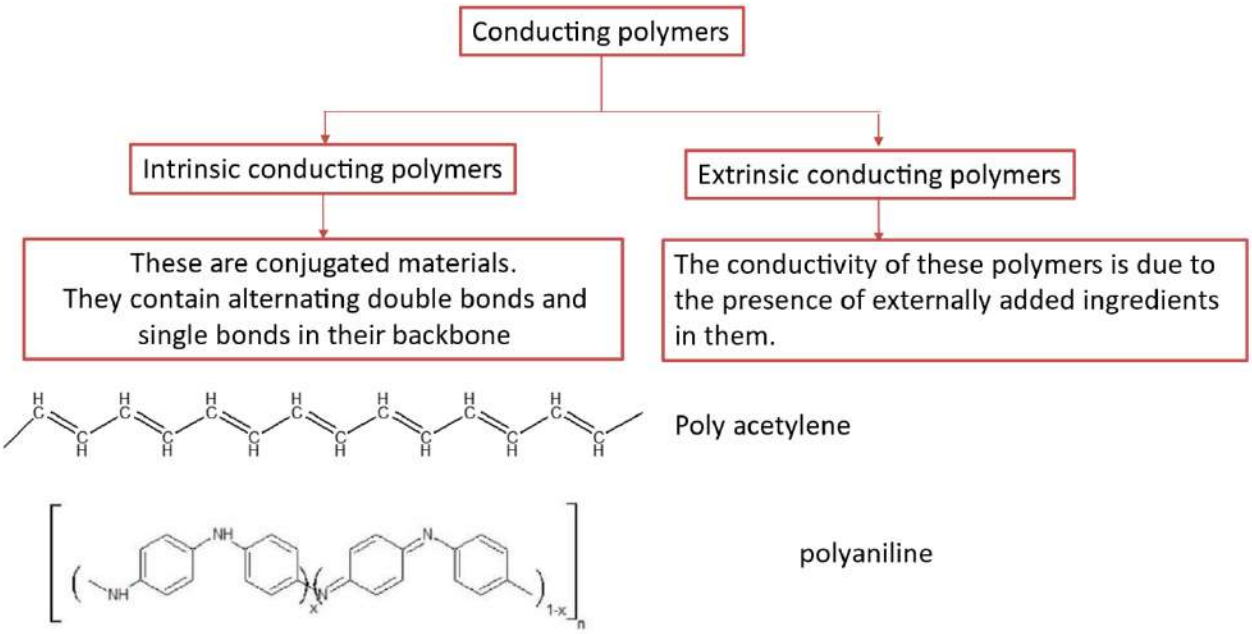


Flow of electrons from atoms to atoms.

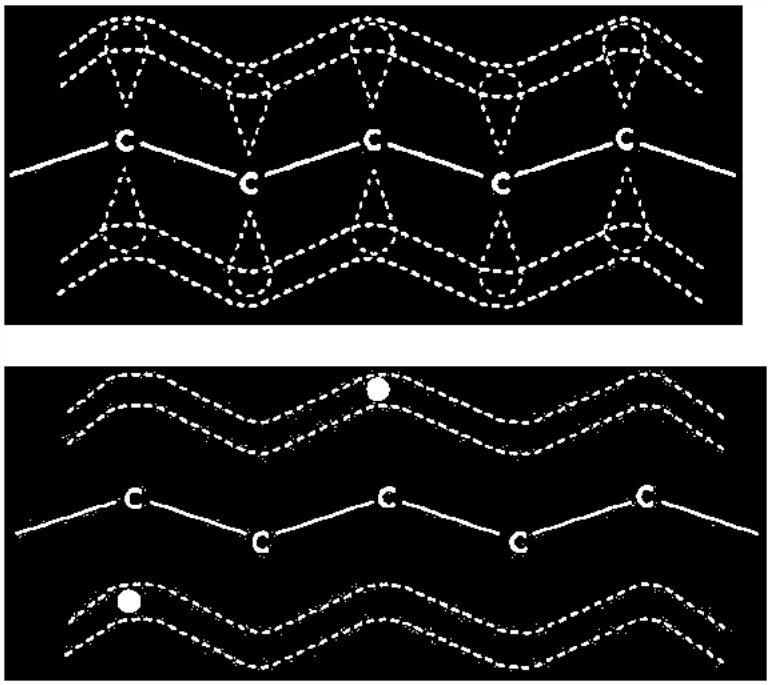
If any material having free electrons which can move will be conducting



4.12.2 Types of Conducting Polymers

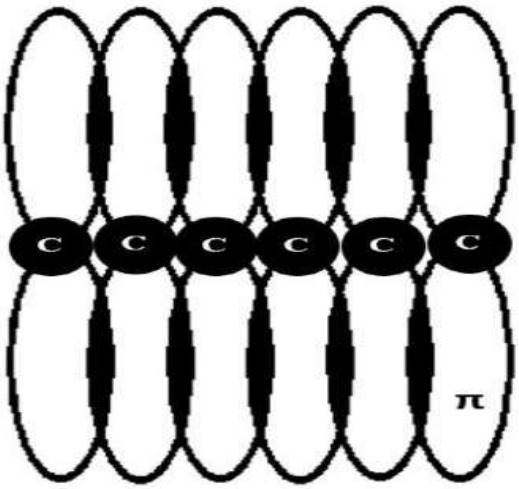


4.12.3 Conduction Mechanism in Polyacetylene:



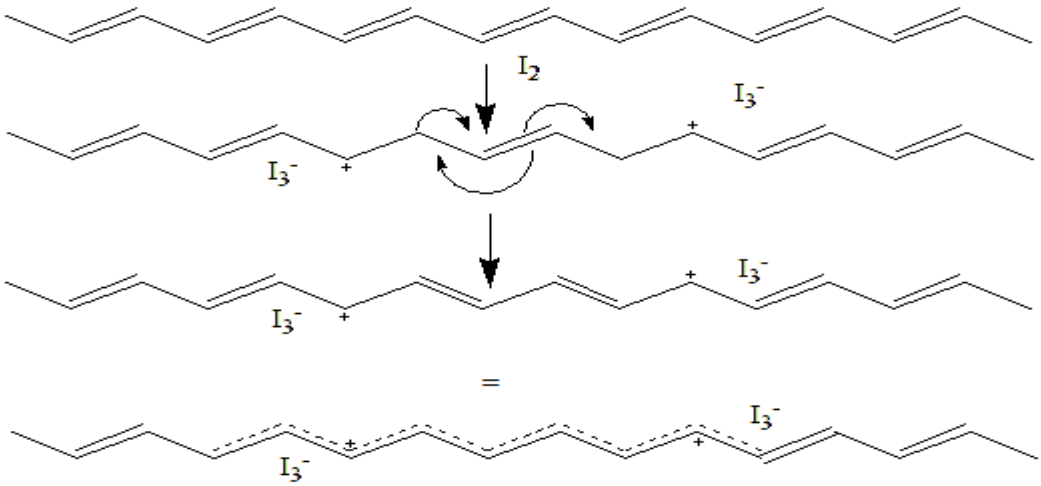
- The bands should be partially filled in order to be electrically conducting, as neither empty nor full bands can carry electricity.
- Owing to the presence of partially filled energy bands metals have high conductivities
- The energy bands of insulators and semiconductors are completely full or completely empty.

- Electrical conduction could occur only after thermal or photolytic activation of electrons give to them sufficient energy to jump the gap and reach into the lower levels of conduction band. (Polyacetylene band gap 1.5 ev and polyaniline 3.2ev)
- Unsaturated polymers consist of hydrogen and SP2 hybridized carbon atoms
- The localized electrons from the backbone of polymer chain and dominate the mechanical properties.
- The electrons in the pi bond are delocalized along the chain and are responsible for optical and electrical properties of conjugated polymers.



σ and π bonds of a poly carbon-carbon backbone

- The current along the polymer chains can only flow if one or more electrons are removed or added it.

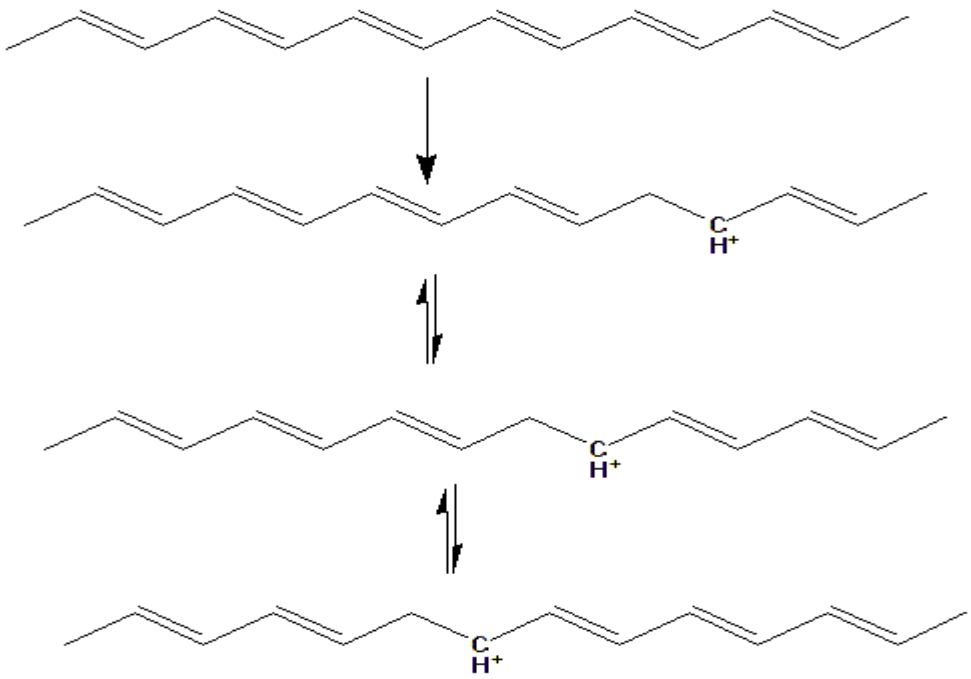


Oxidation with halogen (p-doping)

$$[CH]_n + 3x/2\ I_2 \longrightarrow [CH]_n^{x+} + xI_3^-$$

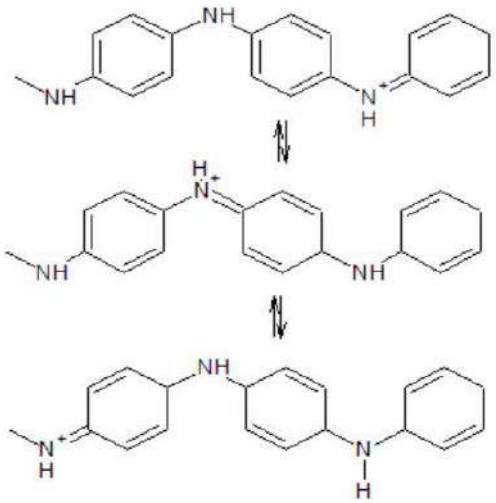
- The lone electron of pi bond from which an electron was removed can move easily.

Due to this, the double bond move successively moves along the chain.



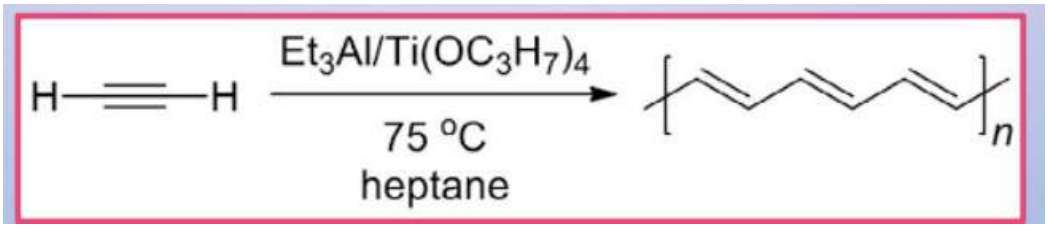
4.12.4 Conduction Mechanism in Polyaniline:

The doping process occurs through the protonation of the nitrogen atoms in the amine groups of the polyaniline by an acidic solution, oxidizing the structure. Thus, positive charges are formed in the polyaniline structure that moves as the structure resonates, shown in figure.



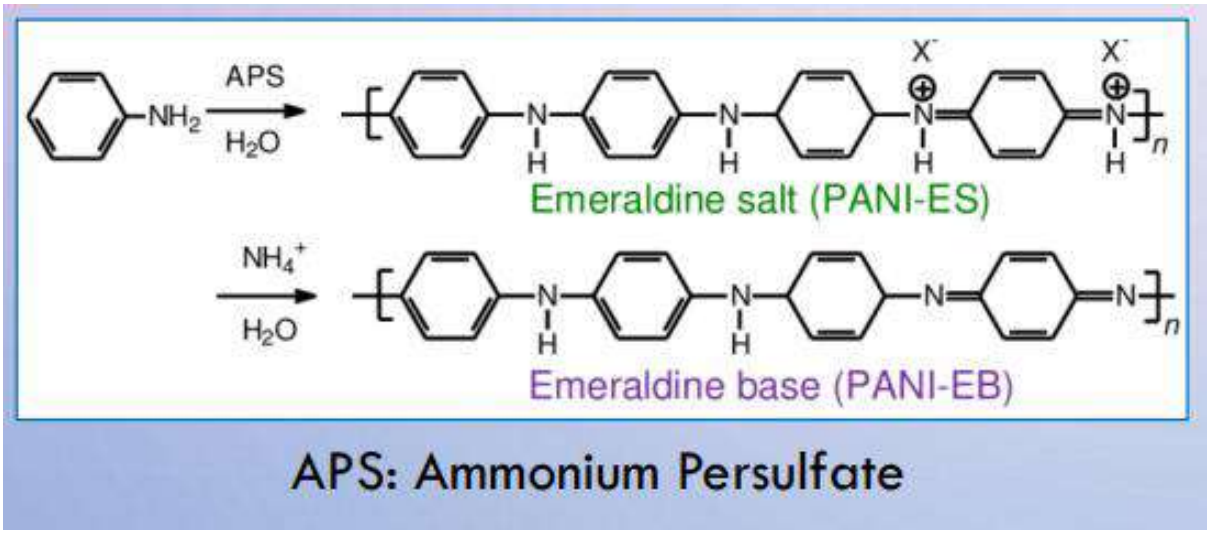
4.12.5 Preparation of Polyacetylene

The polyacetylene film forms at the gas-liquid interface when acetylene gas passes through a heptane solution of the Ziegler-Natta catalyst.

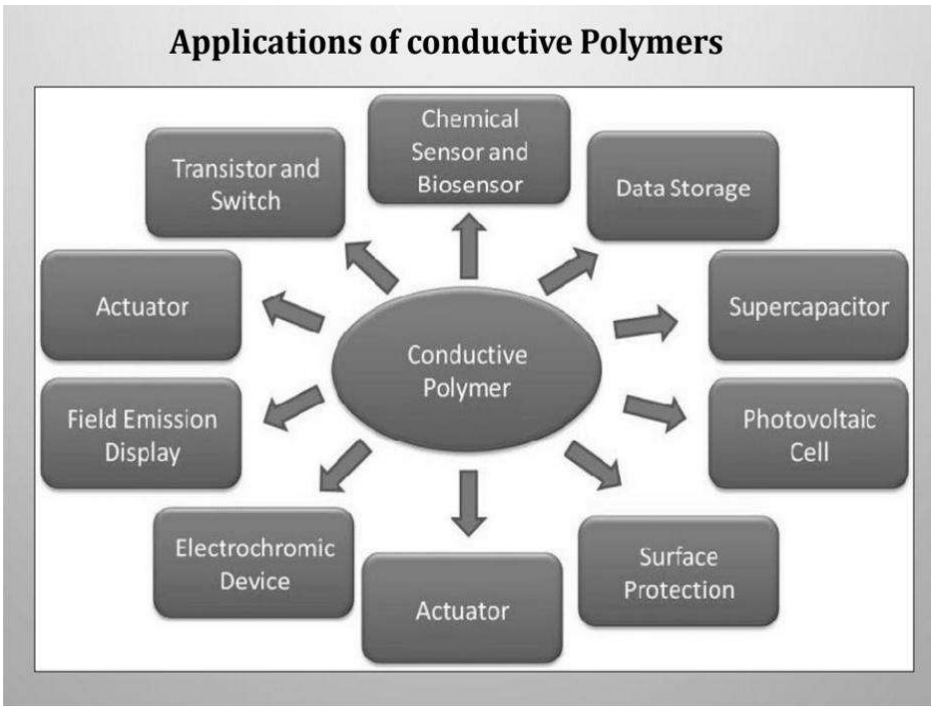


4.12.6 Preparation of Polyaniline

Polyaniline prepared by free radical mechanism of chain growth polymerization by using ammonium persulfate as initiator.



4.12.7 Applications of Conductive Polymers



Applications of Conducting Polymers

- Conducting polymers find their applications in rechargeable batteries.
- Conducting polymers also used for making analytical sensors for pH, O_2, NO_X, SO_2, NH and glucose.
- Conducting polymers also used for making ion exchangers.
- Conducting polymers are used for making electronic displays and optical fibres.
- Conducting polymers are applicable in photovoltaic devices.

4.12.8 Factors Affecting Conductivity

1. **Band Gap:** Band gap increases, conductivity decreases.
2. **Doping Level:** Conductivity increases with increase in doping level.
3. **Temperature:** The conductivity of a conducting polymer decreases with decrease in temperature.
4. **Conjugation Length:** Conductivity increases with increase in conjugation (or) chain length.

4.13 Biodegradable Polymers

Biodegradable plastics are plastics that are capable of being decomposed by bacteria or other microorganisms.

Example: Aliphatic polyesters are biodegradable due to their enzymatically hydrolysable ester bond.

These are classified as

1. **Natural**
   - 1. Polysaccherides- Eg: starch and cellulose.
   - 2. Proteins – Eg: gelatin, casein, silk, wool
   - 3. Polyesters- Eg: polyhydroxyalkanoates
   - 4. Others- Eg: lignin, natural rubber.
2. **Synthetic** E.g: PVA, PLA, PGA etc.,

4.13.1 Properties of Biodegradable Polymers

- Moisture sensitive.
- Sensitive towards acids and bases.
- Dissolves in chlorinated organic solvents.
- Good oxygen permeability.
- PLA is biocompatible.

4.13.2 Applications of Biodegradable Polymers

- It is used in number of biomedical applications such as drug delivery, stents, dialysis media, medical implants like anchors, screws, pins, mesh, surgical threads etc.,
- Disposable table wares, packages for food and flushable sanitary products, diapers etc.
- Used in food industry as a packing material, food storage and compostable bags, chewing gums.

- Adhesive for wood, paper & cloth for book bindings, envelops, wall paper, etc.

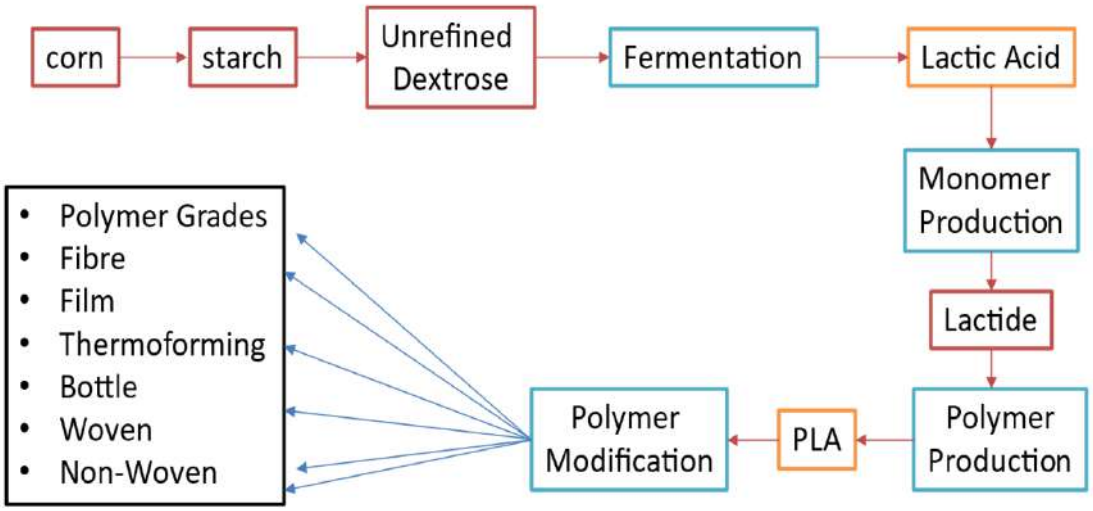
4.13.3 Polylactic Acid

- In 1932 Carother (Dupont) created PLA
- Extremely high cost of manufacturing
- Highly versatile thermoplastic polymer
- Made 100 % from renewable resources
- Sources of lactic acids are corn, starch, sugar beets and wheat.

Properties

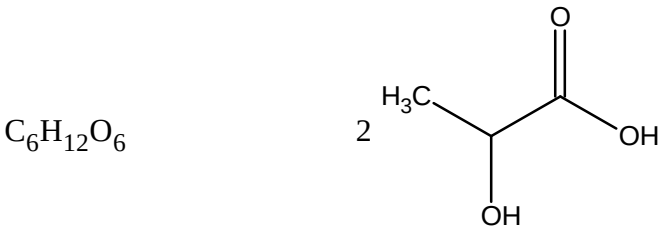
- Insoluble in water, moisture and grease resistant
- Biodegradable and compostable
- Clarity and glossiness similar to PET
- Comparable physical properties to polyethylene terephthalate (PET)

<u>Manufacturing Process of Polylactic Acid Materials</u>

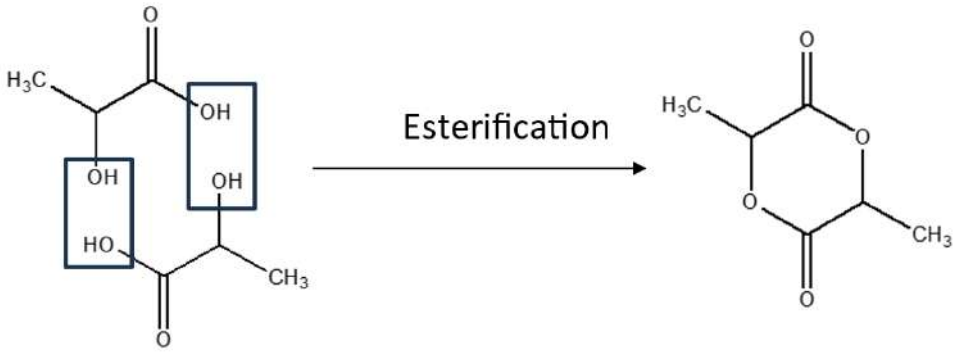


Preparation of Polylactic acid form Lactic Acid:

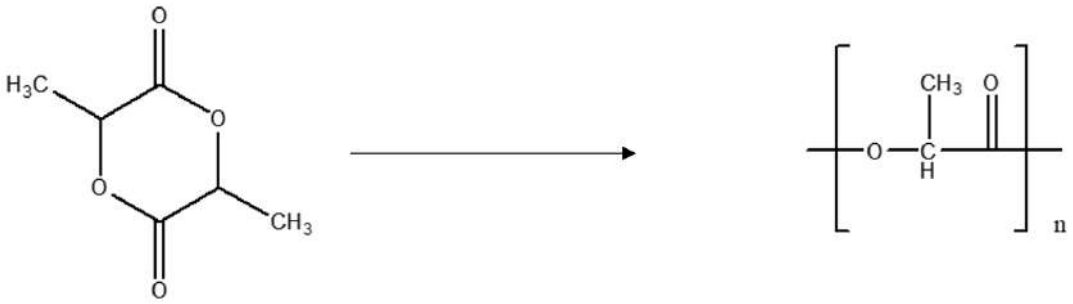
Step 1: Bacteria break down one molecule of dextrose into two molecules of lactic acid.



Step 2: Two molecules of lactic acid combine to form one molecule of lactide.



Step 3: The lactide polymerizes through ring opening polymerization to polylactic acid.

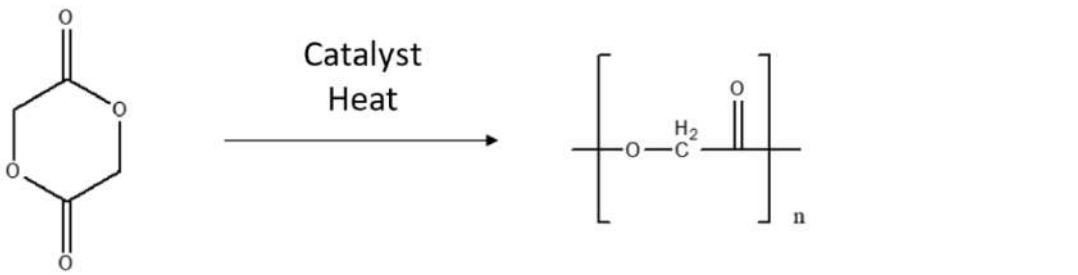


Uses and Applications

1. Single use items: Plates, cups, film wrap
2. Plastic bottling and fast – food companies
3. Textile industry
4. Paper coatings, clothing fibres, compost bags
5. Biomedical field sutures, stents and dialysis
6. Polylactic acid injections for skin rejuvenation

4.13.4 Polyglycolic acid

Preparation



Glycolide

Polyglycolide / polyglycolic acid

Properties

- It has high melting point 225 – 230^{0}C.
- Its highly crystalline nature gives it excellent mechanical strength.
- Biodegradability: It gradually degrades into its monomer units glycolic acids.
- PGA has high tensile strength which is equal to that of polypropylene.
- Highly resistant to hydrolysis.

Applications

- Used in absorbable sutures. Also utilized in drug delivery systems, tissue engineering and orthopaedic devices.
- Finds its utility in eco-friendly applications, such as biodegradable packaging material.
- Its complete degradation into non-toxic compounds makes it a suitable replacement for traditional, non-degradable plastic materials.

Short Answer Questions

1. Explain the functionality of monomers.
2. Interpret the significance of Ziegler Natta Catalyst.
3. Explain the steps of chain growth polymerization.
4. Illustrate the conducting polymers.
5. List out the applications of poly lactic acid.
6. List out the applications of poly aniline.
7. Write a note on the carbon fibers.
8. Differentiate thermo plastic and thermo setting polymers.
9. Explain the preparation method of Teflon.
10. List out the applications of elastomers.

Descriptive Questions

1. Demonstrate the mechanism of coordination polymerization.
2. Demonstrate the mechanism of chain growth polymerization.
3. Demonstrate the mechanism of step growth polymerization.
4. Explain the preparation properties and applications of PVC.
5. Explain the preparation properties and applications of Bakelite.
6. Explain the preparation properties and applications of Nylon 6,6.
7. Explain the preparation properties and applications of Buna-S.
8. Explain the preparation properties and applications of Buna -N.
9. Illustrate the conducting polymers with their mechanism of conduction along with applications.
10. Interpret the bio degradable polymers and write a note on poly glycolic acid.

Unit – 5

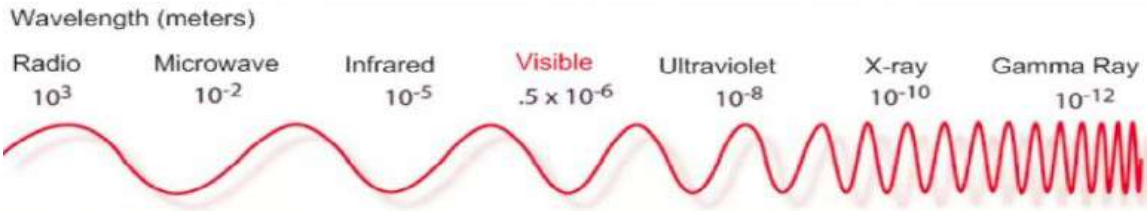
Instrumental Methods and Applications



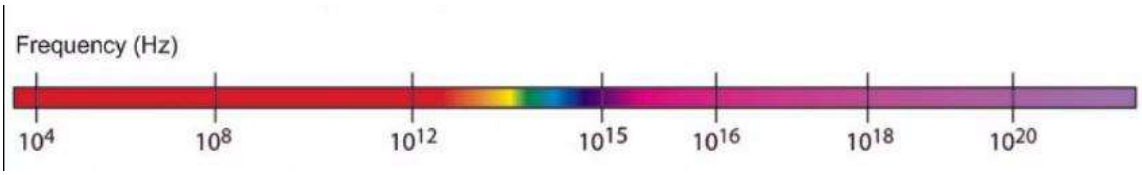
5.1 Electromagnetic Spectrum

The electromagnetic spectrum is the complete spectrum (or continuum) of all form of "light". An electromagnetic wave consists of electric and magnetic fields which vibrate – thus making waves.

The electromagnetic spectrum consists of



Wavelength decreases **Frequency increases** **Energy increases**



All of these electromagnetic waves are similar in a lot of important ways and can travel through empty space.

The speed of these waves at empty space is about 3×10^{8} m/s.

Electromagnetic waves can travel extremely fast in vacuum but differ in their wavelength and frequency.

The frequency of a wave is the number of times it repeats in one second, and frequency is usually measured in units of Hertz (1Hz = 1cycle/second).

The wavelength is the distance between successive wave peaks. For electromagnetic waves, wavelength is usually measured in units of nanometres (1 nm = 1×10^{-9} m).

5.2 Laws of Absorption

Beer's Law: The intensity of a beam of monochromatic light decreases exponentially with the increase in concentration of absorbing substances.

$$- dI / dC \propto I$$

$$\mathbf{I = I_0 e\ KC}$$

Lambert's Law: When a beam of light is allowed to pass through a transparent medium, the rate of decrease of intensity with the thickness of medium is directly proportional to the intensity of the light.

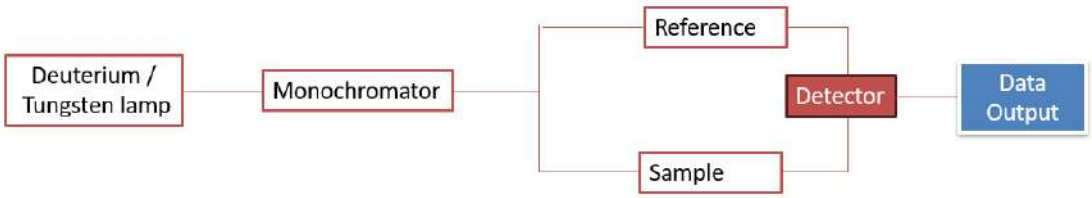
$$A = \varepsilon . C . l$$

Where ε. = molar absorptivity or absorption coefficient; C = concentration; l = path length

Limitations of Laws:

1. The real limitation of the beer's law successfully in describing the absorption behaviour of dilute solutions only.
2. As degree of interaction depends upon the concentration, the occurrence of this phenomenon causes deviations from linear relationship between absorbance and concentrations.

5.3 Instrumentation of UV Spectroscopy



UV spectroscopy obeys the Beer – Lambert law. Instrumentation and working of the UV spectrometers can be studied simultaneously.

UV spectrometers consists the following parts-

Light Source: Tungsten filament lamps and Hydrogen – Deuterium lamps are most widely used and suitable light sources which can cover the whole UV/Vis region.

Monochromator: Monochromators generally composed of prism and slits. The most of the spectrophotometers are double beam spectrophotometers.

The radiation emitted from the primary light source is dispersed with the help of rotating prisms.

The various wavelengths of the light source which are separated by the prism are then selected by the slits such the rotation of the prism results in a series of continuously increasing wavelength to pass through the slits for recording purpose.

The beam selected by the slit is monochromatic and further divided into two beams with the help of another prism.

Sample and Reference Cells: One of the two divided beams is passed through the sample solution and second beam is passes through the reference solution.

Both sample and reference solution are contained in the cells. These cells are made of either silica or quartz. Glass can't be used for the cells as it also absorbs in the UV region.

Detector: Generally, two photocells serve the purpose of detector in UV spectroscopy. One of the photocells receives the beam of sample and second detector receives the beam from the reference.

The intensity of the radiation from the reference cell is stronger than the beam of sample cell. This results in the generation of pulsating or alternating currents in the photocells.

Amplifier: The alternating current generated in the photocells is transferred to the amplifier. The amplifier is coupled to a small servo meter.

Generally current generated in the photocells is of very low intensity, the main purpose of amplifier is to amplify the signals many times so we can get clear and recordable signals.

Recording Devices: Most of the time amplifier is coupled to a pen recorder which is connected to the computer. Computer stores all the data generated and produces the spectrum of the desired compound.

5.3.1 Applications of UV-Vis spectroscopy

1. Detection of functional groups
2. Detection of extent of conjugation
3. Identification of an unknown compound
4. Determination of conjugations of geometrical isomers (cis low and trans high)
5. Determination of the purity of the substance

5.4 Electronic Transitions

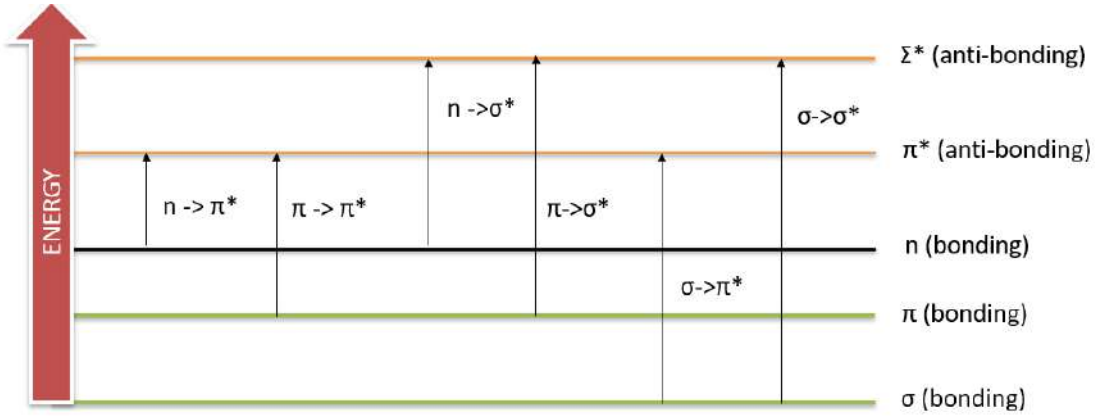
$\sigma \rightarrow \sigma^*$ transition
σ electron from orbital is excited to corresponding anti-bonding orbital σ^*.
The energy required is large for this transition.
e.g. Methane (CH_4) has C – H bond only and can undergo $\sigma \rightarrow \sigma^*$ transition and shows absorbance maxima at 125 nm.

$\pi \rightarrow \pi^*$ transition
π electron in a bonding orbital is excited to corresponding anti-bonding orbital π^*.
Compounds containing multiple bonds like alkenes, alkynes, carbonyl, nitriles, aromatic compounds etc undergo $\pi \rightarrow \pi^*$ transitions.
e.g. Alkenes generally absorb in the region 170 to 205 nm.

$n \rightarrow \sigma^*$ transition
Saturated compounds containing atoms with lone pair of electrons like O, N, S and halogens are capable of $n \rightarrow \sigma^*$ transition.
These transitions usually require less energy than $\sigma \rightarrow \sigma^*$ transitions.
The number of organic functional groups with $n \rightarrow \sigma^*$ peaks in UV region is small (150 – 250 nm)

$n \rightarrow \pi^*$ transition
An electron from non-bonding orbital is promoted to anti-bonding π^* orbital.
Compounds containing double bond involving hetero atoms (C=O, C≡N, N=O) undergo such transitions.
$n \rightarrow \pi^*$ transitions require minimum energy and show absorption at longer wavelength around 300 nm.

$\sigma \rightarrow \pi^*$ transition & $\pi \rightarrow \sigma^*$ transition
These electronic transitions are forbidden transitions and are only theoretically possible.
Thus $n \rightarrow \pi^*$ & $\pi \rightarrow \pi^*$ electronic transitions show absorption in region above 200 nm which is accessible to uv-visible spectrophotometer.

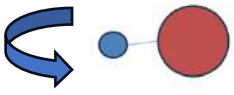


5.5 Infrared Spectroscopy

- Infrared spectroscopy or vibrational spectroscopy is study of absorption of Infrared radiation results in vibrational transitions.
- It is an important analytical technique for determining structure of both organic and inorganic compounds.
- Lies in the wavelength of 0.8 – 100 nm.
- Between visible and microwave region.
- Atoms in a molecule do not remain in fixed positions but vibrate about their mean positions.
- On absorption of IR transition from ground vibrational level to excited vibrational level give rise to closely packed absorption spectrum.
- The energy of the molecule at vibrational level (Hooke's law) $E_v = (V+1/2)\ h\ \nu'.$, v = 0,1,2,3….
  Where v is the vibrational quantum number and ν' is vibrational frequency.
- IR spectrum is obtained by plotting absorbance v/s wavelength.
- Since wavelength is inversely proportional to energy the wavelength in cm^{-1} is used to represent the position of absorption.

5.5.1 Principle of IR Spectroscopy

- Molecules are made up of atoms linked by chemical bonds.
- The movement of atoms and the chemical bonds like spring and balls.
- The characteristic vibration is called natural frequency of vibration.



5.5.2 Selection Rules

The selection rule for a vibrational transition in the di atomic molecule is $\Delta v = \pm 1$
IR-radiation is absorbed only when a change in dipole moment of the molecule takes place. Complete symmetry about a bond may eliminate certain absorption bands.
Dipple moment = $\mu = Q \times r$
Q= charge and r = distance between charges

Therefore, number of absorption bands observed is not exactly equal to the fundamental vibrations, some of the fundamental vibrations are IR active while others are not.
This is governed by selection rule described below.

1. In a molecule with a centre of symmetry, the vibrations symmetrical about the centre of symmetry are IR-inactive.
2. The vibrations which are not symmetrical about the centre of symmetry are IR-active.
   Here are some examples which could explain the selection rule.
   a. All the symmetrical diatomic molecules such as H_2, N_2 and Cl_2 etc. are IR-inactive.
   b. The symmetrical stretching of the C=C bond in ethylene (centre of symmetry) is IR-inactive.
   c. The symmetrical stretching in CO_2 is IR-inactive, whereas asymmetric stretching is IR-active.

5.5.3 Fundamental Modes of Vibration

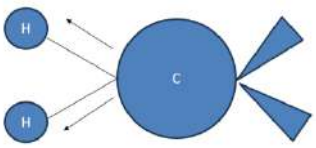
There are two types of vibrations
1. Stretching vibrations
2. Bending vibrations

1. Stretching Vibrations

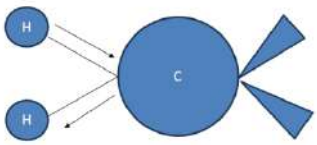
Vibrations or oscillations along the line of bond
Change in bond length resulting from change in interatomic distance (r)
Occurs at higher energy 4000 – 12500 cm^{-1}

These are again 2 types
   a. Symmetrical stretching
   b. Asymmetrical stretching

Symmetrical Stretching
2 bonds increase or decrease in length simultaneously
Symmetrical stretching is IR – inactive (no change in μ)



Asymmetrical Stretching
In this one bond length is increased and other is decreased



2. Bending Vibrations

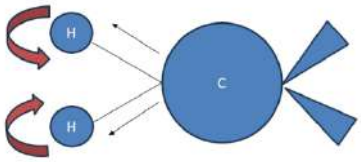
Change in bond angle in the position of a group of atoms with respect to the rest of the molecule.

Bending Modes

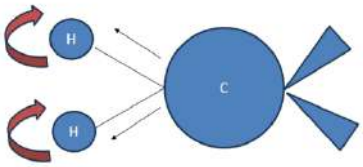
1. Scissoring and Rocking

In-plane bending modes (atoms remain in the same plane)

Scissoring
This is an in-plane bending
Two atoms are approach each other
Bond angles are decrease



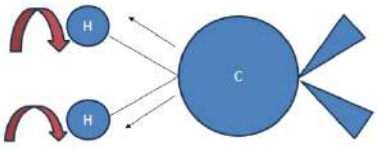
Rocking
Movement of atoms take place in the same direction



2. Wagging and Twisting

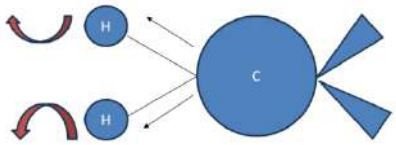
Out-of-plane bending modes (atoms move out of plane)

Wagging
Two atoms move to one side of the plane.
They move up and down the plane

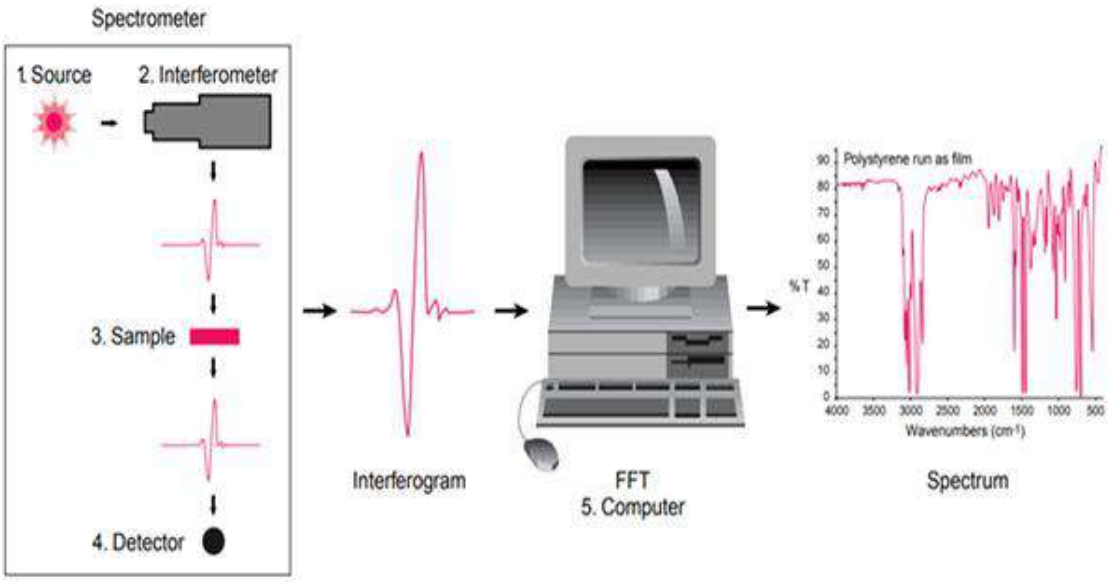


Twisting

One atom moves above the plane and another atom moves below the plane



5.5.4 Instrumentation of IR Spectroscopy



1. In infrared spectroscopy, IR radiation is passed through a sample.
2. Some of the infrared radiation is absorbed by the sample and some it is passed through (transmitted).
3. The resulting spectrum represents the molecular absorption and transmissions, creating a molecular finger print of the sample.
4. The normal instrumentation process is as follows

   a. The Source: Infrared energy is emitted from a glowing black-body source.
   This beam passes through an aperture (slit) which controls the amount of energy presented to the sample.

   b. The Interferometer: The beam enters the interferometer where the "spectral encoding" takes place. The resulting interferogram signal then exists the interferometer.

   c. The Sample: The beam enters the sample compartment where it is transmitted through or reflected off of the surface of the sample, depending on the type of analysis being accomplished (able to perform).

   This is where specific frequencies of energy, which are uniquely characteristic of the sample, are absorbed.

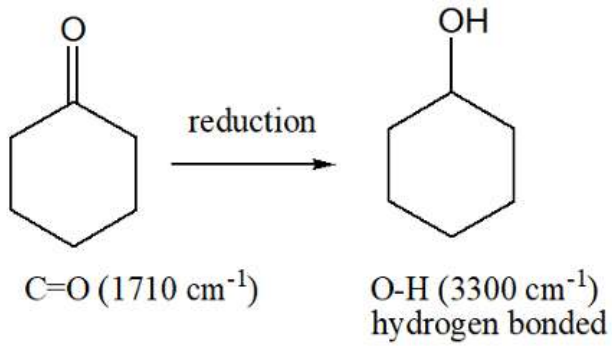
d. The Detector: The beam finally passes to the detector for final measurement. The detectors used are specially designed to measure the special interferogram signal.

e. The Computer: The measured signal is digitized and sent to the computer where the Fourier transformation takes place.

The final infrared spectrum is then presented to the user for interpretation and any further manipulation.

5.5.5 Applications of IR spectroscopy

1. Identification of functional group
2. Distinction between intermolecular and intra-molecular hydrogen bonding
3. Identification of purity of the compound. If compound is impure the additional peaks are observed in the IR spectrum
4. Study of chemical reaction



5. Identification of geometrical isomers (cis – trans)

5.6 Chromatography

Chromatography is a method by which a mixture is separated by distributing its components between two phases.

The stationary phase remains fixed in place while the mobile phase carries the components of the mixture through the medium being used.

The stationary phase acts as a constraint on many of the components in a mixture, slowing them down to move slower than the mobile phase.

The movement of the components in the mobile phase is controlled by the significance of their interactions with the mobile and/or stationary phases.

Because of the differences in factors such as the solubility of certain components in the mobile phase and the strength of their affinities for the stationary phase, some components will move faster than others, thus facilitating the separation of the components within that mixture.

5.6.1 Classification of Chromatography

1. **Liquid Chromatography**
   - **Liquid - Liquid Chromatography (LLC):** Separation involved by using liquid stationary phase and liquid mobile phase
   - **Liquid – Solid Chromatography (LSC):** Stationary phase is solid and mobile phase is liquid
   - **Ion – Exchange Chromatography (IEC):** The counter ions of the stationary phases are selectively exchanged with ionic components of the sample.
   - **Exclusion Chromatography (EC):** The stationary phase is porous, gel structure the solute molecules are separated based on molecular geometry

2. **Gas Chromatography**
   - **Gas – Liquid Chromatography (GLC):** The stationary phase is liquid and the mobile phase is gas
   - **Gas – Solid Chromatography (GSC):** The stationary phase is solid and the mobile phase is gas

Types of Chromatography

Liquid chromatography – Separates liquid samples with a liquid solvent (mobile phase) and a column composed of solid beads (stationary phase)

Gas chromatography – Separates vaporized samples with a carrier gas (mobile gas) and a column composed of a liquid or of solid beads (stationary phase)

Paper chromatography – Separates dried liquid samples with a solvent (mobile phase) and a paper strip (stationary phase)

Thin layer chromatography – separates dried liquid samples with a liquid solvent (mobile phase) and a glass plate covered with a thin layer of alumina or silica gel (stationary phase)

5.6.2 HPLC (High Performance Liquid Chromatography)

High performance liquid chromatography is an advancement of column chromatography to get higher efficiencies and better resolution by using smaller particles of stationary phase.

The mobile phase pumped through column about 300 psi pressure to achieve high flow rate 1-5 cm^3/ minute.

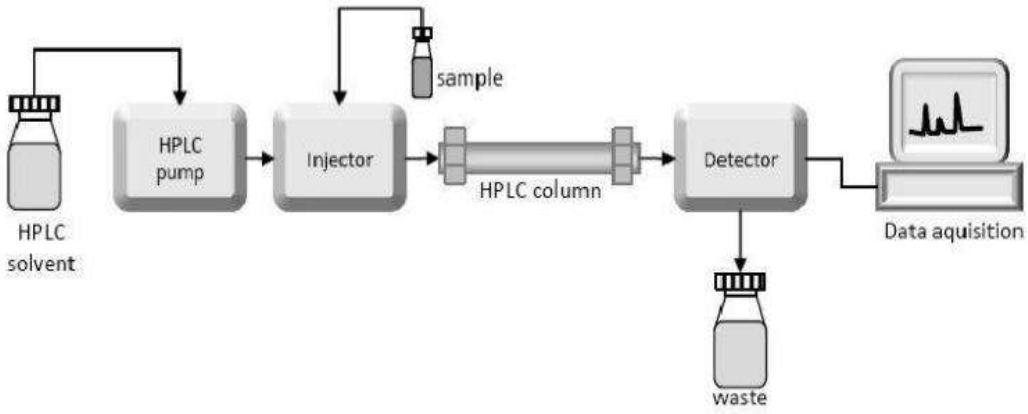
The column is usually packed with smaller size particles of 3μm diameter.

5.6.3 Principle of High-Performance Liquid Chromatography (HPLC)

- The purification takes place in a separation column between a stationary and a mobile phase.
- The stationary phase is a granular material with very small porous particles in a separation column.
- The mobile phase, on the other hand, is a solvent or solvent mixture which is forced at high pressure through the separation column.

- Via a valve with a connected sample loop, i.e. a small tube or a capillary made of stainless steel, the sample is injected into the mobile phase flow from the pump to the separation column using a syringe.
- Subsequently, the individual components of the sample migrate through the column at different rates because they are retained to a varying degree by interactions with the stationary phase.
- After leaving the column, the individual substances are detected by a suitable detector and passed on as a signal to the HPLC software on the computer.
- At the end of this operation/run, a chromatogram in the HPLC software on the computer is obtained.
- The chromatogram allows the identification and quantification of the different substances.

5.6.4 Instrumentation of HPLC



The Pump

- The development of HPLC led to the development of the pump system.
- The pump is positioned in the most upper stream of the liquid chromatography system and generates a flow of eluent from the solvent reservoir into the system.

Injector

- An injector is placed next to the pump.
- The simplest method is to use a syringe, and the sample is introduced to the flow of eluent.
- The most widely used injection method is based on sampling loops.

Column

- The separation is performed inside the column.
- The recent columns are often prepared in a stainless-steel housing, instead of glass columns.

Detector

- Separation of analytes is performed inside the column, whereas a detector is used to observe the obtained separation.

- The composition of the eluent is consistent when no analyte is present. While the presence of analyte changes the composition of the eluent. What detector does is to measure these differences.

Recorder

- The change in eluent detected by a detector is in the form of an electronic signal, and thus it is still not visible to our eyes.
- In older days, the pen (paper)-chart recorder was popularly used. Nowadays, a computer-based data processor (integrator) is more common.

Degasser

- When gas is present in the eluent, this is detected as noise and causes an unstable baseline.
- Degasser uses special polymer membrane tubing to remove gases.

Column Heater

- In order to obtain repeatable results, it is important to keep consistent temperature conditions.
- Thus, columns are generally kept inside the column oven (column heater).

5.6.5 Applications of High-Performance Liquid Chromatography (HPLC)

The HPLC has developed into a universally applicable method so that it finds its use in almost all areas of chemistry, biochemistry, and pharmacy.

- Analysis of drugs
- Analysis of synthetic polymers
- Analysis of pollutants in environmental analytics
- Determination of drugs in biological matrices
- Isolation of valuable products
- Product purity and quality control of industrial products and fine chemicals
- Separation and purification of biopolymers such as enzymes or nucleic acids
- Water purification
- Pre-concentration of trace components
- Ligand-exchange chromatography
- Ion-exchange chromatography of proteins
- High-pH anion-exchange chromatography of carbohydrates and oligosaccharides

Short Answer Questions

1. Represent electromagnetic spectrum with a neat sketch.
2. Explain Beer’s law.
3. Explain Lambert’s law.
4. Explain the principle of IR spectroscopy.
5. Represent the diagram of electronic transitions with a neat sketch.
6. List out the applications of UV-visible spectroscopy.
7. List out the applications of IR Spectroscopy.
8. List out the applications of HPLC.
9. Explain the classification of chromatography.
10. Differentiate wagging and scissoring modes of vibration.

Descriptive Questions

1. Demonstrate the instrumentation of UV-visible spectroscopy.
2. Explain the instrumentation of IR Spectroscopy.
3. Explain the instrumentation of HPLC chromatography.
4. Explain the different electronic transitions occurred in UV spectroscopy.
5. Explain the fundamental modes of vibration in IR Spectroscopy.
6. Illustrate the classification of chromatography.
7. Explain the electromagnetic spectrum with various regions representation.
8. Illustrate about UV Spectroscopy with applications.
9. Illustrate about IR Spectroscopy with applications.
10. Illustrate about High-performance liquid chromatography.

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Dr. R. S. S. Srikanth Vemuri is an accomplished Associate Professor at Vignan's Institute of Engineering for Women in Visakhapatnam, specializing in kinetics, surfactants, and electron transfer reactions. With a Ph.D. in Physical Chemistry from Andhra University and over ten years of teaching and research experience, he is dedicated to delivering high-quality education in Engineering Chemistry and Applied Chemistry. Dr. Vemuri has a robust research portfolio, including 20 published papers, contributions to international and national conferences, and a patent. His notable works include studies on surfactant- complex analysis and sustainable approaches for industrial wastewater treatment. In addition to his academic responsibilities, Dr. Vemuri is committed to professional development, having participated in numerous faculty development programs and obtained certifications in various fields such as chromatography and data science. He has been recognized with several awards, including the Young Scientist Award by VD GOOD Organization in 2021. Dr. Vemuri's passion for teaching is reflected in his focus on fostering interactive classroom discussions and mentoring students. His proficiency in multiple languages, strong communication skills, and strategic planning abilities further enhance his contributions to academia.

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